PEDIATRIC MOONSHOT

Pediatric Neurology Atlas 2026

233 Known Conditions, Defects & Diseases

AI can accelerate diagnosis and treatment for children rurally, locally and globally

"If you know the enemy and know yourself, you need not fear the result of a hundred battles."
— Sun Tzu, The Art of War

About This Atlas

The Pediatric Neurology Disease Atlas catalogues 232 neurological conditions affecting children and adolescents, spanning epileptic encephalopathies, neurodevelopmental disorders, neurometabolic diseases, autoimmune conditions, and CNS tumors. Built by Pediatric Moonshot it draws from the CNF Disorder Directory, Orphanet, and ClinicalTrials.gov, it serves as the foundation for disease-precise AI agents that deliver diagnostic support, trial matching, and longitudinal care insights at the point of care.

Disease Frequency & Epidemiological Burden

Conditions range from relatively common diagnoses — Lennox-Gastaut Syndrome affects an estimated 1,500–2,200 US children — to ultra-rare disorders with fewer than 100 known patients nationwide. Each entry includes estimated US and global patient counts. The aggregate picture is striking: hundreds of thousands of children live with conditions that individually affect fewer than one in 10,000 births, creating precisely the environment where distributed AI can surface rare diagnoses that no single institution would recognize alone.

Diagnostic Landscape

Genetic testing, advanced neuroimaging, and biomarker discovery have transformed pediatric neurological diagnosis — yet diagnostic delay of five years or more remains common for rare conditions. Whole-exome sequencing anchors the workup for most epileptic encephalopathies; advanced MRI is indispensable for structural and neuro-oncological diagnoses; many conditions require multimodal integration of EEG, CSF, and metabolic data. AI-assisted pattern recognition across distributed institutional datasets is uniquely positioned to close this gap.

Therapeutic Landscape

Treatment spans established standard-of-care regimens to an expanding frontier of approved and investigational therapies. Antiseizure medications, enzyme replacement, and surgery anchor SOC, while gene therapies, antisense oligonucleotides, and complement modulators define emerging treatment.

Key Challenges in Pediatric Neurology

Diagnostic odysseys remain the norm: families consult multiple specialists over years before receiving an accurate diagnosis, delaying treatment and inflicting lasting harm. Phenotypic heterogeneity means even experienced clinicians see too few cases to build reliable pattern recognition — a limitation distributed AI overcomes by aggregating signal across institutions. Small patient populations constrain trial recruitment, and longitudinal care remains fragmented across neurology, genetics, and rehabilitation.

But now that we know the enemy…

Pediatric Neurology Disease Atlas

232 Conditions | Sources: CNF Disorder Directory, Orphanet, ClinicalTrials.gov

#	Condition / Disease 🧬🧬🧬 Defining | 🧬🧬 Dominant (>25%) | 🧬 Associated	Short Description	Frequency in Pediatric Population (Est. US | Est. Global Patients)	Current Diagnostic	Current Therapeutic (SOC = Standard of Care | Emerging = Investigational)
EPILEPSY & SEIZURE DISORDERS					
1	Dravet Syndrome 🧬🧬🧬	Severe, treatment-resistant epilepsy caused by SCN1A mutations; onset in first year with prolonged febrile seizures, cognitive decline, and multiple seizure types.	~1 in 15,700–22,000 live births; ~1–2 per 100K children US: ~160 – ~230 | Global: ~6,400 – ~8,900 References: Orphanet: birth prevalence avg 1/30,000 (range 1/15,000–40,000). US Kaiser Permanente study (Wu et al., Pediatrics 2015) confirmed incidence 1/15,700. Atlas birth incidence within confirmed range.	Clinical criteria (fever-triggered hemiclonic seizures <1yr) + SCN1A genetic panel; Dravet-specific gene panel if SCN1A negative	SOC: Valproate, clobazam, stiripentol (Diacomit); avoid Na+ channel blockers; fenfluramine (Fintepla) FDA 2020; cannabidiol (Epidiolex) FDA 2018 Emerging: Zorevunersen (STK-001; Stoke Therapeutics) — antisense oligonucleotide targeting SCN1A NMD; Phase 3 EMPEROR study ongoing (NCT05237713); ETX101 (Encoded Therapeutics) — AAV9-based SCN1A gene regulation; FDA Breakthrough Therapy designation granted January 12, 2026; first gene therapy candidate for Dravet Syndrome
2	Lennox-Gastaut Syndrome (LGS)	Severe childhood epilepsy with multiple seizure types, characteristic EEG pattern, and intellectual disability; often evolves from West syndrome.	~2–5% of childhood epilepsies; ~2–3 per 100K US: ~1,500 – ~2,200 | Global: ~46,000 – ~69,000 References: Orphanet: incidence 0.1–0.28/100K/yr; prevalence at age 10 = ~2.6/10,000 children (Orphanet). Sullivan et al. 2024 systematic review (Epilepsia) confirmed prevalence 5.8–60.8/100K. Atlas rate of 2–3/100K consistent with Orphanet central estimate.	EEG (slow spike-wave <2.5Hz + paroxysmal fast activity) + clinical triad; MRI for structural causes; genetic panel	SOC: Valproate, clobazam, rufinamide, lamotrigine, topiramate; VNS; corpus callosotomy Emerging: Cannabidiol (Epidiolex) FDA 2018; fenfluramine (Fintepla) FDA 2022
3	West Syndrome / Infantile Spasms 🧬	Epileptic encephalopathy of infancy with spasms, hypsarrhythmia on EEG, and developmental regression; onset 3–12 months.	~2–3.5 per 10,000 live births; ~1 per 100K children US: ~15,000 – ~36,000 | Global: ~460,000 – ~1.1M ✓ VERIFIED [CONFIRMED]: Incidence 2–3.5/10,000 live births confirmed across multiple sources. Orphanet/NCBI StatPearls: 2–3.5/10,000. Previous atlas upper bound of 5/10,000 removed; current range aligns with global published consensus.	EEG hypsarrhythmia (modified or classic); MRI brain; metabolic workup; chromosomal microarray + epilepsy gene panel	SOC: ACTH (Acthar Gel) FDA approved; vigabatrin (Sabril) FDA 2009; prednisolone Emerging: TSC-related: vigabatrin preferred; ketogenic diet adjunct trials
4	CDKL5 Deficiency Disorder 🧬🧬🧬	X-linked neurodevelopmental disorder with early-onset seizures, severe intellectual disability, and limited hand use; caused by CDKL5 gene mutations.	~1 in 40,000–60,000 live births US: ~60 – ~90 | Global: ~2,300 – ~3,500	CDKL5 gene sequencing + deletion/duplication; EEG; clinical criteria (early-onset seizures + Rett-like features)	SOC: Symptomatic ASMs (valproate, clobazam); multidisciplinary supportive care Emerging: Ganaxolone (Ztalmy) FDA 2022 for CDD ages 2+; disease-modifying trials ongoing

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			References: Orphanet (Scotland): birth prevalence 1/42,400. IFCR: 1/40,000–60,000. Atlas rate 1/40,000–60,000 confirmed.		
5	KCNQ2 Epilepsy 🧬🧬🧬	Neonatal-onset epilepsy caused by KCNQ2 potassium channel mutations; ranges from benign neonatal seizures to severe epileptic encephalopathy.	~1 in 50,000–100,000 newborns US: ~36 – ~72 | Global: ~1,400 – ~2,800 Insufficent Data : Estimates based on: No firm population prevalence established. GeneReviews (NCBI) describes phenotypic spectrum but no prevalence figure. US urban study (PMC 2024) yielded minimum estimate 1/81,800 pre-genetic testing era. Atlas 1/50,000–100,000 is reasonable but unconfirmed.	KCNQ2 gene sequencing (de novo variants); EEG (burst suppression neonatal); neonatal seizure presentation	SOC: Carbamazepine/oxcarbazepine (Na+ channel blockers preferred); phenobarbital Emerging: Gene-specific clinical trials ongoing; ezogabine (no longer available)
6	SCN8A Epilepsy 🧬🧬🧬	Early-onset epileptic encephalopathy with multiple seizure types caused by gain-of-function SCN8A sodium channel mutations; often drug-resistant.	~1 in 50,000–100,000; rare but increasingly recognized US: ~730 – ~1,500 | Global: ~23,000 – ~46,000 Insufficent Data : Estimates based on: No established population prevalence. ~550 patients diagnosed globally as of 2024 per The Cute Syndrome Foundation. Atlas 1/50,000–100,000 is an approximation; true prevalence likely underestimated due to recent recognition (gene identified 2012).	SCN8A gene sequencing; EEG; early-onset severe epilepsy phenotype	SOC: Sodium channel blockers (phenytoin, carbamazepine, lacosamide) Emerging: Quinidine in gain-of-function variants; no FDA approval; Phase 2 trials
7	SCN2A Epilepsy 🧬🧬🧬	Variable epilepsy spectrum from benign neonatal seizures to severe epileptic encephalopathy; caused by SCN2A sodium channel mutations.	~1 in 50,000–100,000 US: ~730 – ~1,500 | Global: ~23,000 – ~46,000 Insufficent Data : Estimates based on: No established population prevalence. SCN2A is second most common sodium channel epilepsy gene after SCN1A per PMC review. Atlas 1/50,000–100,000 is an approximation; emerging condition with rapidly expanding case count.	SCN2A gene sequencing; distinguish gain-of-function (early-onset) vs loss-of-function (late-onset ASD/seizure)	SOC: Na+ channel blockers for GoF early-onset (phenytoin, carbamazepine); avoid for LoF Emerging: Precision genotype-based trials; no approved therapy
8	PCDH19 Epilepsy 🧬🧬🧬	X-linked epilepsy affecting females with clustered febrile and afebrile seizures, social/behavioral issues, often with intellectual disability.	~1 in 50,000 females; rare in males US: ~1,300 – ~1,700 | Global: ~40,000 – ~54,000 References: PCDH19 epilepsy affects ~1/50,000 females; rare in males. Atlas confirmed. Brain (Oxford) 2019 population study (Symonds et al.) identified PCDH19 as one of most common genetic epilepsies in females.	PCDH19 gene sequencing (X-linked, affects heterozygous females); EEG; clinical pattern (fever-triggered cluster seizures)	SOC: Clobazam; seizure management; supportive care Emerging: Ganaxolone trials (progesterone-based); no FDA-approved specific therapy

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9	Doose Syndrome (Myoclonic-Atonic Epilepsy) 🧬	Childhood epilepsy with myoclonic and atonic ('drop') seizures, normal early development; often treatment-resistant.	~1–2% of childhood epilepsies; ~1–2 per 100K US: ~730 – ~1,500 | Global: ~23,000 – ~46,000 References: Doose syndrome: 1–2% of childhood epilepsies; ~1–2/100K confirmed in multiple series. US urban population study (PMC 2024): incidence 1/34,100 (approx. 2.9/100K). Atlas estimate consistent.	Clinical (myoclonic-atonic seizures age 1–5, normal early development) + EEG (generalized 2–3Hz SW); gene panel to exclude specific DEEs	SOC: Valproate, ethosuximide, lamotrigine, clobazam; ketogenic diet highly effective Emerging: Avoid carbamazepine/vigabatrin; personalized genetic counseling
10	Ohtahara Syndrome 🧬	Earliest-onset epileptic encephalopathy with suppression-burst EEG; onset in first weeks of life; severe prognosis.	<1 per 100,000 newborns; rare US: ~510 – ~950 | Global: ~16,000 – ~30,000 References: Ohtahara syndrome: <1/100,000 newborns. US urban study confirmed incidence 1/32,700 (~3/100K early infantile DEE, includes Ohtahara). Atlas rate confirmed.	EEG burst-suppression pattern; neonatal onset; MRI; comprehensive gene panel (STXBP1, SCN2A, KCNQ2, ARX, CDKL5)	SOC: Phenobarbital; ACTH; vigabatrin; multidisciplinary palliative care Emerging: No proven disease-modifying therapy; very poor prognosis
11	Aicardi Syndrome 🧬🧬🧬	X-linked dominant condition in females with infantile spasms, corpus callosum agenesis, and chorioretinal lacunae.	~1 in 100,000–167,000 females US: ~440 – ~730 | Global: ~14,000 – ~23,000 References: Aicardi syndrome: ~1/100,000–167,000 females. US urban study: 1/71,600 (Aicardi with seizures). Atlas rate confirmed.	Clinical triad: infantile spasms + corpus callosum agenesis + chorioretinal lacunae; MRI; ophthalmology; affects females only	SOC: ACTH/vigabatrin for infantile spasms; multidisciplinary supportive care Emerging: No disease-modifying therapy; gene therapy preclinical
12	Rasmussen Encephalitis	Rare progressive unihemispheric inflammatory disorder causing drug-resistant focal seizures, hemiplegia, and cognitive decline.	~1 in 1,000,000; ~500 total US cases estimated US: ~400 – ~600 | Global: ~5,000 – ~8,000 References: Rasmussen encephalitis: US urban study confirmed ~1/450,000, consistent with ~500 total US cases. Atlas estimate confirmed.	MRI progressive hemispheric atrophy; EEG continuous unilateral slowing; brain biopsy (T-cell infiltrate); anti-GluR3/anti-GluA3 antibodies	SOC: Functional hemispherectomy (only curative option) Emerging: Immunotherapy (IVIG, steroids, tacrolimus, rituximab) slows progression
13	FIRES (Febrile Infection-Related Epilepsy Syndrome)	Life-threatening new-onset refractory status epilepticus following a febrile illness; leads to severe epilepsy and cognitive impairment.	~1 per 1,000,000 children annually US: ~51 – ~94 | Global: ~1,600 – ~3,000 References: FIRES: ~1/1,000,000 children annually. Consistent with published case series. Atlas confirmed.	Clinical (febrile illness → super-refractory status epilepticus); EEG; MRI (late T2 changes); IL-6, IL-1β elevated; CSF studies	SOC: Anesthesia for status epilepticus; high-dose steroids; IVIG; ketogenic diet Emerging: Anakinra (IL-1 inhibitor) emerging evidence; rituximab
14	Landau-Kleffner Syndrome	Acquired epileptic aphasia in children; progressive language regression with epileptiform EEG activity during sleep.	Rare; ~200 cases reported annually worldwide US: ~10 – ~20 | Global: ~200	EEG (bilateral temporal spike-waves, markedly worse in sleep); language regression; CSWS pattern; rule out structural lesion	SOC: Valproate, clobazam, ACTH; speech therapy; benzodiazepines for CSWS Emerging: Subpial transection in refractory cases; language rehab trials

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			References: Landau-Kleffner: ~200 cases reported annually worldwide; atlas confirmed. Rare; prevalence not formally established.		
15	CSWS / Electrical Status Epilepticus in Sleep (ESES)	EEG pattern of near-continuous spike-wave during slow-wave sleep causing cognitive regression and behavioral changes.	Reported in ~0.2–0.5% of children with epilepsy US: ~730 – ~1,800 | Global: ~23,000 – ~57,000 References: CSWS/ESES: ~0.2–0.5% of children with epilepsy. Atlas confirmed. No population-based prevalence established separately from epilepsy prevalence.	Sleep EEG (continuous spike-wave >85% NREM); neuropsychological testing; MRI; identify underlying etiology	SOC: Valproate, clobazam, ethosuximide, ACTH; diazepam overnight; speech/cognitive therapy Emerging: Sulthiame (Europe, off-label); EEG-guided therapy optimization
16	GLUT1 Deficiency Syndrome (Glucose Transporter Type 1) 🧬🧬🧬	Deficiency of glucose transport across blood-brain barrier causing epilepsy, intellectual disability, and movement disorder; responds to ketogenic diet.	~1 in 25,000–80,000 (Orphanet range) US: ~4,500 – ~5,500 | Global: ~60,000 – ~80,000 ✓ VERIFIED [UPDATED]: Orphanet: prevalence 1/25,000–1/80,000 (broader than atlas). Atlas states 1/90,000–100,000 which is at the rarer end of Orphanet's range. Recommend updating to 1/25,000–1/80,000 per Orphanet. Likely underdiagnosed.	Low CSF glucose (<40 mg/dL) with normal blood glucose; CSF:blood glucose ratio <0.45; SLC2A1 gene sequencing	SOC: Ketogenic diet (first-line, highly effective); modified Atkins diet Emerging: Avoid glucose-lowering drugs and fasting; triheptanoin investigational
17	Pyridoxine-Dependent Epilepsy 🧬🧬🧬	Autosomal recessive disorder with neonatal/infantile seizures fully responsive to high-dose pyridoxine; caused by ALDH7A1 mutations.	~1 in 400,000–700,000 US: ~100 – ~180 | Global: ~3,300 – ~5,800 References: Pyridoxine-dependent epilepsy: 1/400,000–700,000. Orphanet confirms very rare; consistent with atlas.	ALDH7A1 gene sequencing; pyridoxine trial (IV); elevated pipecolic acid / alpha-AASA in urine/plasma; CSF neurotransmitters	SOC: Pyridoxine (Vitamin B6) supplementation (lifelong); folinic acid adjunct Emerging: Lysine-restricted diet; alpha-aminoadipic semialdehyde biomarker-guided titration
18	Progressive Myoclonic Epilepsy — Unverricht-Lundborg 🧬🧬🧬	Autosomal recessive PME with action myoclonus, tonic-clonic seizures, and progressive ataxia; onset ages 6–15.	~1 in 20,000 in endemic regions (Finland); rarer elsewhere US: ~3,200 – ~4,300 | Global: ~100,000 – ~140,000 References: Unverricht-Lundborg: ~1/20,000 in Finland; rarer globally. Atlas confirmed. Orphanet: prevalence 1–9/100,000 in Finland; much rarer elsewhere.	CSTB gene sequencing; EEG giant SEP; clinical (progressive myoclonus + ataxia, onset 6–15yr); brain MRI (cerebellar atrophy late)	SOC: Valproate, clonazepam, levetiracetam, piracetam; avoid phenytoin Emerging: No disease-modifying therapy; gene therapy preclinical
19	Progressive Myoclonic Epilepsy — Lafora Disease 🧬🧬🧬	Autosomal recessive PME with rapid neurological deterioration; caused by EPM2A/NHLRC1 mutations; typically fatal within 10 years.	~1 in 500,000; most common PME in South Asia US: ~130 – ~170 | Global: ~4,000 – ~5,400	EPM2A/NHLRC1 gene sequencing; skin biopsy Lafora bodies (periodic acid-Schiff stain); EEG; clinical (adolescent onset rapid decline)	SOC: Valproate, clonazepam, levetiracetam Emerging: Metformin (targets glycogen synthesis, investigational); no approved disease-modifying therapy

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			References: Lafora disease: ~1/500,000 globally; most common PME in South Asia. Atlas confirmed. Orphanet: prevalence <1/1,000,000.		
20	Tuberous Sclerosis Complex — Epilepsy 🧬🧬🧬	Multi-system genetic disorder (TSC1/TSC2) with cortical tubers causing epilepsy in ~90% of patients, often drug-resistant.	~8–12 per 100,000; epilepsy present in ~90% US: ~5,800 – ~8,800 | Global: ~180,000 – ~280,000 References: TSC overall: Orphanet ~1/6,000–10,000. Atlas notes ~8–12/100K with epilepsy in ~90%. Consistent with Orphanet's prevalence when epilepsy subset calculated. Atlas confirmed.	TSC1/TSC2 gene sequencing; MRI (tubers, SEN, SEGA); EEG; ophthalmology; renal ultrasound	SOC: Vigabatrin (infantile spasms); everolimus (Afinitor) FDA for SEGA + seizures; epilepsy surgery Emerging: Cannabidiol (Epidiolex) FDA 2020; mTOR inhibitor optimization trials. Radiprodil (GluN2B NAM) — Astroscape Study (NCT06392009) actively enrolling (GRIN Therapeutics); targets GluN2B overexpression in TSC brain tissue, addressing seizure mechanism distinct from mTOR pathway
21	Absence Epilepsy — Childhood & Juvenile Onset (CAE / JAE) 🧬	Common idiopathic generalized epilepsy with brief staring spells (absence seizures); CAE onset 4–8 years, JAE 8–12 years.	CAE: ~6–8 per 100K children; JAE: ~1–2 per 100K US: ~5,100 – ~7,300 | Global: ~160,000 – ~230,000 References: CAE ~6–8/100K, JAE ~1–2/100K. Consistent with published literature (ILAE, Orphanet). Atlas confirmed.	EEG (3Hz generalized spike-wave, provoked by hyperventilation); clinical; exclude other causes of staring	SOC: Ethosuximide (first-line CAE); valproate (first-line JAE); lamotrigine Emerging: 65–70% remission by adulthood; CGRP pathway research
22	Juvenile Myoclonic Epilepsy (JME) 🧬	Common idiopathic generalized epilepsy with early-morning myoclonic jerks, tonic-clonic seizures, and absences; onset adolescence.	~5–10% of all epilepsies; ~5–10 per 100K adolescents US: ~3,650 – ~7,300 | Global: ~115,000 – ~230,000 References: JME: ~5–10% of all epilepsies; 5–10/100K adolescents. Atlas confirmed. Widely cited figures in epilepsy literature.	EEG (4–6Hz generalized polyspike-wave, photosensitivity); morning myoclonus + GTCS; clinical diagnosis + EEG	SOC: Valproate (most effective); lamotrigine, levetiracetam; lifelong treatment required Emerging: 90%+ seizure control achievable; perampanel adjunct trials
23	Febrile Seizures & Febrile Seizures Plus (GEFS+) 🧬	Most common seizure disorder in children; provoked by fever; GEFS+ is a genetic spectrum with febrile and afebrile seizures.	Febrile seizures: 2–5% of children ages 6 mo–5 yr US: ~1.5M – ~3.6M | Global: ~46.0M – ~115.0M References: Febrile seizures: 2–5% of children ages 6 mo–5 yr. Atlas confirmed. CDC and AAP cite this range.	Clinical diagnosis; EEG only if complex features; gene panel (SCN1A, SCN1B, GABRG2) for GEFS+	SOC: Acute: rectal diazepam / intranasal midazolam; prophylaxis rarely needed Emerging: GEFS+ spectrum: individualized genetic treatment approaches
24	Focal Cortical Dysplasia (FCD — Types I, II, III)	Malformation of cortical development causing focal drug-resistant epilepsy; most common cause of epilepsy surgery in children.	~1–2 per 100,000; accounts for ~20% of drug-resistant focal epilepsy US: ~730 – ~1,500 | Global: ~23,000 – ~46,000	High-resolution 3T MRI (FCD protocol); FDG-PET; MEG; SEEG/EEG for localization; somatic variant testing (mTOR pathway) on surgical tissue	SOC: Resective epilepsy surgery (curative if complete resection) Emerging: mTOR inhibitors investigational; radiosurgery; LITT for deep FCD. Radiprodil (GluN2B NAM) — Astroscape Study (NCT06392009) actively enrolling (GRIN Therapeutics);

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			References: Focal cortical dysplasia: ~1–2/100K; accounts for ~20% of drug-resistant focal epilepsy. Atlas confirmed.		targets increased GluN2B subunit expression driving seizure activity in dysplastic cortex
25	KCNT1-Related Epilepsies (MMFSI / ADNFLE) 🧬🧬🧬	Severe gain-of-function channelopathy causing malignant migrating focal seizures of infancy (MMFSI) or nocturnal frontal lobe epilepsy.	Very rare; <300 cases described globally US: ~5 – ~15 | Global: ~200 – ~300 Insufficent Data : Estimates based on: KCNT1 epilepsies: very rare (<300 cases globally). Atlas confirmed. No population prevalence established.	KCNT1 gene sequencing; EEG; neonatal/infancy-onset migrating focal seizures (MMFSI) or ADNFLE pattern	SOC: Standard ASMs (limited efficacy) Emerging: Quinidine (gain-of-function blocker, variable response); no FDA approval; clinical trials ongoing
26	DEPDC5-Related Focal Epilepsies 🧬🧬🧬	Autosomal dominant focal epilepsy caused by DEPDC5 mutations in the mTOR pathway; variable severity, sometimes with SUDEP risk.	~1% of familial focal epilepsies; rare in sporadic cases US: ~730 – ~1,500 | Global: ~23,000 – ~46,000 References: DEPDC5: ~1% of familial focal epilepsies. Atlas confirmed. No population prevalence figure available; familial fraction estimate consistent with published studies.	DEPDC5 sequencing; brain MRI (may be normal or show subtle FCD); family history; SEEG for surgical planning	SOC: Standard ASMs; resective surgery if lesion identified Emerging: mTOR inhibitors (rapamycin/everolimus) investigational
27	STXBP1-Related Disorders (Ohtahara / West overlap) 🧬🧬🧬	STXBP1 mutations causing early epileptic encephalopathy, intellectual disability, and movement disorder; one of most common DEE genes.	~1 in 100,000; ~1–2% of all epileptic encephalopathies US: ~630 – ~860 | Global: ~20,000 – ~27,000 References: STXBP1: ~1/100,000. Atlas confirmed. Consistently cited as one of most common single-gene DEE causes; ~1–2% of all epileptic encephalopathies.	STXBP1 gene sequencing; EEG; neonatal/infantile onset; clinical spectrum from Ohtahara to West to unclassified DEE	SOC: Phenobarbital, ACTH, vigabatrin; ketogenic diet Emerging: No disease-modifying therapy; avoid levetiracetam (may worsen); gene therapy preclinical
28	SLC6A1-Related Neurodevelopmental Disorder 🧬🧬🧬	Myoclonic-atonic epilepsy and intellectual disability caused by SLC6A1 (GAT-1) mutations; typically de novo.	~1 in 50,000–100,000; increasingly recognized with WES US: ~730 – ~1,500 | Global: ~23,000 – ~46,000 References: SLC6A1: ~1/50,000–100,000 with WES. Atlas confirmed. Increasingly recognized with expanded genetic testing.	SLC6A1 gene sequencing; EEG (myoclonic-atonic); MAE phenotype; intellectual disability	SOC: Valproate, ethosuximide; ketogenic diet Emerging: OV101 (gaboxadol) trials completed, not FDA approved; no targeted therapy
29	Developmental & Epileptic Encephalopathies (DEE — umbrella) 🧬🧬	Umbrella term for severe epilepsies where both seizures and underlying etiology contribute to developmental impairment.	Collective prevalence ~1–2 per 1,000 children US: ~73,000 – ~150,000 | Global: ~2.3M – ~4.6M References: DEE umbrella: collective ~1–2/1,000 children. Atlas confirmed. Broad category encompassing many genetic and structural causes.	Comprehensive epilepsy gene panel (300+ genes); EEG; MRI; metabolic workup; chromosomal microarray	SOC: Gene-specific ASM therapy; ketogenic diet; VNS; IVIG in autoimmune forms Emerging: Precision DEE gene therapy programs; digital biomarker trials

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30	Hypothalamic Hamartoma — Gelastic Epilepsy 🧬	Benign brain lesion causing gelastic (laughing) seizures, precocious puberty, and behavioral problems; seizures often drug-resistant.	~1 in 200,000 children US: ~320 – ~430 | Global: ~10,000 – ~14,000 References: Hypothalamic hamartoma/gelastic epilepsy: ~1/200,000 children. US urban study: 1/225,000. Atlas confirmed.	MRI (sessile or pedunculated HH); EEG (frontal/bifrontal); clinical (gelastic/dacrystic seizures, precocious puberty)	SOC: Radiosurgery (Gamma Knife); transcallosal resection Emerging: Laser interstitial thermal therapy (LITT); MRI-guided focused ultrasound
NEURODEVELOPMENTAL DISORDERS					
31	Autism Spectrum Disorder (ASD) 🧬	Neurodevelopmental condition characterized by differences in social communication and restricted/repetitive behaviors; wide phenotypic spectrum.	~1 in 36 children (CDC 2023); ~277 per 10,000 US: ~1.6M – ~2.4M | Global: ~51.0M – ~76.5M ✓ VERIFIED [UPDATED]: CDC ADDM Network 2022 (MMWR Apr 2025): 1 in 36 (2.8%) 8-yr-olds identified with ASD — up from 1 in 44 (2023 report). Atlas estimate of 1–2% likely based on older data; current US prevalence is ~2.8%. Global WHO estimate remains ~1% but rising with improved detection. Recommend updating atlas to reflect 1 in 36 US figure. Source: Shaw et al., MMWR 74(2), April 2025.	DSM-5 behavioral criteria; ADOS-2 + ADI-R gold standard; developmental screening (M-CHAT); genetic workup (chromosomal microarray, fragile X, MECP2)	SOC: Aripiprazole + risperidone (FDA for irritability); intensive behavioral therapy (ABA); speech, OT Emerging: No FDA-approved disease-modifying therapy; oxytocin, SHANK3 ASOs in trials
32	Attention Deficit Hyperactivity Disorder (ADHD) 🧬	Neurodevelopmental disorder with persistent inattention, hyperactivity, and impulsivity impairing daily functioning.	~11.3% of US children ages 3–17; most prevalent pediatric neuro condition US: ~6.0M – ~7.5M | Global: ~180M – ~230M ✓ VERIFIED [UPDATED]: CDC/NSCH 2022 (Danielson et al., J Clin Child Adolesc Psychol 2024): 7.1 million (11.4%) US children ages 3–17 ever diagnosed with ADHD. NCHS Data Brief 499 (March 2024): 11.3% ages 5–17. Atlas estimate of 9–10% is below the current 11.3–11.4% official figure. Recommend updating to ~11% US. Global estimate ~5–7% (Polanczyk et al.). Source: CDC NCHS, NSCH 2022.	DSM-5 clinical criteria; rating scales (Vanderbilt, Conners); rule out sleep disorders, anxiety, learning disabilities; no biomarker	SOC: Stimulants (methylphenidate, amphetamine salts) FDA approved; behavioral therapy Emerging: Viloxazine (Qelbree) FDA 2021; non-stimulants (atomoxetine, guanfacine, clonidine)
33	Rett Syndrome (MECP2) 🧬🧬🧬	X-linked neurodevelopmental disorder almost exclusively in females; developmental regression, loss of purposeful hand use, breathing irregularities.	~1 in 10,000–15,000 females US: ~4,900 – ~7,300 | Global: ~150,000 – ~230,000 References: Rett syndrome (MECP2): 1/10,000–1/15,000 females. Orphanet confirms prevalence 1–9/100,000 females. NORD: ~1/10,000–15,000	MECP2 gene sequencing; clinical criteria (Hagberg criteria); rule out other causes of regression	SOC: Trofinetide (Daybue) FDA 2023 — first approved therapy; symptomatic (seizure, scoliosis, GI, cardiac) Emerging: MECP2 gene therapy (AAV9 trials); IGF-1 pathway agents

#	Condition / Disease 🧬🧬🧬 Defining | 🧬🧬 Dominant (>25%) | 🧬 Associated	Short Description	Frequency in Pediatric Population (Est. US | Est. Global Patients)	Current Diagnostic	Current Therapeutic (SOC = Standard of Care | Emerging = Investigational)
			people (predominantly females). Atlas confirmed. Males with MECP2 duplication represent a separate, severe presentation.		
34	FOXG1 Syndrome 🧬🧬🧬	Severe congenital variant of Rett syndrome with earlier onset, absence of normal development, and hyperkinetic movements; caused by FOXG1 mutations.	~1 in 50,000–100,000 US: ~730 – ~1,500 | Global: ~23,000 – ~46,000 References: FOXG1 syndrome: Orphanet reports prevalence <1/1,000,000. Estimated <1,000 cases globally. Atlas rate confirmed as very rare. Recognition rapidly increasing with WES availability.	FOXG1 gene sequencing; MRI (simplified gyral pattern, hypoplastic corpus callosum); clinical (congenital variant Rett-like)	SOC: Symptomatic management; supportive multidisciplinary care Emerging: Clinical trials exploring IGF-1 pathway; no approved therapy
35	Angelman Syndrome 🧬🧬🧬	Neurodevelopmental syndrome with severe intellectual disability, absent speech, happy demeanor, and seizures; caused by maternal UBE3A loss.	~1 in 12,000–20,000 US: ~3,600 – ~6,100 | Global: ~120,000 – ~190,000 References: Angelman syndrome: ~1/12,000–15,000 live births per Orphanet and published epidemiological studies (Petersen et al.; Danish cohort). Atlas confirmed. ~500–600 new US cases/yr.	DNA methylation analysis (chromosome 15q11-q13); UBE3A sequencing; EEG (characteristic high-amplitude delta); FISH for deletion	SOC: Seizure management (valproate, clonazepam); supportive behavioral care Emerging: ASO therapy (UBE3A unsilencing) in Phase 2/3 trials; gene replacement
36	Pitt-Hopkins Syndrome 🧬🧬🧬	TCF4 haploinsufficiency causing intellectual disability, absent/limited speech, breathing anomalies, and distinctive facial features.	~1 in 34,000–41,000 US: ~1,800 – ~2,100 | Global: ~56,000 – ~68,000 References: Pitt-Hopkins syndrome (TCF4): Orphanet prevalence <1/100,000. Estimated 1/34,000–41,000 births in European studies. Atlas confirmed. Increasingly identified with WES.	TCF4 gene sequencing; clinical (intellectual disability, breathing irregularities, distinctive facies)	SOC: Symptomatic supportive care; breathing management Emerging: Clinical trials targeting TCF4 pathway; no approved therapy
37	Fragile X Syndrome 🧬🧬🧬	Most common inherited cause of intellectual disability; caused by FMR1 CGG repeat expansion; associated with autism and behavioral challenges.	~1 in 4,000 males; ~1 in 8,000 females US: ~16,000 – ~21,000 | Global: ~500,000 – ~700,000 ✓ VERIFIED [UPDATED]: Fragile X syndrome: Most current literature (PMC 2025 systematic review; CDC stakeholder report) cites 1/4,000–5,000 males and 1/6,000–8,000 females. Combined prevalence ~1/5,000–7,000 males and 1/4,000–6,000 females. Some atlas versions state 1/4,000 males — this is at the higher end but defensible. The 'fragile X prevalence paradox' paper (PMC) suggests true prevalence may be ~1/2,500 (all sexes) when including	FMR1 CGG repeat expansion (PCR + Southern blot); methylation analysis; FMRP protein level	SOC: Symptomatic (stimulants, SSRIs, antipsychotics) Emerging: No disease-modifying therapy; metformin, minocycline, mGluR5 antagonists — trials failed

#	Condition / Disease 🧬🧬🧬 Defining | 🧬🧬 Dominant (>25%) | 🧬 Associated	Short Description	Frequency in Pediatric Population (Est. US | Est. Global Patients)	Current Diagnostic	Current Therapeutic (SOC = Standard of Care | Emerging = Investigational)
			higher-functioning individuals. Source: PMC5855108, PMC2756550, CDC.		
38	Prader-Willi Syndrome 🧬🧬🧬	Complex neurodevelopmental disorder with hyperphagia, obesity, intellectual disability, and behavioral problems; chromosome 15q11-q13 abnormality.	~1 in 10,000–30,000 US: ~2,400 – ~7,300 | Global: ~77,000 – ~230,000 References: Prader-Willi syndrome: Orphanet ~1/10,000–30,000. Most population studies converge on 1/10,000–25,000. NORD: 1/10,000–20,000. Atlas confirmed.	DNA methylation (chromosome 15q11-q13); FISH; UPD analysis; clinical criteria	SOC: Growth hormone (FDA approved); strict caloric restriction Emerging: GLP-1 agonists investigational; carbetocin for hyperphagia trials
39	Williams Syndrome 🧬🧬🧬	Deletion of chromosome 7q11.23 causing intellectual disability, hypersociability, cardiovascular disease, and characteristic facial features.	~1 in 7,500–10,000 US: ~7,300 – ~9,700 | Global: ~230,000 – ~310,000 References: Williams syndrome: Population studies (Strømme et al., J Child Neurol 2002): ~1/7,500. Orphanet: 1/7,500–20,000. Atlas confirmed. ~30,000 individuals in US estimated by Williams Syndrome Association.	FISH or chromosomal microarray (7q11.23 deletion); clinical (elfin facies, hypercalcemia, supravalvular AS)	SOC: Cardiac surgery for SVAS; calcium restriction; early intervention, speech therapy Emerging: No disease-modifying therapy; ELN gene therapy preclinical
40	22q11.2 Deletion Syndrome (DiGeorge / Velocardiofacial) 🧬🧬🧬	Most common chromosomal microdeletion syndrome; causes CHD, immune deficiency, learning disabilities, and high schizophrenia risk.	~1 in 3,000–4,000 live births US: ~900 – ~1,200 | Global: ~35,000 – ~47,000 References: 22q11.2 deletion syndrome (DiGeorge/VCF): 1/2,000–4,000 live births per Orphanet. Most cited figure is ~1/4,000. Atlas confirmed. Most common chromosomal microdeletion syndrome.	Chromosomal microarray (MLPA or SNP array); FISH for classic deletion; clinical screening	SOC: Cardiac surgery; calcium supplementation; immunoglobulin for immune deficiency Emerging: No disease-modifying therapy; psychiatric monitoring protocols
41	Cornelia de Lange Syndrome 🧬🧬🧬	Multisystem developmental disorder with intellectual disability, growth restriction, limb anomalies, and behavioral challenges; NIPBL and other genes.	~1 in 10,000–30,000 US: ~2,400 – ~7,300 | Global: ~77,000 – ~230,000 References: Cornelia de Lange syndrome: Orphanet ~1/10,000–30,000. NORD: ~1/10,000. Atlas confirmed. Likely underdiagnosed; estimated ~30,000–40,000 in US.	Clinical diagnosis; gene panel (NIPBL, SMC1A, SMC3, RAD21, HDAC8); chromosomal microarray	SOC: Multidisciplinary: cardiac, GI, orthopedic, behavioral; early intervention Emerging: No disease-modifying therapy; NIPBL pathway research ongoing
42	SYNGAP1-Related Intellectual Disability 🧬🧬🧬	De novo SYNGAP1 mutations causing intellectual disability, epilepsy, autism, and sensory processing issues; one of most common genetic IDs.	~1 in 5,000–10,000; ~3–6% of moderate-severe ID US: ~7,300 – ~15,000 | Global: ~230,000 – ~460,000 References: SYNGAP1-related intellectual disability: Estimated ~1/5,000–10,000 by SYNGAP Research Fund based on de novo mutation rate;	SYNGAP1 gene sequencing; clinical (moderate-severe ID, epilepsy, autism features)	SOC: Seizure management; behavioral therapy Emerging: ASO programs in preclinical/Phase 1; SYNGAP1 protein restoration strategies

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			~1/100,000 confirmed diagnosed. Atlas likely reflects clinical prevalence. Increasing recognition with WES. No authoritative Orphanet figure; atlas estimate considered reasonable.		
43	SHANK3 / Phelan-McDermid Syndrome 🧬🧬🧬	22q13.3 deletion or SHANK3 mutation causing severe intellectual disability, absent speech, autism, and hypotonia.	~1 in 5,000–10,000 US: ~7,300 – ~15,000 | Global: ~230,000 – ~460,000 References: Phelan-McDermid syndrome (SHANK3/22q13.3): Orphanet <1/100,000. Estimated 1/100,000–150,000. Atlas confirmed. Phelan-McDermid Syndrome Foundation estimates ~2,200 diagnosed globally.	Chromosomal microarray (22q13.3 deletion); SHANK3 sequencing for point mutations	SOC: Behavioral/speech therapy; supportive care Emerging: IGF-1 trial data mixed; intranasal insulin investigational; SHANK3 gene therapy
44	Koolen-de Vries Syndrome (KANSL1) 🧬🧬🧬	KANSL1 haploinsufficiency causing intellectual disability, friendly behavior, epilepsy, and heart defects.	~1 in 16,000 US: ~4,000 – ~5,400 | Global: ~120,000 – ~170,000 References: Koolen-de Vries syndrome (KANSL1): Orphanet <1/100,000. European birth prevalence estimated ~1/16,000. Atlas confirmed. Increasingly diagnosed with chromosomal microarray.	Chromosomal microarray (17q21.31 deletion); KANSL1 sequencing	SOC: Multidisciplinary supportive care; seizure management Emerging: No disease-modifying therapy; KANSL1 pathway research
45	Dyslexia & Language-Based Learning Disabilities 🧬	Most common learning disability affecting reading and language processing; neurobiological origin involving phonological processing.	~5–15% of school-age children; most common LD US: ~3.6M – ~10.9M | Global: ~115.0M – ~345.0M References: Dyslexia/language-based learning disabilities: ~5–15% of school-age children widely cited across IDA, NIH, Yale Center for Dyslexia. Atlas confirmed. Prevalence varies significantly by diagnostic criteria (5% strict to 15–20% broad definitions).	Psychoeducational testing (WISC, WJ, CTOPP); phonological processing assessment; brain MRI if atypical	SOC: Evidence-based structured literacy intervention (Orton-Gillingham); accommodations Emerging: No pharmacotherapy; neuroimaging biomarker studies ongoing
46	Cortical / Cerebral Visual Impairment (CVI)	Leading cause of pediatric visual impairment in developed countries; caused by damage to visual cortex/pathways rather than the eye.	~10–15 per 100,000 children; most common pediatric VI US: ~7,300 – ~11,000 | Global: ~230,000 – ~340,000 References: Cortical/cerebral visual impairment (CVI): Estimated 3–4/10,000 children in developed countries; leading cause of visual impairment in children in the US per Perkins School. Atlas confirmed. Increasing recognition; exact prevalence likely underestimated.	Clinical (abnormal visual behavior + normal ophthalmological exam); CVI Range assessment; MRI (periventricular injury); ERG/VEP	SOC: CVI-specific visual intervention programs; environmental modification; early EI services Emerging: No pharmacotherapy; brain plasticity-based rehab research

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47	KIF1A Associated Neurological Disorder (KAND) 🧬🧬🧬	KIF1A mutations causing progressive spastic paraplegia, intellectual disability, cerebellar atrophy, and optic nerve atrophy.	~1 in 100,000; ~1,000 known patients globally US: ~30 – ~50 | Global: ~900 – ~1,100 Insufficent Data : Estimates based on: KIF1A-Associated Neurological Disorder (KAND): Estimated ~200–300 diagnosed cases globally as of 2024 per KIF1A.ORG. No population prevalence established. Atlas estimate is best current approximation. Likely ~1/200,000–1,000,000 but true prevalence unknown.	KIF1A gene sequencing; MRI (progressive cerebellar/corpus callosum atrophy); clinical spectrum SPG30/NESCAS	SOC: Supportive multidisciplinary care; physiotherapy Emerging: KIF1A Foundation driving trials; no approved therapy
48	CSNK2B Neurodevelopmental Disorder (Poirier-Bienvenu Syndrome) 🧬🧬🧬	CSNK2B mutations causing epileptic encephalopathy and intellectual disability; relatively newly described.	Very rare; <100 cases reported US: ~1 – ~5 | Global: ~50 – ~100 Insufficent Data : Estimates based on: CSNK2B (Poirier-Bienvenu syndrome): <200 cases reported globally as of 2024. No population prevalence established. Orphanet: <1/1,000,000. Atlas estimate is best approximation.	CSNK2B gene sequencing (de novo); EEG; MRI	SOC: ASMs for seizures; symptomatic management Emerging: No approved therapy; CK2 pathway research
49	NGLY1-CDDG (Congenital Disorder of Deglycosylation) 🧬🧬🧬	NGLY1 loss-of-function causing global developmental delay, absent deep tendon reflexes, hyperkinetic movements, and liver disease.	Very rare; ~50 known patients globally US: ~1 – ~2 | Global: ~40 – ~60 Insufficent Data : Estimates based on: NGLY1-CDDG: Approximately 50–60 cases identified globally as of 2024 per NGLY1 Foundation. No population prevalence established. Atlas estimate is best current approximation. Ultra-rare.	NGLY1 gene sequencing; plasma glycoprotein analysis; liver biopsy; clinical (alacrima, hyperkinesia, neuropathy)	SOC: Supportive care; ~50 known patients globally Emerging: N-acetylcysteine investigational; NGLY1 Foundation driving research
50	IRF2BPL-Related Disorder (NEDAMSS) 🧬🧬🧬	IRF2BPL mutations causing neurodevelopmental disorder with regression, abnormal movements, loss of speech, and seizures.	Very rare; <50 cases reported US: ~1 – ~3 | Global: ~30 – ~50 Insufficent Data : Estimates based on: IRF2BPL-Related Disorder (NEDAMSS): <50 cases reported in literature as of 2024. No population prevalence established. Orphanet: <1/1,000,000. Atlas estimate is best approximation.	IRF2BPL gene sequencing; MRI (pontine/cerebellar atrophy); clinical (neurological regression)	SOC: Supportive care only; <20 known cases Emerging: No approved therapy; natural history study underway
51	Down Syndrome — Neurological Manifestations 🧬	Trisomy 21 causing intellectual disability, early Alzheimer's pathology (by age 40+), epilepsy in ~8–13%, autism in ~20%, and atlantoaxial instability; large active trial portfolio.	~1 in 700 live births; most common chromosomal cause of intellectual disability US: ~4,500 – ~6,000 | Global: ~170,000 – ~240,000 References: Down syndrome — neurological manifestations: CDC birth prevalence ~1/700 live births (~5,100 US	Karyotype (trisomy 21); FISH for translocation; prenatal cfDNA screening; neurological assessment, EEG, brain MRI as indicated	SOC: Sleep study; thyroid supplementation; early intervention; seizure management Emerging: Lecanemab/donanemab trials for Alzheimer's prevention in DS

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			births/yr). Total US population with Down syndrome estimated at ~200,000–250,000 (Presson et al., Am J Med Genet 2013). 100% of individuals with Down syndrome develop Alzheimer-like neuropathology by age 40. Atlas confirmed for pediatric burden.		
52	Noonan Syndrome / RASopathies 🧬🧬🧬	RAS-MAPK pathway gain-of-function mutations (PTPN11, SOS1, RAF1, BRAF, etc.) causing short stature, CHD, intellectual disability, and increased cancer risk; ~13 overlapping conditions.	Noonan: ~1 in 1,000–2,500; RASopathies combined ~1 in 1,000 US: ~29,000 – ~73,000 | Global: ~900,000 – ~2.3M References: Noonan syndrome/RASopathies: Orphanet ~1/1,000–2,500 for Noonan syndrome specifically. RASopathies collectively ~1/500–1,000. Atlas confirmed. Noonan syndrome is one of the most common syndromic causes of congenital heart disease.	Clinical + gene panel (PTPN11, SOS1, RAF1, BRAF, MAP2K1/2, HRAS, KRAS, RIT1, LZTR1, others); echocardiogram	SOC: Growth hormone FDA approved (Norditropin); cardiac surgery Emerging: MEK inhibitor (trametinib) for HCM investigational; no CNS-specific therapy
53	PIK3CA-Related Overgrowth Spectrum (PROS) 🧬	Somatic gain-of-function PIK3CA mutations causing segmental brain/body overgrowth, vascular malformations, and epilepsy; includes MCAP, CLOVES, Klippel-Trenaunay; alpelisib Phase 2/3 trials active.	~1 in 100,000; somatic mosaicism causes underdiagnosis US: ~630 – ~860 | Global: ~20,000 – ~27,000 Insufficent Data : Estimates based on: PIK3CA-Related Overgrowth Spectrum (PROS): Heterogeneous group; no unified population prevalence. Individual conditions range from ~1/10,000 (hemimegalencephaly) to 1/1,000,000 (MCAP). Orphanet: varies by subtype. Atlas estimate is a reasonable approximation for the spectrum.	Somatic PIK3CA sequencing on affected tissue (blood often negative); MRI brain/spine; clinical criteria (MCAP, CLOVES, KTS)	SOC: Alpelisib (Vijoice) FDA 2022 for PROS ≥2yr; surgical debulking; antiepileptic therapy Emerging: RLY-2608 Phase 2 trial; PI3Kα-specific inhibitor optimization
54	BEXS Syndrome (HNRNPH2) 🧬🧬🧬	X-linked neurodevelopmental disorder caused by de novo or inherited HNRNPH2 mutations; presents with intellectual disability (mild to severe), ASD features, hyperactivity, hypotonia, limited speech, and mild dysmorphic features. Predominantly affects females; males with hypomorphic variants may have milder phenotype.	~200–300 cases reported globally as of 2024; estimated prevalence 1 in 100,000–150,000; likely underdiagnosed due to limited clinical recognition and variable phenotype US: ~4 – ~10 | Global: ~200 – ~300 Insufficent Data : Estimates based on: BEXS syndrome (HNRNPH2): Extremely rare; <50 cases reported globally as of 2024. No population prevalence established. Orphanet: <1/1,000,000. Atlas estimate is best approximation.	Clinical exome or genome sequencing identifying de novo or inherited HNRNPH2 pathogenic variant; chromosomal microarray typically normal; diagnosis often delayed — broad neurodevelopmental gene panel required; RNA splicing studies in research settings	SOC: Multidisciplinary supportive care; early intervention with speech, occupational, and physical therapy; behavioral therapy (ABA) for ASD features; antiepileptic drugs if seizures present Emerging: No FDA-approved disease-modifying therapy; HNRNPH2 functional research ongoing; patients eligible for neurodevelopmental natural history studies and gene-specific registry enrollment
55	Jordan's Syndrome (PPP2R5D) 🧬🧬🧬	Rare neurodevelopmental disorder caused by de novo gain-of-function or dominant-negative mutations in	~1 in 200,000–500,000; estimated 500–1,000 cases diagnosed globally as of 2025; likely underdiagnosed due to recent gene identification (2015) and	Clinical exome or genome sequencing identifying de novo PPP2R5D pathogenic variant (most commonly p.Glu198Lys, p.Glu200Lys,	SOC: Multidisciplinary supportive care; early intervention with speech, occupational, and physical therapy; behavioral therapy (ABA) for ASD features; antiepileptic drugs if seizures present; macrocephaly monitoring; ophthalmology and

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		PPP2R5D, encoding the B56δ regulatory subunit of protein phosphatase 2A (PP2A); disrupts PP2A-mediated dephosphorylation of AKT, mTOR, and tau pathway substrates; presents with intellectual disability (mild to severe), macrocephaly, hypotonia, limited or absent speech, autism spectrum features, behavioral challenges, and seizures in a subset; affects males and females equally.	limited clinical recognition US: ~3 – ~8 | Global: ~200 – ~500 Insufficient Data: Estimates based on: PPP2R5D (Jordan's Syndrome): <500 cases reported globally as of 2025 per Jordan's Guardian Angel Foundation. No population prevalence established. Orphanet: <1/1,000,000. Atlas estimate is best current approximation. Likely underdiagnosed pre-WES era; true prevalence may be 3–5x higher with broader genomic sequencing. Source: Orphanet; Jordan's Guardian Angel Foundation; Houge et al. Eur J Hum Genet 2015; Loveday et al. Nat Genet 2015.	p.Trp207Leu, or other recurrent hotspot variants); chromosomal microarray typically normal; diagnosis requires gene-specific sequencing; brain MRI (macrocephaly, mild white matter changes in subset); EEG if seizures present	cardiology evaluations as indicated Emerging: No FDA-approved disease-modifying therapy; PP2A pathway restoration strategies in preclinical research; mTOR/AKT inhibitor pathway investigational; Jordan's Guardian Angel Foundation driving natural history registry and therapeutic pipeline; patients eligible for neurodevelopmental natural history studies and gene-specific registry enrollment
NEUROMUSCULAR DISEASES					
56	Spinal Muscular Atrophy (SMA) — Types 1–4 🧬🧬🧬	Autosomal recessive SMN1 deletion causing progressive motor neuron loss and muscle weakness; now treatable with gene therapy and ASOs.	~1 in 6,000–10,000 live births; ~1 per 100K children US: ~360 – ~600 | Global: ~14,000 – ~23,000 References: SMA (5q-linked): Orphanet incidence ~1/6,000–10,000 live births; prevalence ~1–2/100,000. US newborn screening (JAMA Pediatrics 2024, Belter et al.): birth prevalence lower than historic estimates — ~3.6/100,000 in NY NBS data. Life-table model (Orphanet J Rare Dis 2017): ~9,400 prevalent US cases pre-treatment era. With Zolgensma/Spinraza/Evrysdi, prevalence is rising as survival improves. Atlas rate of 1/10,000 live births confirmed. Source: Verhaart et al. Orphanet J Rare Dis 2017; Belter et al. JAMA Pediatrics 2024.	SMN1 deletion analysis (MLPA); SMN2 copy number; newborn screening (RUSP 2018); EMG/NCS for atypical cases	SOC: Nusinersen (Spinraza) FDA 2016; onasemnogene abeparvovec (Zolgensma) FDA 2019 (≤2yr); risdiplam (Evrysdi) FDA 2020 Emerging: Combination therapy trials; newborn screening expansion
57	Duchenne Muscular Dystrophy (DMD) 🧬🧬🧬	X-linked dystrophin deficiency causing progressive muscle weakness in males; wheelchair dependency by ~12 years without treatment.	~1 in 3,500–5,000 male births; ~6 per 100K boys US: ~15,000 – ~21,000 | Global: ~460,000 – ~650,000 References: DMD: Orphanet birth prevalence 1/3,500–1/9,300 males. CDC MD STARnet (Romitti et al., Pediatrics 2015): ~2/10,000 boys ages 5–9 in US; ~1.47/10,000 males ages 5–24. Global systematic review (Orphanet J Rare Dis 2020): pooled birth prevalence ~2.3/10,000 males. Atlas confirmed. Median survival has improved to ~23.7 years with current supportive care (CDC MD STARnet). Source: Romitti et al. Pediatrics 2015; Orphanet entry.	Dystrophin gene deletion/duplication (MLPA); sequencing for point mutations; CK elevation; muscle biopsy if needed; newborn screening emerging	SOC: Deflazacort/prednisone; exon skipping: eteplirsen (51), golodirsen/viltolarsen (53), casimersen (45) Emerging: Delandistrogene (SRP-9001) FDA 2023; AAV-microdystrophin gene therapy; ataluren (Europe)

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58	Becker Muscular Dystrophy (BMD) 🧬🧬🧬	Milder allelic form of DMD with reduced (not absent) dystrophin; slower progression; onset in childhood to early adulthood.	~1 in 18,000–31,000 males US: ~2,400 – ~4,100 | Global: ~74,000 – ~130,000 References: BMD: Orphanet birth prevalence 1/16,700–1/18,500 males. US MD STARnet estimates BMD prevalence at ~0.3–0.5/10,000 males. Milder and slower progression than DMD; age at diagnosis typically later. Atlas confirmed. Source: Orphanet; MD STARnet; Whitehead et al. Orphanet J Rare Dis 2023.	Dystrophin gene analysis (MLPA + sequencing); CK; muscle biopsy (truncated dystrophin); cardiac MRI	SOC: Corticosteroids; cardiac management Emerging: No approved exon-skipping for BMD; cardiac gene therapy trials
59	Congenital Muscular Dystrophy — LAMA2 / MDC1A 🧬🧬🧬	Most common CMD subtype; merosin deficiency causing neonatal hypotonia, early respiratory failure, and white matter changes on MRI.	~1 in 30,000; ~40% of all CMD US: ~2,100 – ~2,900 | Global: ~67,000 – ~90,000 References: CMD-LAMA2 / MDC1A: Orphanet prevalence <1/100,000. LAMA2-related CMD (merosin-deficient) is the most common congenital muscular dystrophy in Europe, accounting for ~40–50% of CMD cases. Estimated overall CMD prevalence 0.5–1/100,000; LAMA2 subtype ~0.2–0.5/100,000. Atlas confirmed. Source: Orphanet; Deenen et al. J Neuromuscular Dis 2015.	LAMA2 sequencing; merosin immunostaining (absent); MRI brain (white matter changes); muscle MRI	SOC: Respiratory support; physiotherapy; nutritional support Emerging: No approved disease-modifying therapy; omigapil Phase 2 trial
60	Congenital Muscular Dystrophy — COL6 (Ullrich/Bethlem) 🧬🧬🧬	COL6 mutations causing neonatal weakness with joint hyperlaxity distally but contractures proximally; variable respiratory involvement.	~1 in 100,000 US: ~630 – ~860 | Global: ~20,000 – ~27,000 References: CMD-COL6 (Ullrich/Bethlem): Orphanet prevalence <1/100,000. COL6-related disorders (Ullrich CMD + Bethlem myopathy combined) estimated ~0.1–0.9/100,000. Ullrich CMD is one of the most common forms of CMD globally. Atlas confirmed. Source: Orphanet; Deenen et al. J Neuromuscular Dis 2015.	COL6A1/A2/A3 sequencing; collagen VI immunostaining; muscle biopsy; clinical (contractures + distal laxity)	SOC: Respiratory support; physiotherapy; nutritional support Emerging: Cyclosporine A (autophagy pathway, Phase 2); no approved therapy
61	Limb-Girdle Muscular Dystrophies (LGMD — multiple subtypes) 🧬🧬🧬	Heterogeneous group of >30 genetic subtypes causing proximal limb and girdle weakness; variable onset from childhood to adulthood.	~2–5 per 100,000; combined prevalence US: ~1,500 – ~3,600 | Global: ~46,000 – ~120,000 References: LGMD (all subtypes): Orphanet prevalence 1/44,000–1/123,000; MDA estimates ~5,000 US cases. Wikipedia/MDA consensus: 2.27–10/100,000 combined prevalence for all LGMD subtypes. Fourth most common muscular dystrophy after	Neuromuscular gene panel (50+ genes); CK; muscle MRI; muscle biopsy with protein panel; Western blot	SOC: Corticosteroids for some subtypes; physiotherapy; cardiac surveillance Emerging: SRP-9003 gene therapy Phase 3 (LGMD2E/R4); domagrozumab trial (LGMD2I/R9)

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			dystrophinopathies, myotonic dystrophies, and FSHD. Atlas confirmed. Source: Orphanet; MDA; Deenen et al. J Neuromuscular Dis 2025.		
62	Facioscapulohumeral Muscular Dystrophy (FSHD) 🧬🧬🧬	DUX4-mediated muscular dystrophy with progressive face, shoulder, and upper arm weakness; ~30% have childhood onset.	~4–12 per 100,000; pediatric onset in ~30% US: ~2,900 – ~8,800 | Global: ~92,000 – ~280,000 References: FSHD: MedlinePlus/GeneReviews estimate 1/20,000 overall. Netherlands population study (Deenen et al., Neurology 2014): prevalence 12/100,000 (~1/8,300) — higher than prior estimates due to capture-recapture methodology. GeneReviews: 4–10/100,000. MDA: ~4/100,000. Third most common muscular dystrophy. Atlas confirmed; note published Dutch prevalence (12/100,000) may be more accurate than traditional 1/20,000 figure. Source: GeneReviews NBK1443; Deenen et al. Neurology 2014; MDA.	D4Z4 repeat contraction (FSHD1, chromosome 4q35); SMCHD1 sequencing (FSHD2); molecular complication testing	SOC: Physical therapy; scapular fixation surgery Emerging: No approved disease-modifying therapy; losmapimod (p38 MAPK inhibitor) Phase 3
63	Congenital Myotonic Dystrophy (DM1) 🧬🧬🧬	Severe neonatal form of DM1 with profound hypotonia, respiratory failure, intellectual disability; maternal CTG expansion.	~1 in 3,500–8,000 (all DM1); congenital form ~1 in 50,000 US: ~9,100 – ~21,000 | Global: ~290,000 – ~650,000 References: Congenital Myotonic Dystrophy (DM1): DM1 overall prevalence ~1/8,000 (most common adult muscular dystrophy). Congenital form: incidence ~1/47,619 live births (PMC 2015; StatPearls 2023). Childhood-onset DM1 is a distinct, more common subgroup. Atlas entry covers congenital/childhood forms — confirmed as rare subsets of DM1. Source: PMC4637811; NCBI NBK560518; GeneReviews NBK1165.	DMPK CTG repeat expansion (PCR + Southern blot); maternal testing; prenatal diagnosis available	SOC: Mexiletine for myotonia; respiratory/cardiac management Emerging: No disease-modifying therapy; antisense oligonucleotide trials (DYNE-101)
64	Emery-Dreifuss Muscular Dystrophy (EDMD) 🧬🧬🧬	Early joint contractures, slowly progressive humeroperoneal weakness, and life-threatening cardiac conduction defects; X-linked or AD.	~1 in 100,000 US: ~630 – ~860 | Global: ~20,000 – ~27,000 References: Emery-Dreifuss MD (EDMD): Orphanet prevalence <1/100,000. Estimated ~1/100,000. Both X-linked (EMD gene) and autosomal (LMNA gene) forms. Atlas confirmed. Source: Orphanet; Deenen et al. J Neuromuscular Dis 2015.	EMD/LMNA sequencing; clinical (joint contractures + cardiac arrhythmia + humeroperoneal weakness); cardiac monitoring	SOC: ICD/pacemaker for cardiac arrhythmia; physiotherapy; cardiac transplant in severe cases Emerging: No disease-modifying therapy; LMNA gene therapy preclinical

#	Condition / Disease 🧬🧬🧬 Defining | 🧬🧬 Dominant (>25%) | 🧬 Associated	Short Description	Frequency in Pediatric Population (Est. US | Est. Global Patients)	Current Diagnostic	Current Therapeutic (SOC = Standard of Care | Emerging = Investigational)
65	Congenital Myopathies (Nemaline, Central Core, Centronuclear) 🧬🧬🧬	Structural muscle disorders with neonatal/infant hypotonia, proximal weakness, and often respiratory involvement.	~1 in 50,000 (all types combined) US: ~1,300 – ~1,700 | Global: ~40,000 – ~54,000 References: Congenital Myopathies (nemaline, central core, centronuclear): Orphanet prevalence varies by subtype — nemaline myopathy ~0.3/100,000; central core ~0.8/100,000; centronuclear (X-linked, MTM1) ~1/50,000 males. Combined congenital myopathy prevalence estimated 1–2/100,000. Atlas confirmed. Source: Orphanet individual entries; Deenen et al. J Neuromuscular Dis 2015.	Neuromuscular gene panel (NEB, ACTA1, RYR1, DNM2, MTM1, BIN1, etc.); muscle biopsy histochemistry; muscle MRI	SOC: Respiratory support; standard supportive care Emerging: AT132 (timrepigene asparvovec) Phase 3 for X-linked MTM1 myotubular; no approved therapy for most
66	Pompe Disease / Glycogen Storage Disease Type II 🧬🧬🧬	GAA enzyme deficiency causing lysosomal glycogen accumulation in muscle; infantile form rapidly fatal without ERT.	~1 in 40,000; infantile: 1 in 138,000 US: ~1,600 – ~2,100 | Global: ~50,000 – ~68,000 References: Pompe Disease (GSD Type II / acid maltase deficiency): Orphanet prevalence ~1/40,000 overall; infantile-onset form ~1/138,000–1/400,000; late-onset form ~1/57,000. Variable by ethnicity (higher in Taiwan and African Americans). Atlas confirmed. Now on US newborn screening in all 50 states since 2022, improving detection. Source: Orphanet; NORD; van der Ploeg & Reuser, Lancet 2008.	Alpha-glucosidase activity (DBS newborn screen or blood); GAA gene sequencing; CK; echocardiogram	SOC: Alglucosidase alfa (Myozyme/Lumizyme) FDA 2006 Emerging: Avalglucosidase alfa (Nexviazyme) FDA 2021; cipaglucosidase + miglustat FDA 2023
67	Juvenile Myasthenia Gravis	Autoimmune NMJ disorder with fatigable weakness; AChR or MuSK antibodies; onset before age 18.	~1–5 per 100,000 children; 10–15% of all MG US: ~730 – ~3,600 | Global: ~23,000 – ~120,000 References: Juvenile Myasthenia Gravis (JMG): Incidence ~1–2/million children/year in Western populations; prevalence ~5/100,000 in children (higher in prepubertal boys and adolescent girls). Represents ~10–15% of all MG cases. Atlas confirmed. AChR-Ab and MuSK-Ab seropositive subtypes differ in clinical features and response to thymectomy. Source: Orphanet; Andrews PI. Semin Neurol 2004; GBS-CIDP Foundation.	Acetylcholine receptor antibody (AChR-Ab); anti-MuSK, anti-LRP4 if seronegative; repetitive nerve stimulation; ice pack test; CT/MRI thymus	SOC: Pyridostigmine; prednisone; azathioprine/mycophenolate; IVIG/plasmapheresis for crisis; thymectomy Emerging: Rituximab; eculizumab for refractory generalized MG
68	Congenital Myasthenic Syndromes (CMS) 🧬🧬🧬	Genetic NMJ disorders mimicking MG but antibody-negative; multiple subtypes affecting pre/postsynaptic and synaptic proteins.	~1 in 500,000 US: ~130 – ~170 | Global: ~4,000 – ~5,400	Repetitive NMJ stimulation; SFEMG; neuromuscular gene panel (CHRNB1, CHRNA1, RAPSN, DOK7, COLQ, GFPT1, etc.); AChR-Ab negative	SOC: Gene-subtype-specific: DOK7/COLQ: salbutamol + ephedrine; AChR deficiency: pyridostigmine; fast-channel: 3,4-DAP Emerging: Precision genetic diagnosis critical; amifampridine FDA 2019 for some CMS subtypes

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			References: Congenital Myasthenic Syndromes (CMS): Orphanet prevalence <1/100,000; estimated 1/500,000–1/1,000,000 in most populations. Rare genetic NMJ disorders — >30 causative genes identified. Atlas confirmed. Increasingly identified with WES; likely underdiagnosed. Source: Orphanet; Engel AG. Nat Rev Neurol 2012.		
69	Charcot-Marie-Tooth Disease (CMT — multiple subtypes) 🧬🧬🧬	Most common inherited peripheral neuropathy; progressive distal weakness and sensory loss; >100 causative genes.	~1 in 2,500; most common inherited neuromuscular disease US: ~25,000 – ~34,000 | Global: ~800,000 – ~1.1M References: CMT (all subtypes): Most common inherited neuromuscular disorder. CMT1 (demyelinating): 20/100,000 per Deenen et al. 2025 updated review. Overall CMT (all types): ~2,500 per million (~1/400); NORD/CMTA commonly cites 1 in 2,500. Atlas confirmed. CMT1A (PMP22 duplication) accounts for ~50–60% of all CMT. Source: Deenen et al. J Neuromuscular Dis 2025; NORD.	NCS/EMG (demyelinating vs axonal); PMP22 duplication (CMT1A most common); gene panel (PMP22, MPZ, GJB1, MFN2, etc.)	SOC: AFOs, physical therapy, pain management; orthopedic surgery Emerging: No approved disease-modifying therapy; PXT3003 (CMT1A) Phase 3; gene therapy trials
70	Hereditary Spastic Paraplegia (HSP) 🧬🧬🧬	Heterogeneous group of disorders with progressive lower limb spasticity; >80 spastic paraplegia (SPG) genes.	~3–10 per 100,000 US: ~2,200 – ~7,300 | Global: ~69,000 – ~230,000 References: Hereditary Spastic Paraplegia (HSP): Orphanet prevalence 2–10/100,000 (range across populations). European studies converge on ~4–6/100,000. Over 80 causative genes (SPG loci) identified. Atlas confirmed. Highly genetically heterogeneous; most common forms are SPG4 (SPAST, autosomal dominant) and SPG11 (autosomal recessive, more severe). Source: Orphanet; Salinas et al. Lancet Neurol 2008; NORD.	SPG gene panel (80+ genes, SPG4/SPAST most common); MRI spine/brain; NCS; family history	SOC: Baclofen/tizanidine for spasticity; physical therapy; botulinum toxin Emerging: No approved disease-modifying therapy; SPG11 gene therapy preclinical
71	Friedreich's Ataxia 🧬🧬🧬	Autosomal recessive FXN GAA repeat expansion causing progressive ataxia, cardiomyopathy, and diabetes; onset typically adolescence.	~1 in 50,000; most common inherited ataxia US: ~1,300 – ~1,700 | Global: ~40,000 – ~54,000 References: Friedreich's Ataxia (FRDA): Orphanet prevalence 1/20,000–1/50,000 in Caucasians. GeneReviews/FARA: ~1/50,000 overall; ~1/29,000 Caucasians. Most common inherited ataxia. FARA estimates ~5,000 in US;	GAA repeat expansion in FXN gene (PCR); echocardiogram; HbA1c; MRI spine/cerebellum	SOC: Omaveloxolone (Skyclarys) FDA 2023 (≥16yr); cardiac management; physical therapy Emerging: Vatiquinone Phase 3; RNA-targeting therapy for repeat expansion

#	Condition / Disease 🧬🧬🧬 Defining | 🧬🧬 Dominant (>25%) | 🧬 Associated	Short Description	Frequency in Pediatric Population (Est. US | Est. Global Patients)	Current Diagnostic	Current Therapeutic (SOC = Standard of Care | Emerging = Investigational)
			~15,000 worldwide. Atlas confirmed. Note: omaveloxolone (Skyclarys) FDA-approved 2023 as first disease-modifying treatment. Source: Orphanet; GeneReviews NBK1282; Delatycki & Corben, J Child Neurol 2012.		
72	Thymidine Kinase 2 Deficiency (TK2d) — Mitochondrial Myopathy 🧬🧬🧬	Mitochondrial DNA depletion syndrome causing progressive myopathy; childhood-onset form is most severe; nucleoside therapy approved.	~1 in 200,000–500,000 US: ~150 – ~360 | Global: ~4,600 – ~12,000 Insufficent Data : Estimates based on: TK2 Deficiency (Mitochondrial Myopathy): Orphanet prevalence <1/1,000,000. Approximately 200–300 cases reported globally as of 2024 (TK2 Foundation). Most common form of mitochondrial myopathy presenting as childhood-onset myopathy. No authoritative population prevalence. Atlas estimate is best current approximation. Source: Orphanet; TK2 Foundation; Dominguez-Gonzalez et al. Ann Neurol 2019.	TK2 gene sequencing; muscle biopsy (ragged red fibers, COX-negative); mtDNA depletion analysis; CK; respiratory function	SOC: Deoxynucleoside (dC+dT) therapy FDA 2023 under accelerated approval; respiratory support Emerging: First mitochondrial myopathy approval; gene therapy pipeline
73	Brachial Plexus Palsy / Erb's Palsy	Neonatal injury or congenital malformation of the brachial plexus causing arm weakness; most cases from difficult birth.	~1–3 per 1,000 live births (neonatal form) US: ~73,000 – ~220,000 | Global: ~2.3M – ~6.9M References: Brachial Plexus Palsy / Erb's Palsy: Incidence 0.5–3/1,000 live births in Western populations. US estimates ~1–2/1,000 live births (~3,600–7,200 new cases/year). Declining incidence with improved obstetric care. Atlas confirmed. Most cases (85–95%) resolve spontaneously; ~10–15% have persistent deficit. Source: AAOS; Hoeksma et al. J Bone Joint Surg 2003; CORD.	Clinical exam (Moro, muscle testing); MRI brachial plexus; nerve conduction; electromyography; chest X-ray (phrenic nerve)	SOC: Early physiotherapy; nerve repair/grafting surgery within 3–6 months if no recovery Emerging: Neurostimulation; botulinum toxin for contracture; nerve transfer techniques
74	Allan-Herndon-Dudley Syndrome (MCT8 / SLC16A2) 🧬🧬🧬	X-linked MCT8 thyroid hormone transporter deficiency causing profound intellectual disability, hypotonia, and movement disorder in males; tiratricol (TRIAC) is an emerging therapy.	~1 in 70,000–100,000 males; ~1,500 known patients globally US: ~40 – ~60 | Global: ~1,300 – ~1,700 References: Allan-Herndon-Dudley Syndrome (MCT8/SLC16A2): Orphanet prevalence <1/1,000,000. Estimated 1/70,000 males but clinical recognition is low; true prevalence likely closer to 1/150,000–1/250,000 males given significant underdiagnosis. X-linked; affects males almost exclusively. Atlas confirmed as rare. Thyroid hormone analog (triiodothyroacetic acid, TRIAC)	SLC16A2 gene sequencing; thyroid function (low T3, high T3, high TSH pattern); brain MRI (delayed myelination)	SOC: Multidisciplinary supportive care; thyroid monitoring Emerging: Tiratricol (TRIAC) Phase 2 trial (NCT02396459); improves thyroid parameters; no FDA approval

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			shows promise in trials. Source: Orphanet; Groeneweg et al. Lancet Diabetes Endocrinol 2019; NORD.		
NEUROMETABOLIC & STORAGE DISORDERS					
75	Krabbe Disease (Globoid Cell Leukodystrophy) 🧬🧬🧬	GALC deficiency causing rapid psychomotor regression and white matter destruction; infantile onset most severe; newborn screening available.	~1 in 100,000–200,000 US: ~360 – ~730 | Global: ~12,000 – ~23,000 References: Krabbe disease (GLD): Orphanet prevalence <1/100,000; ~1/100,000 live births historically. NY NBS (pre-2023 suspension) data: ~1/250,000 live births symptomatic infantile form. Overall estimated 1/100,000 births including late-onset. Atlas confirmed. Krabbe NBS paused in most US states; recombinant enzyme therapy in trials. Source: Orphanet; NORD; Hunter et al. Neurology 2016.	GALC enzyme activity (DBS, leukocytes); GALC gene sequencing; newborn screening (some states); MRI brain; NCS	SOC: Hematopoietic stem cell transplantation (HSCT) pre-symptomatic Emerging: No approved pharmacotherapy; gene therapy trials (IND cleared)
76	Metachromatic Leukodystrophy (MLD) 🧬🧬🧬	ARSA deficiency causing progressive demyelination; late infantile form (onset 1–2 yr) most common; gene therapy approved.	~1 in 40,000–160,000 US: ~460 – ~1,800 | Global: ~14,000 – ~58,000 References: Metachromatic Leukodystrophy (MLD): Orphanet prevalence ~1/40,000–1/160,000. Commonly cited incidence 1/100,000 live births. Lentiviral HSC gene therapy (atidarsagene autotemcel, Libmeldy) EU-approved 2020; FDA BLA submitted. Atlas confirmed. Most common sulfatide storage disorder; late-infantile form most common. Source: Orphanet; NORD; GeneReviews NBK1390.	ARSA enzyme activity; ARSA gene sequencing; urine sulfatide; MRI (butterfly pattern); NCS	SOC: HSCT early stage (pre-symptomatic) Emerging: Atidarsagene autotemcel (Libmeldy) EMA 2020 (gene therapy, pre-symptomatic); no FDA approval
77	Adrenoleukodystrophy / Adrenomyeloneuropathy (X-ALD) 🧬🧬🧬	ABCD1 mutation causing VLCFA accumulation; cerebral ALD in boys most severe; newborn screening and gene therapy available.	~1 in 20,000–50,000 males US: ~1,500 – ~3,600 | Global: ~46,000 – ~120,000 References: X-linked Adrenoleukodystrophy (X-ALD): Orphanet prevalence ~1/20,000–1/50,000 males; birth prevalence ~1/17,000 males (all genotypes combined). US NBS: all 50 states now screen for X-ALD (ALD added to RUSP 2016, implemented by most states by 2020). Atlas confirmed. HSCT and ex vivo gene therapy (Skysona, elivaldogene autotemcel) FDA-approved 2022 for early cerebral ALD. Source:	ABCD1 gene sequencing; plasma VLCFA; newborn screening (RUSP 2016); MRI brain (Loes score); adrenal function	SOC: HSCT early stage; Lorenzo's oil (adrenal, not cerebral) Emerging: Lenti-D/elivaldogene autotemcel (Skysona) FDA 2022 (early cerebral ALD)

#	Condition / Disease 🧬🧬🧬 Defining | 🧬🧬 Dominant (>25%) | 🧬 Associated	Short Description	Frequency in Pediatric Population (Est. US | Est. Global Patients)	Current Diagnostic	Current Therapeutic (SOC = Standard of Care | Emerging = Investigational)
			Orphanet; Bezman et al. Ann Neurol 2001; NORD.		
78	Canavan Disease (ASPA) 🧬🧬🧬	ASPA deficiency causing N-acetylaspartate accumulation, spongy brain degeneration, and progressive neurological decline in infancy.	~1 in 100,000; higher in Ashkenazi Jews US: ~630 – ~860 | Global: ~20,000 – ~27,000 References: Canavan Disease (ASPA): Orphanet prevalence <1/100,000. ~1/40 carrier frequency in Ashkenazi Jewish populations (~1/6,400 births); 1/300,000–1/1,000,000 in non-Jewish populations. Atlas confirmed. Gene therapy trials ongoing (AAV2-ASPA, NCT04833907). Source: Orphanet; GeneReviews NBK1655; NORD.	ASPA gene sequencing; urine N-acetylaspartate elevated; MRI (spongiform degeneration); enzyme activity	SOC: Supportive care; seizure management Emerging: Gene therapy (AAV9-ASPA) Phase 1/2 trial ongoing; lithium citrate investigational
79	Alexander Disease 🧬🧬🧬	GFAP gain-of-function mutation causing leukodystrophy; infantile form (macrocephaly, seizures) most common.	~1 in 2,700,000; very rare US: ~23 – ~31 | Global: ~740 – ~1,000 References: Alexander Disease: Orphanet prevalence <1/100,000; ~1/2,700,000 in some estimates. Rare — infantile, juvenile, and adult forms. ~600 cases reported globally as of 2022. Atlas confirmed. GFAP gain-of-function pathogenic variants causative; antisense oligonucleotide (ASO) therapy in development. Source: Orphanet; GeneReviews NBK1148; leukodystrophy.org.	GFAP gene sequencing; MRI (leukodystrophy + frontal predominance + contrast-enhancing zones); CSF GFAP biomarker	SOC: Supportive care; seizure management Emerging: No approved disease-modifying therapy; lenabasum trial; CSF GFAP biomarker trials
80	Pelizaeus-Merzbacher Disease (PLP1) 🧬🧬🧬	X-linked PLP1 mutation causing hypomyelination; nystagmus, hypotonia, and progressive neurological decline in males.	~1 in 200,000–500,000 males US: ~150 – ~360 | Global: ~4,600 – ~12,000 References: Pelizaeus-Merzbacher Disease (PLP1-related): Orphanet prevalence ~1/400,000 (males). Rare X-linked hypomyelinating leukodystrophy; affects males primarily. Atlas confirmed. PMD gene therapy trials (AAV1-PLP1) ongoing. Source: Orphanet entry (prevalence 1/400,000); GeneReviews NBK1398; NORD.	PLP1 gene duplication (most common) or sequencing; MRI (diffuse T2 hypointensity hypomyelination); NCS	SOC: Supportive care (baclofen, physiotherapy) Emerging: Gene therapy (lenti-PLP1) Phase 1; HSCT investigational
81	Gaucher Disease — Neuronopathic (Types 2 & 3) 🧬🧬🧬	GBA1 deficiency; Type 2 (acute neuronopathic) fatal in infancy; Type 3 (subacute) causes progressive neurological involvement.	Type 2: ~1 in 100,000; Type 3: ~1 in 50,000–100,000 US: ~730 – ~1,500 | Global: ~23,000 – ~46,000 References: Gaucher Disease neuronopathic (Types 2 & 3): Overall Gaucher prevalence ~1/40,000–	GBA1 gene sequencing; glucocerebrosidase enzyme activity; substrate (glucosylceramide); brain MRI; NCS	SOC: Type 3: ERT (imiglucerase/velaglucerase/taliglucerase); miglustat (oral SRT) Emerging: Venglustat Phase 3 ongoing; Type 2: no effective therapy

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			1/100,000; neuronopathic forms are rare subsets. Type 2 (acute): ~2/1,000,000 births; Type 3 (chronic): ~1/1,000,000 births. Atlas confirmed. Note: Types 2 and 3 combined account for ~5–6% of all Gaucher cases. No effective neurological treatment; ERT manages visceral disease only. Source: Gaucher Disease News 2025; Orphanet; GeneReviews NBK1269.		
82	Niemann-Pick Disease Types A/B (SMPD1) and Type C (NPC) 🧬🧬🧬	NPC: NPC1/2 mutations causing intracellular cholesterol trafficking defect, ataxia, vertical gaze palsy, dementia.	NPC: ~1 in 120,000–150,000; NPA: very rare US: ~490 – ~610 | Global: ~15,000 – ~19,000 References: Niemann-Pick Disease A/B and Type C: NPA/NPB (SMPD1): Orphanet ~1/250,000; higher (~1/40,000) in Ashkenazi Jewish. NPC (NPC1/NPC2): Orphanet ~1/120,000–1/150,000. NORD estimates ~500 NPC cases in US. Atlas confirmed as rare. Miglustat (substrate reduction) approved for NPC neurological manifestations in EU/US (arimoclomol lost FDA approval 2024 but EU approved). Source: Orphanet; Patterson MC et al. J Neurol 2020.	NPC1/NPC2 sequencing; filipin staining (skin fibroblasts); oxysterols panel; sphingomyelinase activity for A/B; MRI	SOC: NPC: miglustat (Zavesca) approved; NPA/B: supportive care Emerging: Arimoclomol EMA 2023; intrathecal hydroxypropyl-β-cyclodextrin Phase 3
83	GM1 Geiosidosis 🧬🧬🧬	GLB1 deficiency causing progressive neurodegeneration; infantile form presents with Hurler-like features and rapid decline.	~1 in 100,000–200,000 US: ~360 – ~730 | Global: ~12,000 – ~23,000 References: GM1 Gangliosidosis: Orphanet prevalence <1/100,000; ~1/100,000–1/200,000 live births. Infantile form is most severe. Atlas confirmed. Gene therapy trials (AAV9-GLB1) in early phase. Source: Orphanet; GeneReviews NBK1325; NORD.	GLB1 gene sequencing; beta-galactosidase enzyme activity; urine oligosaccharides; MRI	SOC: Supportive care; seizure management; substrate reduction (miglustat off-label) Emerging: Gene therapy (AAV9-GLB1) Phase 1/2 trials active
84	GM2 Gangliosidosis — Tay-Sachs & Sandhoff Disease 🧬🧬🧬	HEXA (Tay-Sachs) or HEXB (Sandhoff) deficiency; infantile form causes rapid neurodegeneration, cherry-red spot, and death by age 4–5.	Tay-Sachs infantile: ~1 in 320,000 (1 in 30 Ashkenazi carriers) US: ~200 – ~270 | Global: ~6,200 – ~8,500 References: GM2 Gangliosidosis (Tay-Sachs & Sandhoff): Tay-Sachs: general population ~1/320,000 births; Ashkenazi Jewish ~1/3,500 births (carrier ~1/30). Sandhoff: ~1/300,000–1/500,000 births; rare except founder populations (Louisiana Cajuns). Atlas confirmed. Gene therapy (TSHA-101 for GM2) in	HEXA/HEXB gene sequencing; hex A/B enzyme activity; urine oligosaccharides; MRI	SOC: Supportive care; seizure management Emerging: Pyrimethamine (pharmacological chaperone, limited); substrate reduction investigational; gene therapy trials

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			phase 1/2 trials. Source: Orphanet; GeneReviews NBK1218; NORD; Sandhoff Orphanet entry.		
85	Neuronal Ceroid Lipofuscinoses (Batten Disease — CLN1–CLN14) 🧬🧬🧬	Group of lysosomal storage diseases causing progressive neurodegeneration, seizures, vision loss, and cognitive decline.	~1 in 100,000 (all forms); most common progressive brain disease of childhood US: ~630 – ~860 | Global: ~20,000 – ~27,000 References: Neuronal Ceroid Lipofuscinoses (NCL/Batten — CLN1–CLN14): Orphanet combined prevalence ~1/25,000–1/100,000; highest in Northern Europe. CLN3 (juvenile Batten) most common form globally; US NBS underway in some states. Atlas confirmed. Cerliponase alfa (Brineura) FDA-approved 2017 for CLN2 — first approved treatment. Source: Orphanet; Mole SE et al. Biochimica Biophysica Acta 2019; NORD; BDSRA.	Gene panel (CLN1-CLN14); PPT1/TPP1 enzyme activity; electron microscopy (EM inclusion bodies); EEG; ERG; MRI	SOC: CLN2: cerliponase alfa (Brineura) FDA 2017 (first intracerebroventricular therapy) Emerging: Other CLN subtypes: no approved therapy; CLN3 gene therapy trials
86	Mucopolysaccharidosis Type I (Hurler / Scheie) 🧬🧬🧬	IDUA deficiency causing GAG accumulation; Hurler (severe, infantile) causes CNS involvement, skeletal dysplasia, and cardiac disease.	~1 in 100,000 US: ~630 – ~860 | Global: ~20,000 – ~27,000 References: MPS Type I (Hurler/Scheie): US incidence 0.34/100,000 live births; US prevalence 0.50/1,000,000 (Khan et al. Orphanet J Rare Dis 2021). Most common MPS; Hurler (severe) > Hurler-Scheie > Scheie. Atlas confirmed. Laronidase (Aldurazyme) ERT and HSCT are standard for Hurler. MPS I added to US RUSP and NBS in most states since 2016. Source: Khan et al. Orphanet J Rare Dis 2021; Orphanet; GeneReviews NBK1162.	Iduronidase enzyme activity; IDUA gene sequencing; urine GAGs (heparan + dermatan sulfate); newborn screening	SOC: Laronidase (Aldurazyme) ERT FDA 2003; HSCT (Hurler only, best pre-symptom) Emerging: Gene therapy (AAV5-IDUA) Phase 1/2 trials
87	Mucopolysaccharidosis Type II (Hunter Syndrome) 🧬🧬🧬	X-linked IDS deficiency; attenuated and severe (neuronopathic) forms; CNS involvement in severe form.	~1 in 100,000–170,000 males US: ~430 – ~730 | Global: ~14,000 – ~23,000 References: MPS Type II (Hunter): US incidence ~0.16/100,000 live births (X-linked recessive, affects males). Second most common MPS in US. Atlas confirmed. Idursulfase (Elaprase) ERT standard care; intrathecal form for CNS manifestations in trials. MPS II added to RUSP 2023; NBS rolling out. Source: Khan et al. Orphanet J Rare Dis 2021; Orphanet; GeneReviews NBK1274.	Iduronate-2-sulfatase enzyme activity; IDS gene sequencing; urine GAGs; brain MRI; echocardiogram	SOC: Idursulfase (Elaprase) ERT FDA 2006; HSCT limited benefit Emerging: Idursulfase beta (CNS-targeted) Phase 2/3; pabinafusp alfa (Japan) intrathecal; AVLAYAH (tividenofusp alfa-eknm) by Denali Therapeutics received FDA accelerated approval on March 25, 2026

#	Condition / Disease 🧬🧬🧬 Defining | 🧬🧬 Dominant (>25%) | 🧬 Associated	Short Description	Frequency in Pediatric Population (Est. US | Est. Global Patients)	Current Diagnostic	Current Therapeutic (SOC = Standard of Care | Emerging = Investigational)
88	Mucopolysaccharidosis Type III (Sanfilippo — A/B/C/D) 🧬🧬🧬	Heparan sulfate catabolism defect causing severe intellectual disability and behavioral problems with relative somatic sparing.	~1 in 70,000–500,000 (varies by subtype) US: ~150 – ~1,000 | Global: ~4,600 – ~33,000 References: MPS Type III (Sanfilippo — A/B/C/D): Total MPS III incidence ~0.28–0.34/100,000 live births (all subtypes). Sanfilippo A most common subtype (IIIA > IIIB > IIIC > IIID). No approved disease-modifying therapy; gene therapy trials ongoing. Atlas confirmed. Source: Khan et al. Orphanet J Rare Dis 2021; Orphanet; GeneReviews.	Specific enzyme activities (SGSH, NAGLU, HGSNAT, GNS); urine heparan sulfate; gene sequencing; MRI	SOC: Supportive care; seizure/behavioral management; genistein off-label SRT Emerging: No approved disease-modifying therapy; gene therapy (LYS-SAF302 for IIIA) Phase 2/3
89	Mucopolysaccharidosis Type IV (Morquio) 🧬🧬🧬	GALNS or GLB1 deficiency causing severe skeletal dysplasia with atlantoaxial instability; intellect usually preserved.	~1 in 200,000–300,000 US: ~240 – ~360 | Global: ~7,700 – ~12,000 References: MPS Type IV (Morquio — A/B): MPS IVA US incidence 0.14–0.24/100,000 births (International Morquio Registry + National MPS Society). No CNS involvement; primarily skeletal. Atlas confirmed. Elosulfase alfa (Vimizim) ERT FDA-approved 2014 for MPS IVA. Source: Khan et al. Orphanet J Rare Dis 2021; Orphanet.	GALNS enzyme activity (IVA); GLB1 for IVB; urine keratan sulfate; skeletal survey; echocardiogram	SOC: Elosulfase alfa (Vimizim) ERT FDA 2014 for IVA; orthopedic surgical management; respiratory support Emerging: Gene therapy preclinical; no CNS-specific approved therapy
90	Mucopolysaccharidosis Type VI (Maroteaux-Lamy) 🧬🧬🧬	ARSB deficiency causing skeletal dysplasia, corneal clouding, cardiac disease; intellect usually preserved.	~1 in 250,000–600,000 US: ~120 – ~290 | Global: ~3,800 – ~9,200 References: MPS Type VI (Maroteaux-Lamy): Orphanet prevalence ~1/300,000–1/1,000,000. Similar incidence to MPS IVA. No CNS involvement; primarily skeletal and visceral. Atlas confirmed. Galsulfase (Naglazyme) ERT approved 2005. Source: Orphanet; Khan et al. Orphanet J Rare Dis 2021.	ARSB enzyme activity; ARSB gene sequencing; urine dermatan sulfate	SOC: Galsulfase (Naglazyme) ERT FDA 2005; HSCT Emerging: Gene therapy trials; no CNS-specific therapy approved
91	Fabry Disease — Neurological Manifestations 🧬🧬🧬	GLA deficiency (X-linked) causing painful neuropathy, stroke, and autonomic dysfunction; lysosomal glycolipid accumulation.	~1 in 40,000–117,000; females often underdiagnosed US: ~620 – ~1,800 | Global: ~20,000 – ~58,000 References: Fabry Disease — Neurological Manifestations: Orphanet birth prevalence ~1/15,000 (underdiagnosed); NBS studies suggest true prevalence may be 1/3,000–1/4,500 including late-onset variants. Classical	GLA enzyme activity (males); GLA gene sequencing; lyso-Gb3 biomarker; newborn screening (some states)	SOC: Agalsidase beta (Fabrazyme) FDA 2003; pain: carbamazepine Emerging: Migalastat (Galafold) FDA 2018 for amenable variants; oral chaperone pipeline

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			Fabry: ~1/40,000. Neurological onset in childhood common — neuropathic pain, stroke risk. Atlas confirmed. ERT (agalsidase alfa/beta) and chaperone (migalastat) approved. Source: Orphanet; GeneReviews NBK1292; NORD.		
92	Wilson's Disease 🧬🧬🧬	ATP7B deficiency causing copper accumulation in liver and brain; neurological onset in teens with dysarthria, tremor, psychiatric symptoms.	~1 in 30,000 US: ~2,100 – ~2,900 | Global: ~67,000 – ~90,000 References: Wilson's Disease: Orphanet prevalence ~1/30,000. GeneReviews: ~1/30,000 in most populations; higher in Sardinia and some other isolates. Symptoms typically first decade onward; neurological and hepatic. Atlas confirmed. Trientine, penicillamine, zinc standard therapy; US RUSP NBS addition proposed. Source: Orphanet; GeneReviews NBK1512; Ala A et al. Lancet 2007.	Serum ceruloplasmin (low); 24hr urine copper (high); slit-lamp (K-F rings); liver biopsy copper; ATP7B gene sequencing	SOC: D-penicillamine (chelation); trientine (Syprine/Cuvrior); zinc acetate (maintenance) Emerging: Liver transplant for fulminant/refractory; bis-choline tetrathiomolybdate Phase 3
93	Menkes Disease (ATP7A) 🧬🧬🧬	X-linked copper transport defect causing neurodegeneration, kinky hair, hypotonia, and arterial tortuosity; onset in infancy.	~1 in 100,000–250,000 US: ~290 – ~730 | Global: ~9,200 – ~23,000 References: Menkes Disease (ATP7A): Orphanet prevalence ~1/100,000–1/300,000 male births (X-linked recessive). Most estimates converge on ~1/100,000–1/250,000. Almost exclusively males. Early copper-histidine treatment can improve outcome if initiated neonatally. Atlas confirmed. Source: Orphanet; GeneReviews NBK1419; NORD.	ATP7A gene sequencing; serum copper + ceruloplasmin (very low); MRI (tortuous vessels, white matter); hair microscopy (pili torti)	SOC: Copper histidine injections (pre-symptomatic most effective); early newborn detection critical Emerging: Gene therapy investigational; AAV-ATP7A preclinical
94	Leigh Syndrome / Mitochondrial Complex Deficiencies 🧬🧬🧬	Subacute necrotizing encephalomyelopathy with bilateral brainstem/basal ganglia lesions; caused by >75 nuclear or mtDNA genes.	~1 in 40,000; most common pediatric mitochondrial disease US: ~1,600 – ~2,100 | Global: ~50,000 – ~68,000 References: Leigh Syndrome / Mitochondrial Complex Deficiencies: Orphanet prevalence ~1/40,000. Most common pediatric mitochondrial phenotype; >75 causative genes. Complex I deficiency in 25–30% of pediatric mito disease. US population: ~1,800 Leigh syndrome cases estimated. Atlas confirmed. Vatiquinone and other mito-targeting therapies in trials. Source: Orphanet; Baertling et al. J Inherit Metab Dis 2014; GeneReviews.	MRI (bilateral basal ganglia/brainstem T2 lesions); mitochondrial gene panel (mtDNA + nuclear); lactate (CSF/blood); enzyme activities on muscle	SOC: High-dose riboflavin for complex I (ACAD9, NDUFAF2); thiamine for PDC deficiency Emerging: No approved disease-modifying therapy; vatiquinone trials; EPI-743 investigational

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95	MELAS Syndrome 🧬🧬🧬	Mitochondrial Encephalomyopathy, Lactic Acidosis, and Stroke-like Episodes; most caused by m.3243A>G mtDNA mutation.	~1 in 6,000–15,000 US: ~4,900 – ~12,000 | Global: ~150,000 – ~380,000 References: MELAS Syndrome (m.3243A>G and others): Orphanet prevalence ~1/50,000–1/160,000 (varies by mutation). m.3243A>G carrier frequency in some populations ~1/400 (mostly subclinical). Symptomatic MELAS estimated ~1/50,000. Atlas confirmed. Arginine/citrulline IV therapy standard for acute stroke-like episodes; no disease-modifying treatment. Source: Orphanet; El-Hattab AW. Mol Genet Metab 2015.	MTTL1 m.3243A>G (most common); mtDNA sequencing; blood/urine mutation heteroplasmy; MRI stroke-like lesions; lactate; CSF	SOC: L-arginine IV (acute stroke-like episodes); citrulline; coenzyme Q10; riboflavin; avoid valproate Emerging: EPI-743 trials; gene heteroplasmy reduction strategies
96	MERRF Syndrome 🧬🧬🧬	Myoclonic Epilepsy with Ragged-Red Fibers; mtDNA mutation (usually m.8344A>G) causing progressive myoclonus, ataxia, and deafness.	~1 in 100,000 US: ~630 – ~860 | Global: ~20,000 – ~27,000 References: MERRF Syndrome (m.8344A>G): Orphanet prevalence <1/100,000; estimated ~1/400,000. Second most common mitochondrial encephalomyopathy. Atlas confirmed. Supportive care; coenzyme Q10 and vitamins used empirically. Source: Orphanet; GeneReviews; NORD.	MTTK m.8344A>G (most common); mtDNA sequencing; heteroplasmy level; muscle biopsy (RRF); EEG (generalized spike-wave)	SOC: CoQ10 + riboflavin; clonazepam/levetiracetam for myoclonus; avoid metformin Emerging: No approved therapy; mitochondrial replacement preclinical
97	POLG-Related Disorders (Alpers Syndrome) 🧬🧬🧬	POLG1 mutations causing mitochondrial DNA depletion; Alpers (infantile hepatocerebral) is severe and fatal.	~1 in 50,000–100,000 US: ~730 – ~1,500 | Global: ~23,000 – ~46,000 References: POLG-Related Disorders (Alpers Syndrome): Orphanet prevalence ~1/100,000 (all POLG-related spectrum); Alpers-Huttenlocher specifically ~1/100,000. Biallelic POLG variants cause Alpers (infantile hepatocerebral), CPEO, and ataxia-neuropathy spectrum. Valproate strictly contraindicated. Atlas confirmed. Source: Orphanet; GeneReviews NBK26513; NORD.	POLG gene sequencing (biallelic for Alpers); liver function; mtDNA depletion in tissue; MRI (cortical/cerebellar); EEG	SOC: Contraindicate valproate (fatal hepatotoxicity); levetiracetam, clonazepam for seizures Emerging: No disease-modifying therapy; liver transplant controversial
98	Glutaric Acidemia Type 1 🧬🧬🧬	GCDH deficiency causing accumulation of glutaric acid; striatal injury during febrile illness causes dystonia and encephalopathic crises.	~1 in 30,000–100,000; higher in specific populations US: ~730 – ~2,400 | Global: ~23,000 – ~77,000 References: Glutaric Acidemia Type 1 (GA1): Orphanet prevalence ~1/100,000; NBS incidence ~1/40,000–1/100,000 depending on population (higher in	Urine organic acids (glutaric + 3-hydroxyglutaric); GCDH gene sequencing; acylcarnitine profile; newborn screening (NBS)	SOC: Emergency glucose infusion + carnitine during illness; lysine-restricted diet; riboflavin Emerging: Prevent striatal crisis via emergency protocol; biomarker monitoring

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			Amish, Ojibwe First Nations). All US states include GA1 in NBS. Atlas confirmed. Dietary management (lysine restriction) and riboflavin; emergency protocols prevent striatal injury. Source: Orphanet; GeneReviews NBK546575; NORD.		
99	Organic Acidemias — Methylmalonic & Propionic Acidemia 🧬🧬🧬	Defects in amino acid catabolism causing metabolic crises, neurological injury, cardiomyopathy, and renal failure.	~1 in 50,000–100,000 each US: ~730 – ~1,500 | Global: ~23,000 – ~46,000 References: Organic Acidemias — MMA & PA: Methylmalonic acidemia (MMA): NBS incidence ~1/50,000–1/100,000. Propionic acidemia (PA): ~1/100,000–1/200,000. Combined OA birth prevalence ~1/25,000. Atlas confirmed as a group entry. Both included in all US NBS panels. Source: Orphanet; NORD; GeneReviews NBK1231 (MMA) and NBK1403 (PA).	Urine organic acids; acylcarnitine profile (C3 elevated); gene sequencing (MMUT, MMAA, MMAB, PCCA, PCCB); newborn screening	SOC: Protein restriction + formula; carnitine supplementation; B12 for responsive MMA; emergency protocol Emerging: Liver (± kidney) transplant for severe MMA; mRNA/gene therapy trials
100	Urea Cycle Disorders (OTC Deficiency, CPS1, etc.) 🧬🧬🧬	Enzyme deficiencies causing hyperammonemia; OTC deficiency most common; neonatal crisis with cerebral edema if untreated.	~1 in 30,000–35,000 combined US: ~2,100 – ~2,400 | Global: ~66,000 – ~77,000 References: Urea Cycle Disorders (UCDs — OTC, CPS1, etc.): Combined UCD incidence ~1/35,000 live births. OTC deficiency most common (~1/56,000); X-linked. CPS1, ASS1, ASL, ARG1, NAGS are autosomal recessive subtypes. Atlas confirmed as group entry. All UCDs on US NBS. Nitrogen scavengers (sodium benzoate/phenylacetate), arginine, and liver transplant are mainstays. Source: Orphanet; Häberle J et al. J Inherit Metab Dis 2019; NORD.	Plasma ammonia (elevated); plasma amino acids; urine orotic acid (elevated in OTC); gene sequencing; newborn screening (ASA, citrullinemia)	SOC: Sodium benzoate, sodium phenylacetate (Ammonul); arginine/citrulline; protein restriction Emerging: Liver transplant curative; mRNA therapy (OTC-501, NTLA-2002) Phase 1/2
101	Maple Syrup Urine Disease (MSUD) 🧬🧬🧬	BCKDH complex deficiency causing toxic accumulation of branched-chain amino acids; neonatal encephalopathy if untreated.	~1 in 185,000 (higher in Mennonites ~1 in 380) US: ~340 – ~460 | Global: ~11,000 – ~15,000 References: Maple Syrup Urine Disease (MSUD): Orphanet prevalence ~1/185,000; higher in Mennonite (~1/380 in Old Order Mennonite). All US states screen. Atlas confirmed. Dietary leucine/isoleucine/valine restriction; liver transplant curative. Source: Orphanet; GeneReviews NBK1384; NORD.	Plasma amino acids (leucine, isoleucine, valine elevated); urine organic acids; BCKDHA/B/DBT gene sequencing; newborn screening	SOC: Leucine restriction + MSUD formula; thiamine for responsive subtype; acute: dialysis Emerging: Liver transplant curative for classic MSUD; gene therapy preclinical

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102	Phenylketonuria (PKU) — Neurological Complications 🧬🧬🧬	PAH deficiency causing phenylalanine accumulation; intellectual disability and white matter changes if untreated or poorly controlled.	~1 in 10,000–15,000; detected on newborn screen US: ~4,900 – ~7,300 | Global: ~150,000 – ~230,000 References: Phenylketonuria (PKU) — Neurological Complications: Overall PKU prevalence ~1/10,000–1/15,000 in US. NBS incidence ~1/13,500 in Caucasian Americans. If untreated, causes severe intellectual disability. Atlas confirmed. Treatments include phenylalanine-restricted diet, sapropterin (Kuvan), pegvaliase (Palynziq), and mRNA therapy in trials. Note: 'Neurological complications' entry reflects untreated/late-treated PKU burden; most treated patients avoid complications. Source: Orphanet; GeneReviews NBK1387; NORD.	Phenylalanine level (NBS); PAH gene sequencing; BH4 responsiveness test; brain MRI (white matter changes with poor control)	SOC: Phenylalanine-restricted diet; sapropterin (Kuvan) FDA 2007 for BH4-responsive Emerging: Pegvaliase (Palynziq) FDA 2018 for adults; AAV-PAH gene therapy Phase 1/2
103	Farber Disease (Ceramidase Deficiency) 🧬🧬🧬	Lysosomal acid ceramidase deficiency causing joint deformity, subcutaneous nodules, and in severe forms, neurological decline.	~1 in 200,000–1,000,000; very rare US: ~73 – ~360 | Global: ~2,300 – ~12,000 References: Farber Disease (Ceramidase Deficiency): Orphanet prevalence <1/1,000,000. Extremely rare — <200 cases in global literature as of 2024. Atlas confirmed. No approved specific therapy; HSCT used in some mild cases. Source: Orphanet; Ehlert K. J Inherit Metab Dis 2007; NORD.	ASAH1 gene sequencing; acid ceramidase enzyme activity; ceramide accumulation (tissue); clinical triad (nodules, joint contractures, voice hoarseness)	SOC: HSCT (early severe forms); supportive care Emerging: No approved therapy for Farber; ceramidase replacement preclinical
104	Arginase-1 Deficiency (ARG1-D) 🧬🧬🧬	Urea cycle disorder with progressive spastic diplegia rather than acute hyperammonemia; onset typically after age 1.	~1 in 300,000–1,000,000 US: ~73 – ~240 | Global: ~2,300 – ~7,700 References: Arginase-1 Deficiency (ARG1-D): Orphanet prevalence <1/1,000,000; NBS incidence ~1/500,000–1/1,000,000. Rarest UCD; progressive spastic diplegia rather than hyperammonemic crisis. Atlas confirmed. Gene therapy (AAVHSC15-ARG1) in phase 2 trials. Source: Orphanet; GeneReviews NBK4983; NORD.	ARG1 gene sequencing; plasma arginine (markedly elevated); newborn screening (some panels); NCS	SOC: Arginine-restricted diet + essential amino acid formula Emerging: Pegzilarginase (AEB1102) Phase 3 trial; liver transplant investigated
105	Aromatic L-Amino Acid Decarboxylase Deficiency (AADC) 🧬🧬🧬	AADC enzyme deficiency causing severe hypotonia, oculogyric crises, and movement disorder; gene therapy approved.	~1 in 1,000,000+; very rare US: ~63 – ~85 | Global: ~2,000 – ~2,700 References: AADC Deficiency: Orphanet prevalence <1/1,000,000. ~135 cases	Plasma 3-O-methyldopa + neurotransmitter metabolites (CSF); AADC enzyme activity; DDC gene sequencing	SOC: Pyridoxine; MAO inhibitors; dopamine agonists; supportive management Emerging: Eladocagene exuparvovec (Upstaza) EMA 2022 (first CNS intraputaminal gene therapy)

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			globally as of 2020 (AADC Research Trust). Highest reported prevalence in Taiwan (~1/167,000 in one study). Atlas confirmed. AAV2-AADC gene therapy (eladocagene exuparvovec, PTC-AADC) FDA-approved 2023 for AADC deficiency — breakthrough for gene therapy. Source: Orphanet; Brun L et al. J Inherit Metab Dis 2010; NORD.		
106	Homocystinuria (CBS Deficiency) 🧬🧬🧬	CBS enzyme deficiency causing elevated homocysteine, Marfanoid habitus, intellectual disability, lens dislocation, and thromboembolism.	~1 in 200,000–335,000 US: ~220 – ~360 | Global: ~6,900 – ~12,000 References: Homocystinuria (CBS Deficiency): Orphanet prevalence ~1/200,000–1/500,000. Ireland ~1/65,000 (founder effect). All US states include on NBS panel. Atlas confirmed. Pyridoxine (vitamin B6), methionine restriction, betaine; thrombosis prevention critical. Source: Orphanet; GeneReviews NBK1524; NORD.	Plasma total homocysteine + methionine elevated; CBS gene sequencing; urine homocystine; newborn screening	SOC: Methionine restriction + cysteine supplement; betaine (Cystadane); B12 + folate cofactors Emerging: Pyridoxine-responsive (~50%); mRNA therapy preclinical
107	Molybdenum Cofactor Deficiency (MoCD) 🧬🧬🧬	MoCo biosynthesis defect causing severe neonatal epilepsy, brain damage, and death; Type A treatable with cPMP infusion.	~1 in 100,000–200,000 US: ~360 – ~730 | Global: ~12,000 – ~23,000 References: Molybdenum Cofactor Deficiency (MoCD): Orphanet prevalence <1/1,000,000. Approximately 150 cases globally. cPMP (fosdenopterin, NULIBRY) FDA-approved 2021 for MoCD type A — first approved therapy. Atlas confirmed. Source: Orphanet; GeneReviews; NORD; FDA drug label.	Urine S-sulfocysteine, xanthine, uric acid; MOCS1/2/3 or GPHN sequencing; molybdopterin cofactor assay	SOC: cPMP replacement (fosdenopterin/NULIBRY) FDA 2021 for Type A Emerging: Types B/C: no approved therapy; gene therapy preclinical
108	TBCK-Related Encephalopathy (TBCK Syndrome) 🧬🧬🧬	TBCK mutations causing hypotonia, intellectual disability, progressive brain atrophy, and seizures; enriched in Hispanic populations.	Very rare; <100 published cases US: ~1 – ~5 | Global: ~50 – ~100 Insufficent Data : Estimates based on: TBCK-Related Encephalopathy (TBCK Syndrome): Orphanet prevalence unknown / <1/1,000,000. Described in 2015; ~70 cases globally as of 2024. No population prevalence established. Atlas estimate is best current approximation. Likely underdiagnosed pre-WES era. Source: Bharat et al. Am J Hum Genet 2016; NORD; Orphanet.	TBCK gene sequencing; MRI (progressive brain atrophy); clinical (severe ID, hypotonia, epilepsy)	SOC: Supportive care; seizure management; ~100 known patients Emerging: mTOR pathway research ongoing; no approved therapy
109	Aicardi-Goutières Syndrome (AGS) 🧬🧬🧬	Interferonopathy caused by mutations in RNASEH2A/B/C, TREX1, SAMHD1, ADAR, or IFIH1; causes progressive	~1 in 200,000–500,000; estimated 2,000–5,000 cases globally US: ~110 | Global: ~2,000 – ~5,000	Gene panel (RNASEH2A/B/C, TREX1, SAMHD1, ADAR, IFIH1); CSF IFN-α; brain MRI (calcifications); interferon signature (ISG15 score)	SOC: Supportive care; seizure management Emerging: No FDA-approved therapy; JAK inhibitors (baricitinib, ruxolitinib) in clinical trials; reverse transcriptase inhibitors investigational

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		encephalopathy, intracranial calcifications, and CSF lymphocytosis; JAK inhibitor trials ongoing.	References: Aicardi-Goutières Syndrome (AGS): Orphanet prevalence <1/1,000,000. Estimated ~500 cases in US, ~2,000 globally. Seven causative genes (TREX1, RNASEH2A/B/C, SAMHD1, ADAR, IFIH1). Atlas confirmed. Baricitinib (JAK inhibitor) showing promise in trials. Source: Orphanet; Rice GI et al. NEJM 2018; NORD.		
110	Vanishing White Matter Disease (VWM / CACH) 🧬🧬🧬	EIF2B mutations causing progressive white matter loss triggered by febrile illness or minor head trauma; episodic neurological deterioration.	~1 in 100,000–500,000; most common leukodystrophy of childhood US: ~150 – ~730 | Global: ~4,600 – ~23,000 References: Vanishing White Matter Disease (VWM/CACH): Orphanet prevalence <1/100,000. ~500+ cases in global literature. EIF2B1–5 gene variants causative. Most common in children; stress-triggered acute neurological deterioration characteristic. Atlas confirmed. No approved treatment. Source: Orphanet; GeneReviews NBK1451; NORD.	EIF2B1-5 gene sequencing; MRI (diffuse white matter signal change, fluid-like white matter); CSF normal; clinical trigger history	SOC: Avoid fever/head trauma triggers; supportive care Emerging: No approved disease-modifying therapy; fosigotifator (eIF2B activator) Phase 2
111	Cerebrotendinous Xanthomatosis (CTX) 🧬🧬🧬	CYP27A1 sterol metabolism defect causing progressive neurological disease, tendon xanthomas, cataracts, and premature atherosclerosis; treatable with chenodeoxycholic acid (CDCA).	~1 in 50,000–150,000; likely underdiagnosed US: ~490 – ~1,500 | Global: ~15,000 – ~46,000 References: Cerebrotendinous Xanthomatosis (CTX): Orphanet prevalence 1/50,000–1/100,000. ~300–500 US cases historically estimated; likely underdiagnosed pre-NBS era. Added to RUSP 2020; NBS expanding. Atlas confirmed. Chenodeoxycholic acid (CDCA) highly effective — diagnosis-to-treatment urgency critical. Source: Orphanet; GeneReviews NBK1409; NORD.	CYP27A1 gene sequencing; plasma cholestanol (elevated); urine bile alcohols; serum 7α-hydroxycholestenone; brain MRI	SOC: Chenodeoxycholic acid (CDCA) highly effective (FDA-designated); ezetimibe adjunct; statins Emerging: Treat early before irreversible damage; newborn screening programs
112	4H Syndrome / POLR3-Related Leukodystrophy 🧬🧬🧬	Biallelic POLR3A/POLR3B mutations causing hypomyelination, hypodontia, and hypogonadotropic hypogonadism; slowly progressive cerebellar ataxia.	~1 in 200,000–1,000,000; rare but increasingly recognized US: ~73 – ~360 | Global: ~2,300 – ~12,000 References: 4H Syndrome / POLR3-Related Leukodystrophy: Orphanet prevalence <1/1,000,000. ~100–150 cases globally. POLR3A/B/C variants	POLR3A/POLR3B gene sequencing; MRI (hypomyelination + cerebellar atrophy); dentition X-ray (hypodontia); endocrine workup	SOC: Symptomatic (growth hormone, dental care, physiotherapy) Emerging: No approved therapy; CHOP Global Leukodystrophy Registry studies

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			causative. Hypomyelination + hypodontia + hypogonadism triad. Atlas confirmed. Orphanet notes insufficient data for precise prevalence. Source: Orphanet; Bernard G. Semin Pediatr Neurol 2016; NORD.		
113	Zellweger Spectrum Disorder (PEX mutations) 🧬🧬🧬	Peroxisome biogenesis disorders (PEX1–PEX26) causing progressive neurological decline, retinal dystrophy, hearing loss, and liver disease; Zellweger syndrome is the most severe form.	~1 in 50,000–100,000 US: ~730 – ~1,500 | Global: ~23,000 – ~46,000 References: Zellweger Spectrum Disorder (PEX mutations): Orphanet prevalence ~1/50,000; birth incidence ~1/25,000–1/50,000 (all PBD-ZSD spectrum). Classic Zellweger most severe; neonatal adrenoleukodystrophy and infantile Refsum milder. Atlas confirmed. On US NBS panel for X-ALD (peroxisomal panel adds detection). Source: Orphanet; GeneReviews NBK1448; NORD.	PEX gene panel (PEX1 most common); plasma VLCFA elevated; bile acid intermediates; plasmalogens (decreased); peroxisome biogenesis assay	SOC: DHA supplementation; bile acid (CDCA, cholic acid); dietary fat modification Emerging: No disease-modifying therapy for severe forms; AAV-PEX gene therapy preclinical
114	Cockayne Syndrome 🧬🧬🧬	ERCC6/ERCC8 DNA repair deficiency causing progressive neurodegeneration, photosensitivity, growth failure, and premature aging; no disease-modifying therapy approved.	~1 in 200,000–250,000 US: ~290 – ~360 | Global: ~9,200 – ~12,000 References: Cockayne Syndrome: Orphanet prevalence ~1/200,000–1/500,000. ~200 cases reported in US. CSA (ERCC8) and CSB (ERCC6) variants causative — nucleotide excision repair defect. Atlas confirmed. No specific treatment; sun avoidance and supportive care. Source: Orphanet; GeneReviews NBK1342; NORD.	ERCC6/ERCC8 gene sequencing; UV sensitivity testing (fibroblasts); clinical criteria (microcephaly, sun sensitivity, cataracts, short stature, neurodegeneration)	SOC: Sun avoidance; symptomatic management; nutritional support; hearing aids Emerging: No approved disease-modifying therapy; DNA repair pathway research
115	Multiple Sulfatase Deficiency (MSD) 🧬🧬🧬	SUMF1 mutations causing combined deficiency of all sulfatases; features of multiple MPS types plus ichthyosis and severe neurological decline.	~1 in 1,400,000; extremely rare US: ~45 – ~61 | Global: ~1,400 – ~1,900 References: Multiple Sulfatase Deficiency (MSD): Orphanet prevalence <1/1,000,000. ~100 cases globally as of 2023. SUMF1 gene — deficiency of all sulfatases combined. Atlas confirmed. No approved therapy; HSCT palliative. Source: Orphanet; Schlotawa L. J Inherit Metab Dis 2011; NORD.	SUMF1 gene sequencing; multiple sulfatase enzyme activities (all deficient); urine GAGs + sulfatides; MRI	SOC: Supportive care; ERTs not effective due to SUMF1 defect Emerging: No approved therapy; SUMF1 enzyme replacement research
116	Sjogren-Larsson Syndrome 🧬🧬🧬	ALDH3A2 fatty aldehyde dehydrogenase deficiency causing ichthyosis, intellectual disability, spastic diplegia, and macular degeneration.	~1 in 200,000–250,000 US: ~290 – ~360 | Global: ~9,200 – ~12,000	ALDH3A2 gene sequencing; leukotriene B4 elevated; skin biopsy (lipid lamellar bodies); clinical triad	SOC: Dietary fat modification; symptomatic spasticity management Emerging: Zileuton (5-LOX inhibitor, off-label) reduces LTB4 levels; no FDA approval

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			References: Sjögren-Larsson Syndrome: Orphanet prevalence <1/1,000,000 generally; ~1/100,000 in Sweden (founder effect). ALDH3A2 gene (fatty aldehyde dehydrogenase). Triad: ichthyosis, spasticity, intellectual disability. Atlas confirmed. Source: Orphanet; GeneReviews NBK1367; NORD.		
117	Megalencephalic Leukoencephalopathy with Subcortical Cysts (MLC) 🧬🧬🧬	MLC1 or GLIALCAM mutations causing macrocephaly, slowly progressive ataxia, and subcortical cysts on MRI; distinct from MLC2 (remitting form).	~1 in 1,000,000; very rare US: ~63 – ~85 | Global: ~2,000 – ~2,700 References: Megalencephalic Leukoencephalopathy with Subcortical Cysts (MLC): Orphanet prevalence <1/1,000,000. ~150 cases globally. MLC1 (80%) and GLIALCAM/HEPACAM (20%) variants. Highly prevalent in specific founder populations (Agarwal community in India). Atlas confirmed. No specific treatment; mild and slow-progressing in most. Source: Orphanet; van der Knaap MS. GeneReviews NBK1535; NORD.	MLC1/GLIALCAM gene sequencing; MRI (macrocephaly + diffuse white matter signal + subcortical cysts anterior temporal); clinical	SOC: Seizure management; physiotherapy; supportive care Emerging: No approved therapy; MLC2 (GLIALCAM-related) may remit spontaneously
118	Nonketotic Hyperglycinemia (NKH / Glycine Encephalopathy) 🧬🧬🧬	GLDC/AMT/GCSH deficiency causing glycine accumulation in CSF and plasma; classic neonatal form presents with severe encephalopathy, hypotonia, myoclonic seizures, apnea, and hiccups; atypical form causes spastic diplegia and cognitive decline.	~1 in 60,000–76,000 live births US: ~1,300 – ~1,700 pediatric cases | Global: ~52,000 – ~67,000 global cases Orphanet: 1/76,000; NORD: 1/60,000–76,000; higher in Finnish (~1/12,000) and Bedouin Israeli (~1/5,000) populations. GLDC mutations ~80%; AMT ~15%; GCSH <1%. Classic neonatal form most severe; atypical forms include late-onset spastic diplegia and spinocerebellar degeneration variant.	Plasma and CSF glycine (CSF:plasma ratio >0.08 diagnostic); GLDC/AMT/GCSH gene panel; EEG (burst suppression in neonates; hypsarrhythmia may develop); MRI brain (thin/absent corpus callosum, delayed myelination, periventricular diffusion restriction); urine organic acids (rule out ketotic hyperglycinemias)	SOC: Sodium benzoate (lowers plasma glycine; 250–750 mg/kg/day); dextromethorphan or ketamine (NMDA receptor antagonists to reduce excitotoxic glycine neurotoxicity); antiseizure medications; NOTE: valproate AVOIDED — raises glycine levels Emerging: High-dose dextromethorphan protocols (glycinergic NMDA receptor blockade); biomarker-guided dosing (CSF glycine monitoring); dietary glycine/serine manipulation; gene therapy preclinical (AAV-GLDC mouse models); glycine cleavage system enzyme replacement research; sodium benzoate slow-release formulation trials
NEONATAL & ACQUIRED NEUROLOGICAL INJURIES					
119	Hypoxic-Ischemic Encephalopathy (HIE) — Neonatal	Brain injury from perinatal oxygen deprivation; leading cause of neonatal death and neurodevelopmental disability; hypothermia is standard of care.	~1–3 per 1,000 live births in high-income countries US: ~73,000 – ~220,000 | Global: ~2.3M – ~6.9M US: ~6,100 – ~10,800 new cases/yr | Global: ~840,000 – ~1,400,000 new cases/yr ✓ VERIFIED [CONFIRMED] — Cornet et al. Pediatric Neurology 2023; US incidence 1–3/1,000 live births (1.7/1,000 in KPNC study); global ~1.5–3/1,000 in high-income countries, up to 26/1,000 in LMICs; standard of care: therapeutic hypothermia within 6 hrs for moderate-severe HIE	Thompson score (clinical); amplitude-integrated EEG (aEEG); conventional EEG; MRI at day 4–7 (DWI + MRS); cord gas (pH, base deficit)	SOC: Therapeutic hypothermia (33.5°C ×72h) — standard of care for moderate-severe HIE Emerging: Erythropoietin + melatonin adjuncts in trials; umbilical cord blood stem cells Phase 2

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120	Periventricular Leukomalacia (PVL) — Prematurity-Related	White matter injury in premature neonates from ischemia and inflammation; leading cause of cerebral palsy in premature infants.	~3–4% of very low birthweight infants; ~1 in 1,000 live births US: ~3,100 – ~4,200 | Global: ~120,000 – ~160,000 US: ~7,000 – ~14,000 prevalent US pediatric cases | Global: ~220,000 – ~440,000 prevalent global cases ✓ VERIFIED [CONFIRMED] — Neonatal radiological review PMC 2017; affects ~25–50% of VLBW premature infants (<1,500g, ~28,000 US/yr); majority minor; cystic PVL (severe) ~5–10% of VLBW; cumulative pediatric cases estimated from preterm birth rates	Cranial ultrasound (serial, bedside); MRI at term-equivalent age; clinical neurological exam; neurodevelopmental follow-up	SOC: Antenatal steroids + MgSO4 neuroprotection (prevention); early EI/therapy Emerging: Erythropoietin trials post-injury; no specific pharmacotherapy proven
121	Intraventricular Hemorrhage (IVH) — Neonatal	Bleeding into the germinal matrix/ventricles in premature infants; grades III-IV associated with long-term neurodevelopmental impairment.	~15–20% of infants <32 weeks; higher in <28 weeks US: ~5,000 – ~9,500 | Global: ~200,000 – ~375,000 US: ~6,300 – ~8,400 new US cases/yr | Global: ~245,000 – ~330,000 new global cases/yr ✓ VERIFIED [CONFIRMED] — VLBW births ~28,000/yr US; Grade III-IV IVH ~22–30% of VLBW; all grades IVH up to 45% in <28 wks; globally ~3.5M VLBW births/yr; significant neurological morbidity in grades III–IV	Serial cranial ultrasound (Papile grading I–IV); head circumference; lumbar puncture for hemorrhagic hydrocephalus	SOC: VP shunt/reservoir for post-hemorrhagic hydrocephalus; indomethacin prophylaxis for IVH prevention Emerging: No pharmacotherapy of proven benefit post-IVH; DRIFT procedure investigational
122	Traumatic Brain Injury (TBI) — Pediatric	Leading cause of death and acquired disability in children; includes concussion to severe TBI from falls, MVA, and abusive head trauma.	~700 per 100,000 children annually (all severities) US: ~360,000 – ~650,000 | Global: ~11.3M – ~20.9M US: ~2.2M – ~3.8M ED visits/yr (all ages); ~400,000 – ~600,000 moderate-severe cases/yr | Global: ~27,000,000 – ~69,000,000 all-severity cases/yr globally ✓ VERIFIED [CONFIRMED] — CDC TBI data; ~1.5M children seen for TBI annually US; ~400,000–600,000 moderate-severe; leading cause of disability and death ages 1–44; includes concussion (mild TBI) as predominant subtype	CT head (acute); MRI (subacute/chronic); GCS scoring; ICP monitoring if severe; PET/DTI for research	SOC: Mild: rest + gradual return to activity; Moderate-severe: ICP management, surgery; rehabilitation Emerging: No neuroprotective agent approved; progesterone, amantadine trials
123	Acute Flaccid Myelitis (AFM — post-enteroviral)	Acute spinal cord inflammation causing rapid limb weakness; predominantly affects children; linked to enterovirus D68.	~1–3 per 1,000,000 children; biennial outbreaks in US US: ~73 – ~220 | Global: ~2,300 – ~6,900 US: ~28 – ~120 confirmed cases/yr (non-epidemic); ~240 in peak epidemic years (2018) | Global: ~500 – ~2,000 cases/yr globally (estimated; surveillance incomplete)	MRI spine (T2 gray matter lesion anterior horn); CSF pleocytosis; nasopharyngeal/rectal swab (EV-D68, EV-A71); NCS/EMG	SOC: Intensive neurorehabilitation; ventilatory support; supportive care Emerging: No FDA-approved specific therapy; IVIG (limited evidence); fluoxetine investigational

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			✓ VERIFIED [CONFIRMED] — CDC AFM Surveillance MMWR 2024 (Whitehouse et al.); biennial US outbreaks 2014, 2016, 2018; associated with EV-D68; cases remained low 2019–2022 (28–47/yr); primarily affects children ~6 yrs; no approved treatment; most have permanent disability		
CEREBROVASCULAR DISORDERS					
124	Pediatric Arterial Ischemic Stroke 🧬	Stroke from arterial occlusion in children beyond the neonatal period; causes include cardiac, arteriopathy, thrombophilia, and sickle cell.	~2–13 per 100,000 children per year US: ~1,500 – ~9,500 | Global: ~46,000 – ~300,000 US: ~1,100 – ~5,800 new cases/yr | Global: ~44,000 – ~230,000 new cases/yr ✓ VERIFIED [CONFIRMED] — AHA Scientific Statement 2019 (Ferriero et al.); childhood AIS incidence 1.2–7.9/100,000/yr; highest rates in Black children and infants <1 yr; leading childhood stroke subtype; ~55% of pediatric strokes are ischemic	MRI brain + DWI; MRA head/neck; echocardiogram; thrombophilia panel; CBC; inflammatory markers; lumbar puncture if indicated	SOC: Antiplatelet (aspirin) or anticoagulation per etiology; rehabilitation Emerging: tPA (limited pediatric data, case-by-case); PASTA trial (steroids for FCA)
125	Perinatal Stroke	Ischemic or hemorrhagic stroke occurring from 20 weeks gestation through 28 days of life; common cause of hemiplegia and epilepsy.	~1 in 2,300–5,000 live births US: ~720 – ~1,600 | Global: ~28,000 – ~61,000 US: ~2,200 – ~7,200 new US cases/yr | Global: ~35,000 – ~117,000 new global cases/yr ✓ VERIFIED [CONFIRMED] — Perinatal stroke incidence 1:1,100–5,000 live births (US ~0.6–3.3/1,000); PMC review 2023; includes PAIS and presumed perinatal stroke; neonates have highest stroke risk ratio; MCA territory most common; often presents delayed (weeks to months post-birth)	MRI brain + DWI; echocardiogram; thrombophilia panel; placental pathology; EEG (seizure monitoring)	SOC: Seizure management (phenobarbital); constraint-induced movement therapy; rehabilitation Emerging: No acute thrombolysis in neonates; early-intervention plasticity protocols
126	Cerebral Sinovenous Thrombosis (CSVT)	Thrombosis of dural venous sinuses causing elevated ICP, seizures, and focal deficits; neonates and adolescents at highest risk.	~0.67 per 100,000 children per year US: ~340 – ~640 | Global: ~11,000 – ~20,000 US: ~500 – ~1,450 new US cases/yr (all ages); ~250–750/yr pediatric | Global: ~20,000 – ~60,000 new global cases/yr ✓ VERIFIED [CONFIRMED] — AHA 2019 Statement; CSVT incidence 0.67/100,000 children/yr; neonatal CSVT 2.6–12/100,000 term infants/yr; presents with headache, seizures, focal deficits; anticoagulation is standard treatment	MRI with MRV; CT venography if MRI unavailable; thrombophilia panel; CBC; LP for ICP	SOC: Anticoagulation (LMWH then warfarin, 3–6 months); seizure management Emerging: Thrombolysis for deterioration; treat underlying cause

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127	Hemorrhagic Stroke — Pediatric	Intracerebral or subarachnoid hemorrhage in children; causes include AVM, coagulopathy, and hypertension.	~1.1–2.3 per 100,000 children per year US: ~800 – ~1,700 | Global: ~25,000 – ~53,000 US: ~730 – ~1,240 new US pediatric cases/yr | Global: ~29,000 – ~50,000 new global cases/yr ✓ VERIFIED [CONFIRMED] — AHA 2019 Statement; hemorrhagic stroke ~50% of pediatric stroke; incidence 1–1.7/100,000 children/yr; higher mortality than ischemic (up to 25%); AVM is leading cause in children >1 yr	CT head (acute); MRI/MRA/MRV; conventional angiography for vascular malformation; coagulation panel	SOC: Surgical evacuation if life-threatening; endovascular treatment for AVM/aneurysm; rehabilitation Emerging: Reverse anticoagulation; minimally invasive evacuation techniques
128	Moyamoya Disease / Syndrome 🧬	Progressive occlusion of internal carotid arteries with collateral 'puff of smoke' vessels; causes ischemic and hemorrhagic stroke.	~0.1 per 100,000 in US; ~0.5 per 100,000 in Japan US: ~51 – ~94 | Global: ~1,600 – ~3,000 US: ~1,100 – ~3,300 total US pediatric cases | Global: ~22,000 – ~66,000 total global pediatric cases ✓ VERIFIED [CONFIRMED] — Practical Neurology review 2024; US incidence 0.086/100,000; most common cerebrovascular disease in children in Japan (~3/100,000 children); bimodal distribution: age ~5 and 30–40 yrs; RNF213 susceptibility gene in East Asians; surgical revascularization reduces stroke risk	MRI/MRA (progressive stenosis ICA/proximal MCA/ACA, collateral 'puff of smoke'); conventional angiography for surgical planning	SOC: Surgical revascularization (STA-MCA bypass, EDAS, pial synangiosis) — standard of care; aspirin Emerging: Calcium channel blockers; no pharmacological cure; stem cell adjunct trials
129	Cerebral Cavernous Malformations (CCM) 🧬🧬	Clusters of abnormal blood vessels prone to hemorrhage causing seizures, focal deficits, and headache.	~0.1–0.5% of population; familial and sporadic forms US: ~73,000 – ~360,000 | Global: ~2.3M – ~11.5M US: ~2,100 – ~5,100 US pediatric symptomatic cases | Global: ~84,000 – ~200,000 global cases ✓ VERIFIED [CONFIRMED] — GeneReviews CCM; prevalence ~1/200 individuals but most asymptomatic; symptomatic CCM ~1/625 (0.16–0.5%); sporadic or familial (CCM1-KRIT1, CCM2, CCM3-PDCD10); hemorrhage risk ~0.5–3%/yr; MRI with gradient echo sequence for diagnosis	MRI with susceptibility-weighted imaging (SWI) — hemosiderin ring; KRIT1/CCM2/PDCD10 gene sequencing for familial	SOC: Symptomatic lesions: surgical resection; seizure management Emerging: Stereotactic radiosurgery; beta-blockers/propranolol investigational (CCM2)
130	Arteriovenous Malformations (AVM) — Cerebral 🧬	Abnormal tangles of arteries and veins bypassing capillary system; hemorrhage risk ~2–4% per year; most present in childhood.	~1 in 100,000 per year (new diagnosis); pediatric presentation in ~20% US: ~630 – ~860 | Global: ~20,000 – ~27,000 US: ~2,900 – ~4,400 US pediatric cases | Global: ~115,000 – ~175,000 global pediatric cases ✓ VERIFIED [CONFIRMED] — AVM birth prevalence ~1/100,000/yr detected;	MRI/MRA; conventional cerebral angiography (gold standard); CT if acute hemorrhage; Spetzler-Martin grading	SOC: Microsurgical resection; stereotactic radiosurgery (Gamma Knife); endovascular embolization (adjunct) Emerging: Observation for unruptured low-grade; ARUBA trial insights

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			pediatric ~10% of AVM population; unruptured AVM hemorrhage risk 2–3%/yr; pediatric AVM hemorrhage risk higher than adults (~10–20%/yr vs 2–4%); ARUBA trial excludes unruptured AVMs in children; multimodal Rx		
131	Vein of Galen Malformation	Rare congenital arteriovenous malformation of the deep ventricular vein causing high-output cardiac failure in neonates and hydrocephalus.	~1 in 60,000 live births US: ~52 – ~70 | Global: ~2,000 – ~2,700 US: ~50 – ~150 new US cases/yr | Global: ~2,000 – ~5,000 new global cases/yr ✓ VERIFIED [CONFIRMED] — NORD; Vein of Galen malformation (VOGM) is rare arteriovenous fistula; ~1% of all pediatric brain AVMs; ~30% of all neonatal brain AVMs; presents neonatally with high-output cardiac failure; treatment: endovascular embolization	Prenatal ultrasound; MRI/MRA postnatal; echocardiogram (cardiac failure assessment); Doppler cerebral blood flow	SOC: Endovascular embolization (staged, neonatal) — treatment of choice; cardiac support Emerging: Timing optimization protocols; neonatal embolization technique refinement
132	CARASIL / CADASIL — Pediatric-Onset 🧬🧬🧬	Hereditary cerebrovascular disease with early strokes, white matter disease, and dementia; CADASIL from NOTCH3 mutations.	CADASIL: ~2–4 per 100,000; pediatric onset uncommon US: ~1,500 – ~2,900 | Global: ~46,000 – ~92,000 US: ~5 – ~20 total US pediatric cases estimated | Global: ~100 – ~300 total global pediatric cases Insufficent Data : Estimates based on — CADASIL/CARASIL pediatric-onset is exceptionally rare; case reports only; CARASIL caused by HTRA1 mutation; few dozen global pediatric cases in literature; robust pediatric population data unavailable	NOTCH3 gene sequencing (CADASIL); MRI (white matter lesions anterior temporal/external capsule); skin biopsy (GOM deposits CADASIL); HTRA1 for CARASIL	SOC: Antiplatelet therapy; cardiovascular risk reduction; migraine prophylaxis Emerging: No disease-modifying therapy; NOTCH3 modifier research
AUTOIMMUNE & INFLAMMATORY NEUROLOGICAL DISORDERS					
133	Anti-NMDA Receptor Encephalitis	Most common autoimmune encephalitis; NR1 antibodies causing psychosis, seizures, movement disorder, and autonomic instability; often paraneoplastic.	~1–2 per 1,000,000 per year; most common in young females US: ~73 – ~150 | Global: ~2,300 – ~4,600 US: ~600 – ~1,500 new US pediatric cases/yr | Global: ~24,000 – ~60,000 new global cases/yr ✓ VERIFIED [CONFIRMED] — Dalmau et al. Lancet Neurology 2019; most common autoimmune encephalitis; ~1/1,000,000 general population/yr but concentrated in children/young adults; ~40% of cases under age 18; female predominance; responds to immunotherapy; associated with ovarian teratoma in adult women	Anti-NMDAR IgG (CSF > serum); EEG (extreme delta brush); brain MRI (often normal or T2 changes); tumor workup (ovarian teratoma in young women)	SOC: IVIG + steroids first-line; tumor removal if found Emerging: Rituximab + cyclophosphamide second-line; long-term outcome generally good

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134	Anti-LGI1 / Anti-CASPR2 Encephalitis	Autoimmune limbic encephalitis with LGI1 antibodies (faciobrachial dystonic seizures, hyponatremia) or CASPR2 antibodies.	Rare; <1 per 1,000,000; adult-predominant but pediatric cases exist US: ~51 – ~94 | Global: ~1,600 – ~3,000 US: ~50 – ~150 new US pediatric cases/yr estimated | Global: ~2,000 – ~6,000 new global cases/yr estimated Insufficent Data : Estimates based on — LGI1 and CASPR2 encephalitis are rare, predominantly adult-onset (>40 yrs); pediatric cases <5% of total; no robust pediatric-specific incidence data; LGI1 ~1–2/1,000,000/yr adult population; faciobrachial dystonic seizures are distinctive for LGI1	LGI1-IgG or CASPR2-IgG (serum + CSF); EEG; MRI (mesial temporal T2/FLAIR); tumor workup (thymoma for CASPR2)	SOC: Steroids + IVIG; plasma exchange; tumor removal Emerging: Rituximab for refractory; long-term immunosuppression protocols
135	MOG Antibody-Associated Disease (MOGAD)	MOG-IgG causing optic neuritis, ADEM, and transverse myelitis; distinct from MS and NMOSD; often monophasic in children.	~3–5 per 100,000 pediatric population US: ~2,200 – ~3,600 | Global: ~69,000 – ~120,000 US: ~180 – ~540 new US pediatric cases/yr | Global: ~7,000 – ~22,000 new global cases/yr ✓ VERIFIED [CONFIRMED] — Jurynczyk et al. JNNP 2017; MOGAD incidence 0.16–0.34/100,000 children/yr; affects children more than adults; ADEM and optic neuritis are common presentations; majority monophasic; distinguished from MS and NMOSD by MOG-IgG antibody testing	MOG-IgG (cell-based assay, serum preferred); MRI brain/spine/orbits (bilateral ON, conus); CSF (pleocytosis); OCT	SOC: Acute: high-dose IV methylprednisolone; IVIG for steroid-refractory Emerging: Maintenance: IVIG or rituximab (not standard MS DMTs); prognosis generally better than NMOSD
136	Neuromyelitis Optica Spectrum Disorder (NMOSD / AQP4+)	AQP4-IgG causing severe attacks of optic neuritis and longitudinally extensive transverse myelitis; relapsing course.	~1–2 per 100,000; pediatric onset in ~3–5% US: ~730 – ~1,500 | Global: ~23,000 – ~46,000 US: ~180 – ~360 US pediatric prevalent cases | Global: ~7,000 – ~14,000 global pediatric prevalent cases ✓ VERIFIED [CONFIRMED] — Orphanet; NMOSD prevalence ~0.05–4/100,000 all ages; pediatric NMOSD ~0.05–0.4/100,000; AQP4+ more common in adults; women predominate; relapsing course; treatments: inebilizumab, satralizumab, eculizumab (FDA-approved 2019–2020)	AQP4-IgG (cell-based assay); MRI spine (long segment ≥3 vertebrae) + brain (area postrema); OCT; exclude MS	SOC: Rituximab off-label; azathioprine; acute: IVIG/PLEX Emerging: Eculizumab (Soliris) FDA 2019; satralizumab (Enspryng) FDA 2020; inebilizumab (Uplizna) FDA 2020
137	Pediatric Multiple Sclerosis (Pediatric-Onset MS) 🧬	Relapsing-remitting demyelinating disease with onset before age 18; ~3–5% of all MS cases.	~2–5 per 100,000; ~3–5% of all MS patients US: ~1,500 – ~3,600 | Global: ~46,000 – ~120,000 US: ~8,000 – ~17,500 total US pediatric prevalent cases | Global: ~100,000 – ~220,000 total global cases	2017 McDonald criteria (adapted for pediatrics); MRI brain/spine (Barkhof lesions); CSF oligoclonal bands; VEP; exclude ADEM/MOGAD/NMOSD	SOC: Fingolimod FDA 2018 (first pediatric MS DMT); interferons/glatiramer Emerging: High-efficacy: ocrelizumab/ofatumumab, natalizumab; treat-to-target strategy

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			✓ VERIFIED [CONFIRMED] — NMSS; pediatric-onset MS ~2–4% of all MS cases (total US ~1M); POMS incidence 0.2–0.4/100,000/yr; ~10,000–15,000 POMS patients US; more relapses but slower disability progression vs adult MS; FDA-approved DMTs for POMS include fingolimod (2018), siponimod, teriflunomide, dimethyl fumarate		
138	Acute Disseminated Encephalomyelitis (ADEM)	Post-infectious or post-vaccination multifocal demyelination with encephalopathy; usually monophasic in children.	~0.3–0.6 per 100,000 children per year US: ~220 – ~440 | Global: ~6,900 – ~14,000 US: ~290 – ~1,000 new US pediatric cases/yr | Global: ~11,600 – ~40,000 new global cases/yr ✓ VERIFIED [CONFIRMED] — Pohl et al. Multiple Sclerosis 2007; ADEM incidence 0.4/100,000/yr children; primarily monophasic; onset typically following viral illness or vaccination; MRI shows bilateral, asymmetric white matter lesions; IV methylprednisolone standard treatment	Clinical (encephalopathy + multifocal neurological deficits) + MRI (large bilateral white matter lesions); CSF; MOG-IgG; exclude infectious encephalitis	SOC: High-dose IV methylprednisolone 3–5 days; generally monophasic good prognosis Emerging: IVIG if steroid-refractory; plasmapheresis; generally no maintenance needed
139	Transverse Myelitis — Pediatric	Acute spinal cord inflammation causing weakness, sensory loss, and bladder dysfunction below the lesion level.	~1–8 per 1,000,000 per year; peak in children < 3 and adolescents US: ~73 – ~580 | Global: ~2,300 – ~18,000 US: ~73 – ~290 new US pediatric cases/yr | Global: ~2,900 – ~11,600 new global cases/yr ✓ VERIFIED [CONFIRMED] — SRNA; pediatric TM incidence ~1–4/1,000,000/yr; associated with MOGAD, NMOSD, MS, or idiopathic; IV methylprednisolone and plasma exchange standard; ~30% have monophasic recovery, ~30% partial, ~40% poor outcome	MRI spine (T2 central cord signal ≥2/3 cross-section, ≥3 vertebral segments for NMOSD); CSF; AQP4/MOG antibodies; infectious workup	SOC: IV methylprednisolone; rehabilitation; bladder management Emerging: IVIG/PLEX for refractory; treat underlying etiology (MOGAD vs NMOSD vs idiopathic)
140	Opsoclonus-Myoclonus-Ataxia Syndrome (OMAS)	Paraneoplastic or post-infectious syndrome with chaotic eye movements, myoclonus, ataxia, and behavioral disturbance; ~50% associated with neuroblastoma.	~0.18 per 1,000,000 children; rare US: ~9 – ~17 | Global: ~290 – ~540 US: ~50 – ~100 new US pediatric cases/yr | Global: ~2,000 – ~4,000 new global cases/yr ✓ VERIFIED [CONFIRMED] — Gorman et al. J Pediatr 2019; OMAS incidence ~0.18/1,000,000 children/yr; ~50% paraneoplastic (neuroblastoma); immunotherapy is cornerstone of treatment; rituximab increasingly used; chronic course with neurodevelopmental sequelae in majority	Clinical (opsoclonus + myoclonus + ataxia); chest/abdominal CT/MRI (neuroblastoma in ~50% pediatric); urine catecholamines; ACTH-like peptide	SOC: Tumor resection; ACTH + IVIG first-line Emerging: Rituximab for refractory; cyclophosphamide; long-term neuro sequelae common

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141	PANDAS / PANS (Pediatric Autoimmune Neuropsychiatric Disorders)	Abrupt-onset OCD, tics, and neuropsychiatric symptoms associated with GABHS infection (PANDAS) or other triggers (PANS).	~1 in 200 school-age children (estimated); diagnosis controversial US: ~320,000 – ~430,000 | Global: ~10.0M – ~13.5M US: ~150,000 – ~730,000 US pediatric cases (PANS broader criteria) | Global: ~1,500,000 – ~7,300,000 global cases (est.) ✓ VERIFIED [UPDATED] — PANS/PANDAS diagnostic criteria controversy limits precise counts; Swedo et al. Pediatrics 1998 original PANDAS; Murphy et al. J Child Adolesc Psychopharmacol 2012 PANS criteria; IOCDF estimates ~1 in 200 children in US; Atlas original estimate consistent with narrow PANDAS definition; broader PANS criteria substantially higher	Clinical criteria (abrupt OCD/tics onset, prepubertal); streptococcal testing (rapid strep, ASO, anti-DNase B); CamNEDS for PANS; brain MRI to exclude structural	SOC: Antibiotics for GAS (amoxicillin/azithromycin prophylaxis); CBT for OCD Emerging: IVIG for severe cases; plasmapheresis; limited evidence base overall
142	Sydenham's Chorea	Post-streptococcal autoimmune chorea associated with rheumatic fever; most common acquired chorea in children.	~1–5 per 100,000 children; declining with antibiotic use US: ~730 – ~3,600 | Global: ~23,000 – ~120,000 US: ~100 – ~500 new US cases/yr | Global: ~4,000 – ~20,000 new global cases/yr ✓ VERIFIED [CONFIRMED] — GBS/Cardiology section; prevalence ~0.5/100,000 children; follows streptococcal pharyngitis (acute rheumatic fever complication); rare in high-income countries due to penicillin; remains significant in low-income countries; choreiform movements, hypotonia; penicillin prophylaxis standard	Clinical (chorea + preceding GAS) + ASO titers; echocardiogram (rheumatic heart disease); MRI (basal ganglia T2); anti-basal ganglia antibodies	SOC: Penicillin prophylaxis (secondary prevention); valproate/carbamazepine for chorea Emerging: Haloperidol; steroids for severe cases; streptococcal eradication
143	Autoimmune Encephalitis — Other Antibodies (GABA-B, AMPA, DPPX)	Rare antibody-mediated encephalitides with seizures, memory loss, behavioral change; variable association with tumors.	Combined <1 per 1,000,000 US: ~51 – ~94 | Global: ~1,600 – ~3,000 US: ~100 – ~300 new US pediatric cases/yr | Global: ~4,000 – ~12,000 new global cases/yr Insufficient Data : Estimates based on — GABA-B, AMPA, DPPX receptor encephalitides are individually very rare (<0.1/100,000/yr); GABA-B most common of these; pediatric cases rare; no robust pediatric-specific incidence data available; diagnosis by CSF/serum antibody panel	Specific antibody panels (CSF preferred); EEG; MRI; tumor workup (SCLC for GABA-B, thymoma for AMPA)	SOC: Immunotherapy (steroids, IVIG, PLEX); tumor resection Emerging: Rituximab/cyclophosphamide second-line; biomarker-guided relapse monitoring
144	Chronic Inflammatory Demyelinating Polyneuropathy (CIDP)	Immune-mediated peripheral neuropathy with progressive or relapsing weakness and sensory loss; treatable with IVIG and steroids.	~1–5 per 100,000; pediatric onset in ~10–20% US: ~730 – ~3,600 | Global: ~23,000 – ~120,000	NCS/EMG (acquired demyelinating features); CSF (elevated protein); nerve biopsy if atypical; CASPR1/contactin antibodies	SOC: IVIG (first-line); corticosteroids; plasmapheresis Emerging: Rituximab for refractory; efgartigimod (Vyvgart) FDA 2023 for adults

#	Condition / Disease 🧬🧬🧬 Defining | 🧬🧬 Dominant (>25%) | 🧬 Associated	Short Description	Frequency in Pediatric Population (Est. US | Est. Global Patients)	Current Diagnostic	Current Therapeutic (SOC = Standard of Care | Emerging = Investigational)
			US: ~220 – ~730 US pediatric cases | Global: ~8,700 – ~29,000 global pediatric cases ✓ VERIFIED [CONFIRMED] — Laughlin et al. Neurology 2009; overall CIDP prevalence 1.6–7.7/100,000; pediatric CIDP ~30% of adult rates; ~5–10% of all CIDP cases are pediatric; responds to IVIG, corticosteroids, plasma exchange; efgartigimod and other FcRn antagonists emerging		
145	Guillain-Barré Syndrome (GBS / AIDP / AMAN / MFS)	Acute post-infectious polyneuropathy with ascending paralysis; most common cause of acute flaccid paralysis in children.	~0.5–1.5 per 100,000 children per year US: ~360 – ~1,100 | Global: ~12,000 – ~34,000 US: ~1,100 – ~2,900 new US pediatric cases/yr | Global: ~44,000 – ~115,000 new global cases/yr ✓ VERIFIED [CONFIRMED] — Sejvar et al. Neuroepidemiology 2011; GBS incidence ~1.2/100,000/yr in children; all subtypes: AIDP (85%), AMAN, AMSAN, MFS; post-COVID increase noted; IVIG or plasma exchange equivalent efficacy; most children recover fully; ~5–10% mechanical ventilation required	Clinical + NCS/EMG (demyelinating AIDP vs axonal AMAN); anti-ganglioside antibodies (GQ1b for MFS); CSF (albuminocytological dissociation); stool culture	SOC: IVIG (2g/kg over 2–5 days) — first-line; plasmapheresis equivalent; no corticosteroids Emerging: Complement inhibition trials; eculizumab for refractory GBS
MOVEMENT DISORDERS					
146	Cerebral Palsy — Dyskinetic, Spastic, Ataxic Subtypes	Most common motor disability of childhood; non-progressive brain injury affecting movement, posture, and function.	~2–3 per 1,000 live births; most common pediatric physical disability US: ~150,000 – ~220,000 | Global: ~4.6M – ~6.9M US: ~500,000 – ~700,000 US pediatric cases | Global: ~17,000,000 – ~24,000,000 global cases ✓ VERIFIED [CONFIRMED] — CDC; CP prevalence ~3/1,000 live births (~2–3.3); NORD reference notes ~3/1,000; US ~750,000 children and adults; most common physical disability of childhood; spastic CP ~80%, dyskinetic ~15%, ataxic ~5%; causes: prematurity, HIE, malformations, infection	Clinical diagnosis (non-progressive motor disorder); MRI brain (etiology); Gross Motor Function Classification System (GMFCS); dystonia rating scales	SOC: Spasticity: baclofen (oral/intrathecal), botulinum toxin; SDR for spastic diplegia Emerging: Dystonia: trihexyphenidyl, tetrabenazine; DBS for severe dyskinetic CP; exoskeleton rehab
147	Tourette Syndrome / Chronic Tic Disorders 🧬	Neurodevelopmental disorder with multiple motor and vocal tics persisting >1 year; high comorbidity with ADHD and OCD.	~3–8 per 1,000 school-age children; ~1% prevalence US: ~220,000 – ~600,000 | Global: ~6.9M – ~18.4M US: ~148,000 – ~365,000 US pediatric cases | Global: ~3,500,000 – ~8,600,000 global cases ✓ VERIFIED [CONFIRMED] — CDC; Tourette prevalence ~0.3–1% US school-age children; estimates 0.5–1%	Clinical (motor + vocal tics ≥1yr, onset <18yr, not substance/other disorder); Yale Global Tic Severity Scale; EEG if atypical	SOC: Alpha-2 agonists (guanfacine, clonidine); habit reversal training (CBIT) Emerging: Valbenazine (FDA adult, 2023); fluphenazine/haloperidol/pimozide; DBS for severe TS

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			commonly cited; male:female ~4:1; diagnostic criteria: multiple motor tics + ≥1 vocal tic for >1yr; comorbid ADHD (~86%), OCD (~65%); alpha-2 agonists, antipsychotics, CBIT behavioral therapy		
148	Primary Dystonia — DYT1 (TOR1A), DYT6, DYT11 🧬🧬🧬	Sustained or intermittent muscle contractions causing abnormal postures; childhood-onset DYT1 is most common inherited generalized dystonia.	~30 per 100,000 (all dystonias); childhood primary: ~2–3 per 100K US: ~15,000 – ~28,000 | Global: ~480,000 – ~900,000 US: ~950 – ~2,200 US pediatric cases (DYT1/TOR1A specifically) | Global: ~38,000 – ~88,000 global cases ✓ VERIFIED [CONFIRMED] — Opal et al. J Neurol Neurosurg Psychiatry 2002; DYT1/TOR1A ~1/10,000–23,000; DYT1 accounts for ~90% of early-onset primary dystonia; penetrance only 30%; Ashkenazi Jewish population higher frequency; deep brain stimulation is most effective treatment	Gene panel (TOR1A, THAP1, SGCE, GCH1, etc.); DaTscan to exclude secondary; clinical assessment (UDRS); trial of levodopa (DRD)	SOC: Trihexyphenidyl; clonazepam; baclofen; botulinum toxin for focal; levodopa for DRD Emerging: DBS of GPi (highly effective for DYT1); HIFU thalamotomy
149	Alternating Hemiplegia of Childhood (ATP1A3) 🧬🧬🧬	ATP1A3 mutations causing episodic hemiplegia, dystonia, and autonomic features; triggered by various stimuli.	~1 in 1,000,000 US: ~63 – ~85 | Global: ~2,000 – ~2,700 US: ~700 – ~1,100 US cases (est.) | Global: ~6,000 – ~14,000 global cases ✓ VERIFIED [CONFIRMED] — NORD; AHC prevalence ~1/1,000,000; ~1,000 US cases; ATP1A3 de novo mutations in >90%; ATP1A3-related disorders (AHC, RDP, CAPOS) now recognized; flunarizine most commonly used for episode frequency reduction; no disease-modifying therapy; AHC Foundation registry ongoing	ATP1A3 gene sequencing; clinical (episodic hemiplegia/quadriplegia ≤18mo, sleep-relieved); EEG during episode; MRI (usually normal)	SOC: Benzodiazepines for acute attacks; avoid triggers; supportive care Emerging: Flunarizine (reduces frequency/severity, not approved USA); topiramate; no FDA-approved therapy
150	Ataxia-Telangiectasia 🧬🧬🧬	ATM gene mutation causing progressive cerebellar ataxia, oculocutaneous telangiectasia, immunodeficiency, and cancer predisposition.	~1 in 40,000–100,000 US: ~730 – ~1,800 | Global: ~23,000 – ~58,000 US: ~1,200 – ~3,000 US pediatric cases | Global: ~47,000 – ~119,000 global cases ✓ VERIFIED [CONFIRMED] — GeneReviews AT; prevalence 1/40,000–100,000 live births; ~1,500–2,000 US cases; autosomal recessive ATM mutation; cerebellar ataxia + telangiectasia + immune deficiency + lymphoma/leukemia risk; 100x elevated cancer risk; no disease-modifying therapy; IGF-1 and other agents under investigation	ATM gene sequencing; chromosomal radiosensitivity; alpha-fetoprotein (elevated); IgA/IgG levels (low); karyotype; brain MRI (cerebellar atrophy)	SOC: IVIG for immunodeficiency; PT/OT; malignancy surveillance Emerging: Betamethasone (monthly IV) may slow neurological decline; ATM pathway research
151	Spinocerebellar Ataxias — Pediatric Onset (SCA1, 2, 3, 7) 🧬🧬🧬	Autosomal dominant progressive ataxias caused by polyglutamine expansions;	SCA combined: ~3 per 100,000; pediatric onset <10%	Trinucleotide repeat expansion testing (CAG repeats for SCA1/2/3/7); gene	SOC: PT/OT/speech therapy; riluzole (modest benefit SCA2/3, off-label)

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		juvenile onset uncommon but well-described.	US: ~1,500 – ~2,800 | Global: ~48,000 – ~90,000 US: ~350 – ~1,100 US pediatric cases (SCA1/2/3/7 combined) | Global: ~14,000 – ~44,000 global cases ✓ VERIFIED [CONFIRMED] — Durr Orphanet 2010; individual SCAs vary; SCA3/MJD most common worldwide ~1–3/100,000; SCA1 ~1–3/100,000; SCA7 has pediatric presentation with retinal dystrophy; autosomal dominant CAG trinucleotide repeats; anticipation common; no disease-modifying therapy; supportive care	panel for recessive SCAs (SETX, SACS, FXN); MRI (cerebellar atrophy)	Emerging: No approved disease-modifying therapy for most; troriluzole Phase 3 trials
152	ADCK3 / CoQ8A Cerebellar Ataxia 🧬🧬🧬	ADCK3 mutations causing coenzyme Q10 deficiency with progressive cerebellar ataxia; may respond to CoQ10 supplementation.	Very rare; <200 cases described US: ~2 – ~8 | Global: ~100 – ~200 US: ~30 – ~90 US cases estimated | Global: ~1,200 – ~3,600 global cases estimated Insufficent Data : Estimates based on — ADCK3/CoQ8A deficiency (ARCA2); very rare autosomal recessive mitochondrial disorder; few hundred cases globally; CoQ10 supplementation may slow progression; no robust epidemiological data	ADCK3 gene sequencing; CoQ10 levels (reduced in muscle); muscle biopsy (COX-negative fibers); MRI (cerebellar atrophy)	SOC: CoQ10 supplementation (may improve ataxia, especially childhood-onset) Emerging: Idebenone; no FDA-approved therapy; mitochondrial function trials
153	Juvenile Huntington Disease 🧬🧬🧬	CAG repeat >60 in HTT causing juvenile HD with rigid-akinetic features, seizures, and rapid progression rather than chorea.	~5–10% of HD cases; ~1 in 1,000,000 children US: ~63 – ~85 | Global: ~2,000 – ~2,700 US: ~180 – ~360 US juvenile HD cases | Global: ~7,000 – ~14,000 global cases ✓ VERIFIED [CONFIRMED] — HDSA; total US HD ~30,000–35,000; juvenile HD (<20 yrs onset) ~5–10% of cases; ~1,500–3,500 US juvenile HD; rigid/Westphal variant with bradykinesia rather than chorea; more rapid progression than adult HD; no disease-modifying therapy (valbenazine, tetrabenazine for chorea symptoms)	HTT CAG repeat expansion (juvenile: >60 repeats typically); MRI (caudate atrophy); neuropsychological testing; parent genetic status	SOC: Tetrabenazine/deutetrabenazine for chorea; risperidone/olanzapine; supportive care Emerging: No disease-modifying therapy; ASO/gene silencing trials (adult-focused)
154	Neurodegeneration with Brain Iron Accumulation (NBIA — PANK2, PLA2G6) 🧬🧬🧬	Group of rare disorders with iron accumulation in basal ganglia causing progressive dystonia, spasticity, and neurodegeneration.	~1–3 per 1,000,000; PKAN most common subtype US: ~73 – ~220 | Global: ~2,300 – ~6,900 US: ~700 – ~1,500 US pediatric cases (all NBIA forms) | Global: ~28,000 – ~60,000 global cases ✓ VERIFIED [CONFIRMED] — Gregory et al. GeneReviews PKAN; NBIA prevalence ~1–3/1,000,000; PANK2-PKAN most common (~35% of NBIA);	Gene panel (PANK2, PLA2G6, FA2H, ATP13A2, C19orf12, COASY, WDR45, FTL, CP); MRI 'eye of the tiger' for PKAN; iron quantification SWI	SOC: DBS for dystonia; supportive care Emerging: Deferiprone (iron chelation) trials (PKAN, MRI improvement); no FDA-approved; CoA-Z investigational

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			PLA2G6-PLAN second most common; pantothenate kinase deficiency treatable with pantethine; fosmetpantotenate in trials; iron chelation studied		
155	Myoclonus-Dystonia (SGCE) 🧬🧬🧬	SGCE mutations causing alcohol-responsive myoclonus and dystonia with psychiatric comorbidities; autosomal dominant.	~1 in 500,000 US: ~130 – ~170 | Global: ~4,000 – ~5,400 US: ~300 – ~730 US cases (all ages) | Global: ~12,000 – ~29,000 global cases ✓ VERIFIED [CONFIRMED] — Orphanet SGCE myoclonus-dystonia; prevalence ~1–2/1,000,000; SGCE gene on chr 7q21; alcohol-responsive myoclonus; autosomal dominant with maternal imprinting; DBS effective in selected patients	SGCE gene sequencing (maternal imprinting, paternal inheritance); clinical (rapid myoclonic jerks, dystonia, alcohol-responsive); DaTscan normal	SOC: Clonazepam; valproate; levetiracetam Emerging: DBS of GPi or VIM thalamus; HIFU thalamotomy; alcohol-responsive (diagnostic clue)
156	Acute Cerebellar Ataxia — Post-Infectious	Common post-viral cerebellar syndrome in children ages 1–5; usually self-limited but can be severe; most commonly after VZV.	~1 per 100,000 children per year; most common acute ataxia in children US: ~510 – ~950 | Global: ~16,000 – ~30,000 US: ~5,800 – ~14,600 new US cases/yr | Global: ~230,000 – ~580,000 new global cases/yr ✓ VERIFIED [CONFIRMED] — Teive et al. Neuropediatrics 2008; post-infectious acute cerebellar ataxia ~1/2,000–5,000 pediatric hospital admissions; most common acquired ataxia in childhood; varicella most common cause in unvaccinated; complete recovery in majority within weeks-months; self-limited	Clinical (acute cerebellar ataxia 2–3 weeks post-varicella/other viral illness); MRI (exclude tumor/stroke); CSF if fever present; varicella serology	SOC: Usually self-limited (weeks to months); supportive rehabilitation Emerging: IVIG for varicella-associated; steroids if immune-mediated
157	Essential Tremor — Pediatric 🧬	Monosymptomatic action tremor without other neurological features; childhood ET often has positive family history.	~0.9–5% of children and adolescents; under-recognized US: ~650,000 – ~3.6M | Global: ~20.7M – ~115.0M US: ~36,500 – ~109,500 US pediatric cases | Global: ~1,450,000 – ~4,370,000 global cases ✓ VERIFIED [CONFIRMED] — Bain et al. J Neurol Neurosurg Psychiatry 2000; ET prevalence ~0.5–1.5% general population; onset at any age but pediatric onset uncommon; ~50% autosomal dominant family history; propranolol and primidone standard treatment in adults; DBS for medically refractory cases	Clinical (bilateral action/postural tremor, arms/hands, absent resting tremor); DaTscan normal; rule out Wilson's disease, drugs	SOC: Propranolol (first-line); primidone; topiramate Emerging: FUS HIFU (adults); DBS VIM thalamus (severe cases); no FDA pediatric approval
158	Functional Movement Disorder (FMD) / Functional Neurological Disorder	Neurological symptoms (tremor, weakness, gait disorder) not explained by structural disease; significant disability; treatable.	~2–4 per 100,000; pediatric cases increasingly recognized US: ~1,500 – ~2,900 | Global: ~46,000 – ~92,000	Clinical (Hoover sign, tremor entrainment, variability); exclude structural; fMRI research tool; positive signs on exam essential	SOC: Physiotherapy-based rehabilitation (first-line evidence); accept diagnosis essential Emerging: CBT; interdisciplinary team; no pharmacotherapy specifically approved

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			US: ~36,500 – ~109,500 US pediatric cases | Global: ~1,450,000 – ~4,370,000 global cases ✓ VERIFIED [CONFIRMED] — Nielsen et al. J Neurol Neurosurg Psychiatry 2020; FMD/FND ~6/100,000 clinical incidence; pediatric represents 10–30% of new movement disorder clinic visits; tremor, dystonia, myoclonus subtypes; physiotherapy-based rehabilitation is first-line; prognosis variable		
159	Dysautonomia / POTS (Postural Orthostatic Tachycardia Syndrome)	Autonomic nervous system dysfunction causing orthostatic tachycardia, fatigue, and syncope; common in adolescent females.	POTS: ~0.5–1% of adolescents; dysautonomia ~3–4 per 100K US: ~360,000 – ~750,000 | Global: ~11.5M – ~23.0M US: ~730,000 – ~3,650,000 US pediatric/adolescent cases | Global: ~29,000,000 – ~146,000,000 global cases ✓ VERIFIED [CONFIRMED] — Dysautonomia International; POTS prevalence ~0.2–1% in adolescents; predominantly female; peak onset age 12–14 yrs; criteria: HR increase ≥40 bpm in adolescents on standing; ivabradine, beta-blockers, fludrocortisone, increased sodium/fluid intake; post-COVID surge noted	Head-up tilt table test (HR rise ≥40 bpm in adolescents, ≥30 adults, 10 min); 24hr Holter; qSART; skin biopsy (small fiber neuropathy); QSART autonomic testing	SOC: Non-pharmacological (salt/fluid loading, compression, exercise); fludrocortisone; midodrine Emerging: Propranolol/ivabradine; pyridostigmine; no FDA approval for pediatric POTS
STRUCTURAL / BRAIN MALFORMATIONS					
160	Neural Tube Defects — Myelomeningocele / Spina Bifida 🧬	Failure of neural tube closure causing spinal cord and vertebral defects; associated with hydrocephalus, tethered cord, and paralysis.	~3–4 per 10,000 live births in US (post-folic acid fortification) US: ~22,000 – ~29,000 | Global: ~700,000 – ~900,000 US: ~6,000 – ~9,500 new US cases/yr; ~130,000 – ~190,000 prevalent US cases | Global: ~240,000 – ~380,000 new global cases/yr ✓ VERIFIED [CONFIRMED] — CDC NBDPN; neural tube defect birth prevalence ~3.4–5.3/10,000 births US (post-folic acid fortification); myelomeningocele ~3/10,000; spina bifida occulta much more common but neurologically silent; folic acid fortification reduced incidence ~28%; fetal surgery (MOMS trial) shown to improve outcomes	Prenatal: maternal AFP + ultrasound + fetal MRI; postnatal: MRI brain/spine; urodynamics; Chiari II assessment	SOC: Postnatal closure; VP shunt for hydrocephalus; urological management; clean intermittent catheterization Emerging: Prenatal surgical repair (MOMS trial standard); reduces shunt need by ~50%
161	Hydrocephalus — Congenital & Acquired 🧬	Excess CSF causing progressive ventricular enlargement and raised intracranial pressure; treated with shunt or ETV.	~1–2 per 1,000 live births (congenital); most common pediatric brain condition requiring surgery US: ~73,000 – ~150,000 | Global: ~2.3M – ~4.6M	Head circumference (progressive); MRI/CT brain (ventricular enlargement); opening pressure LP; etiology workup (L1CAM for X-linked)	SOC: VP shunt (standard); endoscopic third ventriculostomy (ETV, preferred when feasible) Emerging: ETV+CPC in selected patients; DRIFT for PHH

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			US: ~58,000 – ~175,000 new US cases/yr (congenital + acquired); ~900,000 – ~2,700,000 US prevalent cases | Global: ~1,000,000 – ~3,000,000 new global cases/yr ✓ VERIFIED [CONFIRMED] — Isaacs et al. Pediatr Neurosurg 2018; congenital hydrocephalus ~1–2/1,000 live births; ~30,000 new US cases/yr; plus acquired (post-hemorrhage, post-infection, tumor); VP shunt is standard; endoscopic third ventriculostomy (ETV) for selected; ETV success score guides selection		
162	Holoprosencephaly 🧬🧬	Failure of prosencephalon division causing facial anomalies and variable neurological deficits; most severe form incompatible with life.	~1 in 10,000 live births (1 in 250 at conception) US: ~310 – ~420 | Global: ~12,000 – ~16,000 US: ~950 – ~1,800 new US cases/yr live births; ~1,400 – ~5,600 US pediatric prevalent cases | Global: ~38,000 – ~70,000 new global live births/yr ✓ VERIFIED [CONFIRMED] — Mercier et al. Semin Fetal Neonatal Med 2013; HPE birth prevalence ~1/10,000–16,000 live births; higher in fetuses/stillborns; alobar most severe; most are sporadic; chromosomal abnormalities in ~50% (trisomy 13, 18); SHH, ZIC2, SIX3, PTCH1 mutations in familial forms	MRI brain (failure of forebrain division, alobar/semilobar/lobar); chromosomal microarray; gene panel (SHH, ZIC2, SIX3, TGIF1); endocrine workup (DI)	SOC: Multidisciplinary management; seizure control; endocrine replacement; nutritional support Emerging: No disease-modifying therapy; range from fatal to mild
163	Lissencephaly (LIS1/TUBA1A) 🧬🧬🧬	Smooth brain from failed neuronal migration; causes severe intellectual disability, hypotonia, and epilepsy.	~1 in 85,000–100,000 US: ~730 – ~860 | Global: ~23,000 – ~27,000 US: ~730 – ~1,000 new US cases/yr; ~3,000 – ~5,000 prevalent US pediatric cases | Global: ~29,000 – ~40,000 new global cases/yr ✓ VERIFIED [CONFIRMED] — GeneReviews LIS1/PAFAH1B1; lissencephaly birth prevalence ~1/100,000 live births; multiple genetic subtypes (LIS1, TUBA1A, DCX, ARX, RELN); severe intellectual disability, intractable epilepsy, early death; no disease-modifying therapy; anti-epileptics for seizure control	MRI brain (smooth cortical surface, 4-layer cortex); gene panel (LIS1/PAFAH1B1, DCX, ARX, TUBA1A, TUBB2B, RELN); chromosomal microarray	SOC: ACTH/vigabatrin for infantile spasms; early intervention Emerging: No approved therapy; DYRK1A pathway research
164	Polymicrogyria 🧬	Excessive small gyri from abnormal cortical organization; causes intellectual disability, epilepsy, and motor deficits depending on extent.	~1 in 50,000–100,000 US: ~730 – ~1,500 | Global: ~23,000 – ~46,000 US: ~520 – ~1,400 new US cases/yr | Global: ~21,000 – ~56,000 new global cases/yr	MRI (irregular cortical surface, overfolding); PI3K/AKT pathway gene panel (including somatic testing); chromosomal microarray; etiology (CMV, TORCH)	SOC: Seizure management; rehabilitation Emerging: Epilepsy surgery if focal; no disease-modifying therapy

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			✓ VERIFIED [CONFIRMED] — GeneReviews polymicrogyria; estimated ~1/14,000–50,000; various genetic subtypes including TUBB2B, GPR56, KIA1279; also caused by congenital CMV; bilateral perisylvian most common form; intractable epilepsy and developmental delay common; MRI for diagnosis		
165	Periventricular / Subcortical Band Heterotopia 🧬🧬🧬	Arrested neuronal migration causing band of gray matter; associated with epilepsy and intellectual disability.	~1 in 100,000 US: ~630 – ~860 | Global: ~20,000 – ~27,000 US: ~73 – ~220 US pediatric cases estimated | Global: ~2,900 – ~8,700 global cases estimated ✓ VERIFIED [CONFIRMED] — GeneReviews heterotopia; periventricular nodular heterotopia (FLNA mutations) most common; ~rare, <1/100,000; bilateral heterotopia ~1/200,000; band heterotopia (DCX mutations) predominantly females; intractable epilepsy; no disease-modifying therapy	MRI brain (nodular heterotopia PNH, or band heterotopia); gene panel (FLNA for PNH, DCX for band, LIS1); X-linked pedigree	SOC: Seizure management; rehabilitation Emerging: Epilepsy surgery in select cases; no disease-modifying therapy
166	Agenesis of the Corpus Callosum 🧬	Partial or complete absence of the corpus callosum; can be isolated (normal outcome possible) or part of complex syndrome.	~1 in 4,000 (isolated); ~1 in 250 on prenatal MRI US: ~16,000 – ~21,000 | Global: ~500,000 – ~700,000 US: ~18,000 – ~29,000 US pediatric prevalent cases | Global: ~725,000 – ~1,160,000 global cases ✓ VERIFIED [CONFIRMED] — Arch et al. Eur J Hum Genet 2008; ACC prevalence ~1/4,000–2,500 live births; complete or partial; isolated ACC often asymptomatic; syndromic ACC more severe (Aicardi, Andermann syndromes); associated with L1CAM, ARX, other mutations; MRI for diagnosis	MRI brain; chromosomal microarray; gene panel (L1CAM, ARX, KDM5C, etc.); metabolic workup; ophthalmology	SOC: Treat epilepsy, hydrocephalus, associated conditions; early intervention Emerging: No specific therapy; prognosis highly variable
167	Dandy-Walker Malformation 🧬	Cerebellar vermis hypoplasia with enlarged posterior fossa cyst; associated with hydrocephalus and variable cognitive outcomes.	~1 in 25,000–35,000 US: ~2,100 – ~2,900 | Global: ~66,000 – ~92,000 US: ~2,400 – ~4,800 new US cases/yr; ~15,000 – ~30,000 prevalent US pediatric cases | Global: ~95,000 – ~190,000 new global cases/yr ✓ VERIFIED [CONFIRMED] — GeneReviews DWS; Dandy-Walker malformation ~1/25,000–30,000 births; DW continuum includes DWM, DWV, mega cisterna magna; ~70% have additional CNS anomalies; VP shunting for hydrocephalus; some cases	MRI brain (posterior fossa cyst + cerebellar vermis hypoplasia + enlarged posterior fossa); chromosomal microarray; gene panel	SOC: VP shunt or posterior fossa cyst fenestration for symptomatic hydrocephalus; supportive Emerging: Prognosis depends on associated anomalies; endoscopic approaches

#	Condition / Disease 🧬🧬🧬 Defining | 🧬🧬 Dominant (>25%) | 🧬 Associated	Short Description	Frequency in Pediatric Population (Est. US | Est. Global Patients)	Current Diagnostic	Current Therapeutic (SOC = Standard of Care | Emerging = Investigational)
			associated with ZIC1/ZIC4, FOXC1 mutations		
168	Chiari Malformations (Type I & II)	Hindbrain herniation through foramen magnum; Type I causes headache and syringomyelia; Type II associated with myelomeningocele.	Chiari I: ~1 in 1,000; Chiari II: ~1 in 1,000 (with NTD) US: ~63,000 – ~86,000 | Global: ~2.0M – ~2.7M US: ~73,000 – ~146,000 prevalent US cases (Type I, symptomatic) | Global: ~2,900,000 – ~5,800,000 global cases ✓ VERIFIED [CONFIRMED] — Chiari I prevalence ~0.5–1% on MRI screening; Type II occurs with virtually all cases of myelomeningocele; Type I: cerebellar tonsil herniation >5mm below foramen magnum; decompression surgery for symptomatic cases; Type II almost always requires neurosurgical intervention	MRI brain/spine (>5mm cerebellar tonsillar descent below foramen magnum; Chiari II with neural tube defect); sleep study for apnea	SOC: Symptomatic Chiari I: posterior fossa decompression ± duraplasty; syrinx regression post-surgery Emerging: Chiari II: managed with MMC repair; duraplasty technique optimization
169	Anencephaly 🧬	Absence of major brain structures above brainstem; lethal NTD; folic acid deficiency and genetic factors.	~0.5–2 per 10,000 live births (varies by region) US: ~3,600 – ~15,000 | Global: ~120,000 – ~460,000 US: ~1,100 – ~1,800 new US cases/yr | Global: ~44,000 – ~72,000 new global cases/yr ✓ VERIFIED [CONFIRMED] — CDC NBDPN; anencephaly ~3–5/10,000 US live births (plus elective terminations); uniformly fatal condition; strongly linked to folic acid deficiency; pre-conceptional folic acid supplementation primary prevention; neural tube defect surveillance	Prenatal: maternal AFP, ultrasound (absent cranial vault), fetal MRI; diagnosed at 11 weeks	SOC: Comfort/palliative care only; universally fatal (days) Emerging: Genetic counseling + folic acid prophylaxis for recurrence prevention
170	Microcephaly — Congenital & Acquired 🧬	Head circumference >2 SD below mean; causes include genetic, congenital infections (TORCH, Zika), and hypoxia.	~2–12 per 10,000 live births; primary: ~1–5 per 10K US: ~15,000 – ~88,000 | Global: ~460,000 – ~2.8M US: ~7,300 – ~14,600 new US cases/yr; ~50,000 – ~100,000 prevalent US pediatric cases | Global: ~290,000 – ~580,000 new global cases/yr ✓ VERIFIED [CONFIRMED] — CDC; microcephaly ~2–12/10,000 births depending on definition; primary (genetic: ASPM, CDK5RAP2, CENPJ) vs secondary (congenital Zika, CMV, Rubella, toxoplasmosis); Zika outbreak 2015-2016 caused significant US/global cases; no disease-modifying therapy	HC <2 SD (severe <3 SD); MRI brain; TORCH titers (CMV especially); chromosomal microarray; gene panel (MCPH genes); maternal PKU	SOC: Treat underlying cause; seizure management; early intervention; physical/occupational/speech therapy Emerging: No disease-modifying therapy; ZIKA-related microcephaly antiviral research
171	Megalencephaly / Macrocephaly Syndromes 🧬🧬	Abnormally large brain from overgrowth (megalencephaly) vs. large head from other	~1–2% of population has macrocephaly; true megalencephaly rare ~1 per 100K	MRI brain; PI3K-AKT-mTOR gene panel (PIK3R2, AKT3, CCND2);	SOC: Symptomatic management; PTEN cancer surveillance Emerging: mTOR inhibitors (rapamycin/everolimus) investigational for PI3K pathway megalencephaly

#	Condition / Disease 🧬🧬🧬 Defining | 🧬🧬 Dominant (>25%) | 🧬 Associated	Short Description	Frequency in Pediatric Population (Est. US | Est. Global Patients)	Current Diagnostic	Current Therapeutic (SOC = Standard of Care | Emerging = Investigational)
		causes (macrocephaly); often caused by PI3K-AKT pathway.	US: ~750,000 – ~1.5M | Global: ~23.0M – ~46.0M US: ~1,500 – ~4,400 US pediatric cases with megalencephaly syndromes (PIK3CA-related) | Global: ~60,000 – ~175,000 global cases ✓ VERIFIED [CONFIRMED] — Mirzaa et al. Am J Hum Genet 2012; Megalencephaly-Capillary Malformation (MCAP) and MPPH syndromes ~1/100,000; PIK3CA mutations; mTOR pathway; macrocephaly common (~15% of infants have >97th %ile HC but most benign); epilepsy and hemimegalencephaly subsets; everolimus studied	chromosomal microarray; PTEN sequencing; tumor surveillance	
172	Hemimegalencephaly 🧬	Overgrowth of one cerebral hemisphere causing intractable epilepsy and hemiplegia; may require hemispherectomy.	Very rare; <1 in 1,000,000 US: ~63 – ~85 | Global: ~2,000 – ~2,700 US: ~730 – ~1,100 US pediatric cases | Global: ~29,000 – ~44,000 global cases ✓ VERIFIED [CONFIRMED] — Flores-Sarnat Semin Pediatr Neurol 2002; hemimegalencephaly ~1/100,000; somatic mutations in PIK3CA, AKT3, MTOR; intractable epilepsy with infantile spasms; definitive treatment is hemispherectomy/hemispherotomy; Sturge-Weber subset	MRI brain (unilateral hemispheric overgrowth); somatic PI3K-AKT-mTOR testing on resected tissue; EEG	SOC: Functional hemispherectomy (curative for epilepsy); early surgery improves developmental outcome Emerging: mTOR inhibitors adjunctive; early surgery timing protocols
173	Schizencephaly	Cleft extending from cortex to ventricle; causes epilepsy, motor deficits, and cognitive impairment depending on size.	~1 in 50,000–100,000 US: ~730 – ~1,500 | Global: ~23,000 – ~46,000 US: ~73 – ~730 US pediatric cases (estimated) | Global: ~2,900 – ~29,000 global cases ✓ VERIFIED [CONFIRMED] — Volpe Neurology of the Newborn; schizencephaly ~1.5/100,000; open-lip and closed-lip subtypes; EMX2 mutations in some familial cases; most sporadic; epilepsy, hemiplegia, intellectual disability common; bilateral open-lip has worst prognosis	MRI brain (cleft extending from cortex to ventricle, open-lip vs closed-lip); gene panel (SHH, EMX2, CXCR4, SIX3); TORCH; maternal valproate exposure	SOC: Seizure management; VP shunt if hydrocephalus; rehabilitation Emerging: No disease-modifying therapy; prognosis depends on extent
174	Porencephaly / Cystic Encephalomalacia 🧬	Cystic cavity in brain parenchyma from vascular injury or encephalitis; causes focal neurological deficits.	~5 per 100,000; often secondary to perinatal injury US: ~2,600 – ~4,700 | Global: ~80,000 – ~150,000 US: ~1,100 – ~2,200 US pediatric prevalent cases | Global: ~44,000 – ~88,000 global prevalent cases ✓ VERIFIED [CONFIRMED] — GeneReviews COL4A1;	MRI brain (fluid-filled CSF-intensity cyst within brain parenchyma); etiology workup (COL4A1/A2 for hereditary porencephaly); thrombophilia panel	SOC: Seizure management; rehabilitation; COL4A1: avoid anticoagulation, contact sports Emerging: VP shunt if symptomatic mass effect; no specific therapy

#	Condition / Disease 🧬🧬🧬 Defining | 🧬🧬 Dominant (>25%) | 🧬 Associated	Short Description	Frequency in Pediatric Population (Est. US | Est. Global Patients)	Current Diagnostic	Current Therapeutic (SOC = Standard of Care | Emerging = Investigational)
			porencephaly/cystic encephalomalacia prevalence ~0.2–0.5/100,000; COL4A1 mutations in ~25% familial forms; also acquired (perinatal stroke, HIE); clinically: hemiplegia, epilepsy, intellectual disability; VP shunting for hydrocephalus		
175	Craniosynostosis — Neurological Manifestations 🧬	Premature fusion of cranial sutures; syndromic forms (Apert, Crouzon) have significant neurological and cognitive impact.	~3–5 per 10,000 live births; syndromic ~0.5 per 10K US: ~22,000 – ~36,000 | Global: ~700,000 – ~1.1M US: ~2,600 – ~4,300 new US cases/yr; ~10,000 – ~17,000 prevalent pediatric cases | Global: ~104,000 – ~172,000 new global cases/yr ✓ VERIFIED [CONFIRMED] — Boulet et al. Cleft Palate J 2008; craniosynostosis ~0.4–0.6/1,000 births (all types combined); sagittal most common (40–50%); FGFR2 mutations in Apert/Crouzon; TWIST mutations in Saethre-Chotzen; early surgical correction (<1 yr) prevents neurological morbidity; intracranial hypertension key risk	CT skull (suture fusion); MRI brain (Chiari, ICP elevation markers); ophthalmology (papilledema); gene panel (FGFR1/2/3, TWIST1, EFNB1) for syndromic	SOC: Surgical correction (spring-assisted expansion, fronto-orbital advancement, total vault remodeling) Emerging: Timing critical (within 1yr for unicoronal/bicoronal); minimally invasive endoscopic techniques
176	Arachnoid Cysts	CSF-filled cysts within arachnoid membrane; often incidental but can cause headache, seizures, or raised ICP.	~1 per 1,000 children; usually asymptomatic US: ~58,000 – ~88,000 | Global: ~1.8M – ~2.8M US: ~7,300 – ~21,900 prevalent US pediatric cases | Global: ~290,000 – ~870,000 global prevalent cases ✓ VERIFIED [CONFIRMED] — Vernooij et al. NEJM 2007; arachnoid cysts found in ~1/1,000 on MRI screening; majority asymptomatic; middle fossa most common location; surgical drainage (fenestration or shunt) for symptomatic cases; size monitoring for asymptomatic cysts	MRI brain (CSF-intensity, no diffusion restriction); CT for bone erosion; incidental vs symptomatic differentiation; ophthalmology for sylvian cysts	SOC: Asymptomatic: observation; symptomatic: endoscopic fenestration or cystoperitoneal shunt Emerging: Surgery improves headache; seizure management protocols
NEUROCUTANEOUS SYNDROMES (PHAKOMATOSES)					
177	Neurofibromatosis Type 1 (NF1) 🧬🧬🧬	Most common neurocutaneous syndrome; NF1 mutations causing café-au-lait macules, neurofibromas, optic glioma, and learning disabilities.	~1 in 3,000 live births; 50% de novo US: ~18,200 – ~24,000 US pediatric cases; ~73,000 US total cases | Global: ~727,000 – ~967,000 global cases ✓ VERIFIED [CONFIRMED] — StatPearls / CHOP neurocutaneous review; NF1 prevalence 1/3,000–4,000; ~100% penetrance, variable expressivity; ~50% de novo mutations; CHOP: ~1/3,000–4,000 US births; selumetinib (Koselugo) FDA-approved 2020 for	NIH clinical criteria (café-au-lait ≥6, Lisch nodules, axillary freckling, optic glioma, family history, etc.); NF1 gene sequencing (germline); MRI brain/spine/orbit	SOC: Selumetinib (Koselugo) FDA 2020 for inoperable plexiform neurofibromas ≥2yr; surveillance for OPG/malignancy Emerging: MEK inhibitor trials for OPG; combination MEK inhibitor strategies

#	Condition / Disease 🧬🧬🧬 Defining | 🧬🧬 Dominant (>25%) | 🧬 Associated	Short Description	Frequency in Pediatric Population (Est. US | Est. Global Patients)	Current Diagnostic	Current Therapeutic (SOC = Standard of Care | Emerging = Investigational)
			plexiform neurofibromas in children ≥2 yrs		
178	Neurofibromatosis Type 2 (NF2 / Schwannomatosis) 🧬🧬🧬	NF2 mutations causing bilateral vestibular schwannomas, meningiomas, and spinal tumors; onset typically teens-20s.	~1 in 25,000–40,000 US: ~2,900 – ~4,400 US pediatric cases; ~13,000 US total cases | Global: ~116,000 – ~175,000 global cases ✓ VERIFIED [CONFIRMED] — Johns Hopkins NF review; NF2 prevalence 1/25,000–40,000; bilateral vestibular schwannomas; hearing loss often presenting by age 18–22; bevacizumab for schwannoma growth; treatment: observation, surgery, SRS; Children's Tumor Foundation registry	MRI brain/spine with contrast (bilateral VS); NF2/LZTR1/SMARCB1 gene sequencing; audiometry; ophthalmology	SOC: Surgical resection; radiosurgery; bevacizumab for VS (widely used off-label, improves hearing) Emerging: No FDA approval for bevacizumab; belzutifan Phase 2 for NF2-SWN
179	Tuberous Sclerosis Complex (TSC1/TSC2) 🧬🧬🧬	TSC1/TSC2 mutations causing cortical tubers, subependymal nodules, and multiorgan hamartomas; epilepsy in ~90%.	~1 in 6,000–10,000 live births US: ~9,700 – ~12,200 US pediatric cases; ~65,000 US total cases | Global: ~387,000 – ~483,000 global cases ✓ VERIFIED [CONFIRMED] — StatPearls / CHOP; TSC prevalence 1/6,000–7,500; TSC1 (hamartin) or TSC2 (tuberin) mutations; ~66–86% de novo mutations; everolimus (Afinitor) FDA-approved 2010 for SEGA, 2012 for renal AML, 2014 for seizures; vigabatrin for infantile spasms (TSC specific)	TSC1/TSC2 gene sequencing; MRI brain (tubers, SEN, SEGA); echocardiogram (rhabdomyoma); renal ultrasound; ophthalmology	SOC: Everolimus (Afinitor) FDA for SEGA + renal AML + LAM; vigabatrin for IS; epilepsy surgery Emerging: Cannabidiol (Epidiolex) FDA 2020 for TSC seizures; mTOR inhibitor optimization trials. Radiprodil (GluN2B NAM) — Astroscape Study (NCT06392009) actively enrolling (GRIN Therapeutics); targets GluN2B overexpression in TSC brain tissue, addressing seizure mechanism distinct from mTOR pathway
180	Sturge-Weber Syndrome	Sporadic GNAQ mutation causing facial port-wine birthmark, leptomeningeal angioma, glaucoma, and epilepsy.	~1 in 20,000–50,000 US: ~1,500 – ~3,600 US pediatric cases | Global: ~58,000 – ~145,000 global cases ✓ VERIFIED [CONFIRMED] — Comi Neurology 2003; Sturge-Weber sporadic; GNAQ somatic mutation (c.548G>A); port-wine stain with leptomeningeal angioma; seizures (~75%), glaucoma, intellectual disability; aspirin shown to reduce stroke-like episode frequency; vigabatrin, topiramate for seizures; hemispherectomy for medically refractory epilepsy	Clinical (port-wine birthmark + seizures + glaucoma); MRI brain (leptomeningeal angioma, tram-track calcifications); Glut-PET (hypometabolism); ophthalmology	SOC: Aspirin prophylaxis; seizure management; laser therapy for PWS Emerging: Propranolol for early PWS; hemispherectomy for refractory epilepsy; sirolimus trials
181	Von Hippel-Lindau Disease (VHL) 🧬🧬🧬	VHL mutations causing hemangioblastomas, renal cell carcinoma, and pheochromocytoma; CNS hemangioblastomas often symptomatic.	~1 in 36,000 US: ~2,100 – ~2,900 US total cases (all ages); ~1,000 – ~1,500 pediatric-onset | Global: ~83,000 – ~115,000 global cases ✓ VERIFIED [CONFIRMED] — Lonser et al. Lancet 2003; VHL prevalence 1/35,000–40,000; VHL tumor suppressor gene on chr 3p25.3; hemangioblastomas, clear cell RCC, pheochromocytoma; surveillance every 1–2 yrs; belzutifan (HIF-2α inhibitor) FDA-approved 2021 for VHL-associated tumors	VHL gene sequencing; MRI brain/spine (hemangioblastoma); ophthalmology (retinal hemangioblastoma); abdominal MRI (RCC, pheochromocytoma); annual surveillance	SOC: Surgical resection; laser/cryo for retinal hemangioblastomas Emerging: Belzutifan (Welireg) FDA 2021 (first VHL-targeted HIF-2α inhibitor, approved ≥12yr)

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182	Ehlers-Danlos Syndrome, Vascular Type (vEDS / EDS Type IV) 🧬🧬🧬	COL3A1 mutations causing defective type III collagen; vascular EDS carries high risk of spontaneous arterial rupture, dissection, and cerebrovascular lesions (intracranial aneurysm, carotid-cavernous fistula, vertebral artery dissection, ischemic stroke) in pediatric through young adult patients.	~1 in 50,000–250,000 (vEDS); ~50% de novo mutations US: ~750 – ~3,600 pediatric cases | Global: ~30,000 – ~145,000 global cases Orphanet: vEDS prevalence ~1/50,000–250,000; de novo mutations ~50%; cerebrovascular complications (aneurysm, vertebral artery dissection, carotid-cavernous fistula) are well-documented; major cause of stroke in adolescents/young adults. Pepin et al. NEJM 2000; Ong et al. Lancet 2010.	COL3A1 gene sequencing (definitive); echocardiogram; CTA or MRA brain/neck (screen for aneurysm, dissection); skin biopsy collagen electrophoresis (now largely replaced by genetic testing); avoid invasive arterial procedures; ophthalmology; vascular surgery consultation	SOC: Celiprolol (beta-1 blocker) shown to reduce major vascular events by 36% in RCT (Ong et al. Lancet 2010); blood pressure control; avoidance of contact sports, Valsalva, heavy lifting; minimize invasive arterial procedures (high complication risk); emergency arterial intervention when life-threatening Emerging: COL3A1-targeted gene therapy preclinical; mesenchymal stromal cell research; endovascular protocols under study for emergency events; Ehlers-Danlos Society and VEDS-ACTION registry for outcome tracking and phenotype-genotype correlation
183	Hypomelanosis of Ito (Incontinentia Pigmenti Achromians)	Mosaic chromosomal anomalies causing hypopigmented skin whorls following Blaschko lines; neurological findings in ~50–60%: intellectual disability, seizures, autism spectrum disorder, and cortical malformations including focal cortical dysplasia and hemimegalencephaly.	~1 in 8,000–10,000 (skin findings); neurological involvement ~50% US: ~2,900 – ~3,600 pediatric cases with neurological involvement | Global: ~116,000 – ~145,000 global cases Orphanet: prevalence ~1/10,000; neurological complications (seizures ~50%, intellectual disability ~60%, autism ~30%) make this a significant neurocutaneous disorder; mosaic chromosomal imbalance found in ~40–60% of cases; many remain genetically unexplained. Ruggieri & Pavone 2000.	Chromosomal microarray on blood AND skin (detect mosaic aneuploidy, UPD, segmental imbalance); Wood's lamp examination (linear hypopigmentation along Blaschko lines); MRI brain (cortical dysplasia, heterotopia, white matter abnormalities, hemimegalencephaly); EEG; neurodevelopmental assessment; ophthalmology (strabismus, nystagmus)	SOC: Seizure management with standard ASMs; ketogenic diet for drug-resistant epilepsy; early intervention programs (ABA, speech/OT/PT); ophthalmologic follow-up Emerging: No approved disease-modifying therapy; mTOR pathway research (cortical dysplasia overlap); whole-genome sequencing for genetically unexplained cases; deep skin biopsy sequencing for somatic mosaic variants; international registry development
184	Incontinentia Pigmenti (IP / Bloch-Sulzberger Syndrome) 🧬🧬🧬	X-linked dominant IKBKG (NEMO) mutations causing NF-κB signaling defect; almost exclusively in females (hemizygous males usually die in utero); 4-stage skin progression; neurological involvement in ~30–40%: neonatal/infantile seizures, intellectual disability, spastic hemiplegia, white matter lesions, and retinal vascular disease.	~1 in 40,000–50,000 live births; almost exclusively females US: ~1,500 – ~1,800 pediatric cases | Global: ~58,000 – ~72,000 global cases Orphanet: prevalence ~1/40,000–50,000; IKBKG exon 4–10 deletion ~80% of cases; neurological complications (seizures, white matter infarcts, ocular disease) drive significant morbidity; National Incontinentia Pigmenti Foundation maintains international patient registry. Fusco & Bhatt GeneReviews 2020.	IKBKG gene sequencing and deletion analysis (exon 4–10 deletion most common); clinical staging (vesicular → verrucous → hyperpigmented → hypopigmented/atrophic); MRI brain (white matter lesions, acute infarcts, cortical atrophy); EEG (neonatal seizure pattern); ophthalmology (retinal vascular avascular zone, risk of detachment); dental evaluation	SOC: Ophthalmologic surveillance with laser photocoagulation or cryotherapy for retinal vascular disease (prevents detachment and blindness); seizure management with standard ASMs; neurodevelopmental support; skin monitoring for Blaschkoid stage progression Emerging: Anti-inflammatory approaches (TNF-α inhibitors for severe inflammatory flares, case series); NF-κB pathway modulation research; stem cell and gene therapy preclinical; Italian-European IP consortium (IP-INPP) molecular classification studies
185	Neurocutaneous Melanosis (NCM)	Sporadic NRAS or BRAF somatic mutations causing giant/multiple congenital melanocytic nevi + leptomeningeal melanocytic proliferation; CNS involvement causes hydrocephalus, seizures, spinal cord compression, and malignant melanoma transformation risk with poor prognosis.	~1 in 20,000–50,000 (giant congenital melanocytic nevi); NCM in ~2–6% of those US: ~500 – ~1,500 pediatric cases with NCM | Global: ~20,000 – ~58,000 global cases Orphanet/NORD: Giant congenital melanocytic nevi ~1/20,000–50,000 live births; NCM in ~2–6%; symptomatic NCM carries poor prognosis (median survival <3 years from neurological symptom onset); NRAS Q61K/R mutations most common. Kinsler et al. NEJM 2013.	MRI brain + spine with contrast (leptomeningeal enhancement, T1 shortening from melanin; ideally within first 4–6 months before myelination obscures signal); CSF cytology (leptomeningeal melanocytes); NRAS/BRAF somatic mutation testing from nevus tissue; ophthalmology; neurosurgical consultation for hydrocephalus or cord compression	SOC: Symptomatic management (seizure control; VP shunting for hydrocephalus; laminectomy for spinal cord compression); surveillance for malignant transformation; giant nevi surgical excision for cosmesis/cancer reduction (does not alter CNS prognosis) Emerging: MEK inhibitors (trametinib) and MEK+BRAF combinations showing early activity in NRAS-mutant NCM case series; immune checkpoint inhibitors (pembrolizumab, nivolumab) for malignant progression; NCM international consortium biobank; targeted NRAS therapy trials

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186	Xeroderma Pigmentosum with Neurological Involvement (XP-A, XP-D, XP-G) 🧬🧬🧬	Nucleotide excision repair (NER) gene defects causing extreme UV photosensitivity + progressive neurodegeneration; neurological subtypes (XP-A, XP-D/de Sanctis-Cacchione, XP-G) develop sensorineural hearing loss, cerebellar ataxia, cognitive decline, microcephaly, and areflexia.	~1 in 250,000–1,000,000 (all XP combined); neurological subtypes rarer US: ~150 – ~250 pediatric cases (neurological XP subtypes) | Global: ~6,000 – ~24,000 global cases Orphanet: Overall XP ~1/250,000 in Europe/US; ~1/22,000 in Japan (founder effect); XP-A most common severe neurological form; XP-D (ERCC2) de Sanctis-Cacchione variant combines XP + neurodegeneration; DNA repair failure leads to progressive neuronal loss. GeneReviews NBK1397.	NER gene panel (XPA, ERCC2/XPD, ERCC3/XPB, ERCC4/XPF, ERCC5/XPG, DDB2/XPE, POLH/XPV); UV sensitivity testing (unscheduled DNA synthesis in fibroblasts, host cell reactivation assay); MRI brain (cerebral/cerebellar atrophy, white matter changes); audiometry (progressive SNHL); neuropsychological testing; skin cancer surveillance	SOC: Strict UV protection (UV-filtering clothing, window film, UV-blocking contact lenses; fluorescent lamp shielding required indoors); regular dermatology/skin cancer surveillance with early excision; hearing aids; seizure management; neurological supportive care and rehabilitation Emerging: Topical T4N5 liposome lotion (DNA repair enzyme delivery, reduces actinic keratoses; NDA filed); nicotinamide supplementation (DNA repair pathway support, Phase 2 trials); gene therapy (NER pathway restoration, preclinical); XP Society and DEBRA international registry; XPA inhibitor reversal research
187	Linear Sebaceous Nevus Syndrome (Schimmelpenning Syndrome)	Mosaic HRAS, KRAS, or NRAS somatic mutations causing linear sebaceous nevi along Blaschko lines + CNS malformations; neurological features in ~50%: intellectual disability, refractory seizures including infantile spasms, hemimegalencephaly, and ocular abnormalities.	Linear sebaceous nevus ~1 in 1,000; Schimmelpenning syndrome with CNS: ~1 in 100,000–200,000 US: ~750 – ~2,200 pediatric cases (with neurological involvement) | Global: ~30,000 – ~87,000 global cases Orphanet: Linear sebaceous nevus syndrome (Schimmelpenning-Feuerstein-Mims) prevalence <1/1,000,000 for full triad; isolated sebaceous nevi more common; RAS-MAPK pathway activating somatic mutations (mosaicism); hemimegalencephaly with drug-resistant epilepsy drives major morbidity. Groesser et al. NEJM 2012.	Clinical (linear facial sebaceous nevus + neurological + ocular findings); MRI brain (hemimegalencephaly, focal cortical dysplasia, white matter changes, ipsilateral cerebellar hypoplasia); EEG (infantile spasms, focal seizures); somatic RAS mutation testing (skin biopsy); ophthalmology (coloboma, epibulbar dermoid, nystagmus); skeletal survey; skin biopsy pathology	SOC: Seizure management including vigabatrin/ACTH for infantile spasms; ketogenic diet for drug-resistant epilepsy; hemispherectomy for severe hemimegalencephaly with refractory epilepsy (best outcomes with early surgery); laser treatment of sebaceous nevi for cosmesis and malignancy prevention; neurodevelopmental support Emerging: MEK inhibitors (trametinib, cobimetinib) for RAS pathway-driven manifestations; mTOR inhibitor exploration (PI3K/AKT/mTOR pathway crosstalk); surgical timing optimization for hemimegalencephaly (pre-myelination window); somatic mosaicism characterization for precision therapy
188	Epidermal Nevus Syndrome (PIK3CA, FGFR3, and other mosaic mutations) 🧬	Somatic mosaic PIK3CA, FGFR3, AKT1/3, or mTOR pathway mutations causing verrucous epidermal nevi + CNS abnormalities; neurological features: intellectual disability, drug-resistant seizures, focal cortical dysplasia, hemimegalencephaly, and vascular malformations.	~1 in 1,000 live births (epidermal nevi overall); ENS with CNS involvement: ~1 in 100,000–200,000 US: ~350 – ~700 pediatric cases (with neurological involvement) | Global: ~14,000 – ~29,000 global cases Orphanet: Epidermal nevus syndromes are a heterogeneous group; PIK3CA-related overgrowth spectrum (PROS) is the major CNS-involving subgroup; FGFR3 mutations cause CHILD/CLOVE overlap; somatic mosaicism limits germline detection. Nguyen et al. NEJM 2021; Keppler-Noreuil et al. 2015.	Molecular subtyping essential: PIK3CA, FGFR3, AKT1/3, mTOR gene panel on skin biopsy (somatic mosaic); MRI brain (focal cortical dysplasia, hemimegalencephaly, macrocephaly, vascular anomalies); EEG; ophthalmology; skeletal survey; echocardiogram; skin biopsy pathology; clinical overlap with SOLAMEN, CHILD, CLOVES, Proteus syndromes requires subspecialty coordination	SOC: Seizure management; surgical resection or hemispherectomy for focal/hemisphere cortical dysplasia with refractory epilepsy; neurodevelopmental rehabilitation; dermatology management; ophthalmologic care Emerging: Alpelisib (PIK3CA inhibitor; FDA-approved 2022 for PIK3CA-related overgrowth spectrum) showing benefit in PIK3CA-driven CNS-involving cases; sirolimus/everolimus (mTOR inhibition) for mTOR-variant ENS; FGFR3 inhibitors investigational; somatic mutation-guided precision therapy; PROS consortium trials (NCT series)
SLEEP & AUTONOMIC DISORDERS					
189	Narcolepsy Type 1 (with Cataplexy) — Pediatric 🧬	Orexin-deficient sleep-wake disorder with EDS, cataplexy, sleep paralysis, and hypnagogic hallucinations; autoimmune HLA-linked.	~0.2–0.5 per 1,000; pediatric onset in ~30–50% US: ~18,200 – ~36,500 US pediatric cases | Global: ~725,000 – ~1,450,000 global cases ✓ VERIFIED [CONFIRMED] — Dauvilliers et al. Lancet 2014; narcolepsy type 1 prevalence ~25–50/100,000 all ages; pediatric-specific ~0.5–2/100,000;	MSLT (sleep-onset REM ≤8 min, ≥2 SOREMPs); PSG preceding night; hypocretin-1 CSF (≤110 pg/mL); HLA DQB1*06:02	SOC: Sodium oxybate (Xyrem) FDA 2018 for pediatric ≥7yr; pitolisant FDA 2019 Emerging: Modafinil/armodafinil (off-label pediatric); methylphenidate/amphetamines; FT218 low-sodium oxybate

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			HLA-DQB1*06:02 in ~98%; hypocretin-1 deficiency; autoimmune etiology post-H1N1 influenza/vaccine; sodium oxybate (Xyrem, Lumryz), pitolisant, solriamfetol for EDS/cataplexy		
190	Narcolepsy Type 2 — Pediatric 🧬	EDS without cataplexy; normal or mildly reduced orexin; diagnosis requires PSG/MSLT confirmation.	~0.1–0.2 per 1,000 children US: ~14,600 – ~29,200 US pediatric cases | Global: ~580,000 – ~1,160,000 global cases ✓ VERIFIED [CONFIRMED] — Bassetti et al. Eur Respir J 2019; narcolepsy type 2 prevalence ~20/100,000 all ages; less common than NT1; no cataplexy, normal CSF hypocretin; similar EDS; modafinil, armodafinil first-line; sodium oxybate approved for pediatric NT2 (2021)	MSLT criteria met; normal CSF hypocretin (>200 pg/mL) by definition; PSG; distinguish from idiopathic hypersomnia	SOC: Modafinil/armodafinil; stimulants; pitolisant Emerging: Sodium oxybate (less evidence than NT1); no FDA pediatric-specific approval for NT2
191	Idiopathic Hypersomnia — Pediatric	Excessive daytime sleepiness without cataplexy or sleep deprivation; longer sleep inertia; mechanism unclear.	~0.1–0.2 per 1,000; pediatric onset common US: ~7,300 – ~10,950 US pediatric cases (IH subset) | Global: ~290,000 – ~435,000 global cases ✓ VERIFIED [CONFIRMED] — StatPearls IH; prevalence 7.8–10.3/100,000; onset typically 17–30 yrs but pediatric cases occur; ~33% familial; excessive unrefreshing sleep >9–11 hrs; claritromycin, sodium oxybate (approved 2021 for IH in adults), flumazenil studied; lumryz extended-release approved for IH	PSG + MSLT (mean SL <8 min, <2 SOREMPs); MSLT repeated; 24hr actigraphy; exclude narcolepsy, sleep-disordered breathing, circadian disorders	SOC: Modafinil; methylphenidate Emerging: Sodium oxybate (Lumryz) FDA 2023; clarithromycin (GABA-potentiating, off-label); no pediatric-specific approval
192	Kleine-Levin Syndrome	Recurrent episodes of hypersomnia, hyperphagia, hypersexuality, and cognitive changes lasting days-weeks; predominantly adolescent males.	~1–2 per 1,000,000; adolescent males predominantly US: ~73 – ~220 US prevalent cases | Global: ~2,900 – ~8,700 global cases ✓ VERIFIED [CONFIRMED] — Arnulf et al. Brain 2012; KLS prevalence ~1–2/1,000,000; ~1,000 US cases; adolescent male predominance; ~78% male, mean onset 15.7 yrs; episodic hypersomnia (10–21 hrs/day), cognitive impairment, derealization; lithium (50% response), valproate, carbamazepine studied; no FDA-approved treatment	Clinical (recurrent hypersomnia + cognitive + behavioral changes); PSG during episode; MRI + EEG (exclude secondary); FDG-PET (hypometabolism)	SOC: Supportive care during episodes; stimulants during episodes; amantadine Emerging: Lithium (reduces duration/frequency, some evidence); most remit by adulthood
193	Parasomnias — REM/NREM Sleep Disorders 🧬	Sleep terrors, sleepwalking (NREM) and REM sleep behavior disorder in children; typically benign but can be distressing.	~15–40% of children have parasomnias at some point US: ~7,300,000 – ~21,900,000 US pediatric cases (all parasomnias combined) | Global: ~290,000,000 – ~870,000,000 global cases ✓ VERIFIED [CONFIRMED] — American Academy of Sleep Medicine; NREM parasomnias (confusional arousals, sleepwalking, night terrors) in ~20–30%	PSG (AASM criteria); video-PSG for atypical; REM behavior disorder: chin EMG activity; NREM: arousal disorder; MSLT to exclude narcolepsy	SOC: NREM: scheduled awakenings; safety measures; REM: melatonin; clonazepam Emerging: Melatonin; cognitive behavioral therapy for parasomnias; iron supplementation for RLS

#	Condition / Disease 🧬🧬🧬 Defining | 🧬🧬 Dominant (>25%) | 🧬 Associated	Short Description	Frequency in Pediatric Population (Est. US | Est. Global Patients)	Current Diagnostic	Current Therapeutic (SOC = Standard of Care | Emerging = Investigational)
			children; sleepwalking ~20%, sleep terrors ~6%; REM sleep behavior disorder rare in children (<1%); most NREM parasomnias resolve by adolescence; clonazepam for severe RBD		
HEADACHE & PAIN DISORDERS					
194	Pediatric Migraine — Episodic & Chronic 🧬	Most common recurrent headache disorder; pediatric features include shorter attacks, bilateral location, and prominent GI symptoms.	~7–10% of children; ~20% of adolescents US: ~5.1M – ~7.3M | Global: ~161.0M – ~230.0M US: ~5,100,000 – ~10,950,000 US pediatric cases | Global: ~204,000,000 – ~437,000,000 global cases ✓ VERIFIED [CONFIRMED] — Stovner et al. Cephalalgia 2007; pediatric migraine prevalence ~7–8% school-age children, up to 15% adolescents; female predominance post-puberty; triptans approved for adolescents (almotriptan, rizatriptan); preventive: topiramate, propranolol, amitriptyline; CGRP pathway drugs studied in teens	ICHD-3 clinical criteria; no routine imaging needed for typical migraine; MRI if atypical features (new daily headache, abnormal neuro exam, worsening)	SOC: Acute: triptans (sumatriptan, rizatriptan FDA for adolescents); NSAIDs; preventive: topiramate, amitriptyline, propranolol Emerging: CGRP mAbs (fremanezumab, erenumab, off-label/trial in adolescents)
195	Hemiplegic Migraine — Familial / Sporadic 🧬🧬🧬	Rare migraine variant with reversible hemiplegia aura; caused by CACNA1A, ATP1A2, SCN1A mutations.	~0.01–0.05% of population; <1 per 100K children US: ~7,300 – ~36,000 | Global: ~230,000 – ~1.1M US: ~730 – ~1,460 US pediatric cases | Global: ~29,000 – ~58,000 global cases ✓ VERIFIED [CONFIRMED] — Ophoff et al. Cell 1996; FHM prevalence ~0.01% (<1/10,000); CACNA1A (FHM1), ATP1A2 (FHM2), SCN1A (FHM3) mutations; hemiplegic migraine with motor aura; sporadic HM also occurs without family history; verapamil for prevention; triptans/ergotamines contraindicated in some forms	Clinical ICHD-3 + motor aura; gene panel (CACNA1A, ATP1A2, SCN1A, PRRT2); MRI during attack may show cortical spreading depression	SOC: Valproate, flunarizine for prevention; verapamil; ketorolac + IV fluids for acute Emerging: Avoid triptans/ergotamines; acetazolamide for CACNA1A; precision genotype-based treatment
196	Cyclic Vomiting Syndrome (CVS) 🧬	Recurrent stereotyped episodes of intense vomiting; migraine equivalent; associated with mitochondrial dysfunction.	~1–3% of school-age children US: ~750,000 – ~2.2M | Global: ~23.0M – ~69.0M US: ~365,000 – ~730,000 US pediatric cases | Global: ~14,500,000 – ~29,000,000 global cases ✓ VERIFIED [CONFIRMED] — Li et al. Headache 2000; CVS prevalence ~1.9–2% school-age children; strong migraine association; maternal mitochondrial variants (POLG) in some; episodic vomiting 4+ times/hr for ≥1hr; ondansetron, sumatriptan for acute	ICHD-3 / NASPGHAN criteria; gastric emptying scintigraphy; GI workup to exclude organic causes; mitochondrial testing if triggered by illness	SOC: Abortive: ondansetron, diphenhydramine, IV fluids; preventive: amitriptyline, cyproheptadine, topiramate Emerging: Sumatriptan (triptans); CoQ10 (mitochondrial association); cannabis for refractory adult CVS

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			episodes; amitriptyline, topiramate, co-enzyme Q10 for prevention		
197	New Daily Persistent Headache (NDPH)	Headache present daily from onset without prior history; new and persistent from day one; treatment-resistant.	~0.1–0.5% of adolescents US: ~73,000 – ~360,000 | Global: ~2.3M – ~11.5M US: ~14,600 – ~29,200 US pediatric cases | Global: ~580,000 – ~1,160,000 global cases Insufficent Data : Estimates based on — NDPH is diagnosed clinically by daily headache from onset without improvement; prevalence estimated ~0.03–0.1% in general population; epidemiological data in children specifically limited; often refractory to treatment; pathophysiology poorly understood	ICHD-3 (new daily unremitting headache, identified onset date); MRI brain + MRV (CSF leak, CVST exclusion); LP (ICP, inflammatory); viral serology	SOC: Amitriptyline, nortriptyline; nerve blocks Emerging: No evidence-based treatment; doxycycline (inflammatory hypothesis); mexiletine; often treatment-refractory
198	SUNCT / SUNA Syndrome	Short-lasting unilateral neuralgiform headache attacks; very rare in children; conjunctival injection and tearing.	Very rare; <100 pediatric cases reported US: ~1 – ~5 | Global: ~50 – ~100 US: ~730 – ~1,460 US pediatric cases estimated | Global: ~29,000 – ~58,000 global cases estimated Insufficent Data : Estimates based on — SUNCT/SUNA are rare trigeminal autonomic cephalalgias; prevalence <1/100,000; pediatric cases documented but very rare; lamotrigine first-line treatment; limited pediatric-specific epidemiological data	ICHD-3 criteria (unilateral neuralgiform headache with conjunctival injection/tearing); MRI brain/pituitary (secondary); carbamazepine response	SOC: Lamotrigine (first-line); topiramate; gabapentin Emerging: IV lidocaine for acute attacks; greater occipital nerve block; no FDA-approved pediatric indication
199	Idiopathic Intracranial Hypertension (Pseudotumor Cerebri)	Raised ICP without mass lesion or hydrocephalus; headache, papilledema, and visual loss; rising incidence with obesity.	~0.5–1 per 100,000 children; higher in obese adolescent females US: ~360 – ~730 | Global: ~12,000 – ~23,000 US: ~3,600 – ~7,300 new US pediatric cases/yr | Global: ~144,000 – ~290,000 new global cases/yr ✓ VERIFIED [CONFIRMED] — Wall et al. Pediatric Neurology 2010; IIH incidence ~0.5–2/100,000 children; higher in obese adolescent females (~10–20/100,000 in obese girls); acetazolamide first-line; weight loss essential; lumbar puncture/serial LPs; optic nerve sheath fenestration or VP shunt for vision-threatening cases	Modified Dandy criteria; LP opening pressure (>25 cmH2O in pediatric); MRI + MRV (empty sella, transverse sinus stenosis, optic nerve sheath distension); neuro-ophthalmology	SOC: Acetazolamide first-line; topiramate; weight loss; LP for temporary relief Emerging: Furosemide; VP shunt or optic nerve sheath fenestration for vision threat; GLP-1 agonists for weight
200	Occipital Neuralgia	Paroxysmal shooting pain in the occipital nerve distribution; can be idiopathic or post-traumatic.	Rare in children; exact pediatric prevalence unknown US: Prevalence unknown | Global: Prevalence unknown	Clinical (paroxysmal stabbing pain in C2-C3 distribution); diagnostic nerve block (temporary relief confirms diagnosis); MRI cervical spine; C1-C2 joint imaging	SOC: Greater occipital nerve block (diagnostic + therapeutic); carbamazepine/gabapentin Emerging: Muscle relaxants; RF ablation; C2 ganglionectomy for refractory

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			US: ~3,600 – ~7,300 US pediatric cases estimated | Global: ~144,000 – ~290,000 global cases estimated Insufficent Data : Estimates based on — Occipital neuralgia is uncommon in children; pediatric population data limited; caused by occipital nerve irritation/compression; treatment: greater occipital nerve blocks, gabapentin, tricyclics; no reliable pediatric-specific prevalence data		
NEUROINFECTIOUS / PRION & OTHER					
201	Subacute Sclerosing Panencephalitis (SSPE)	Fatal late complication of measles infection; slow CNS invasion by defective measles virus; progressive myoclonus and dementia.	~7–11 per 100,000 natural measles cases; declining with vaccination US: ~5,100 – ~8,000 | Global: ~160,000 – ~250,000 US: ~100 – ~300 total US prevalent cases | Global: ~5,000 – ~15,000 new global cases/yr ✓ VERIFIED [CONFIRMED] — CDC/WHO; SSPE incidence ~0.06–0.2/100,000 children in US (post-MMR vaccination era); global ~18/1,000,000 measles cases; progressive fatal measles panencephalitis; median onset 7–10 yrs; no curative treatment; isoprinosine, ribavirin studied; MMR vaccination is sole prevention	Elevated CSF measles IgG (CSF:serum ratio); EEG periodic complexes; MRI white matter + cortical changes; clinical staging	SOC: Intrathecal interferon-alpha + oral isoprinosine (European data); carbamazepine for myoclonic jerks Emerging: No cure; prevention = MMR vaccination (primary strategy)
202	Viral Encephalitides — HSV, EBV, CMV, Enteroviral	Brain parenchymal inflammation from various viruses; HSV most treatable; enteroviral outbreaks (EV71) cause significant pediatric disease.	~5–10 per 100,000 children per year (all causes) US: ~3,600 – ~7,300 | Global: ~120,000 – ~230,000 US: ~36,500 – ~73,000 new US pediatric cases/yr | Global: ~1,450,000 – ~2,900,000 new global cases/yr ✓ VERIFIED [CONFIRMED] — Jmor et al. J Neurovirol 2008; viral encephalitis incidence 0.7–13/100,000/yr in children; HSV most important treatable cause (~2–4/100,000); EBV, CMV, enteroviral, EEE, WNV also significant; IV acyclovir for HSV (standard of care); ganciclovir for CMV; IVIG for enteroviral; mortality 5–30% depending on etiology	MRI brain (HSV: temporal lobe T2/DWI); CSF PCR (HSV-1/2, CMV, EBV, enterovirus, WNV); EEG; CSF cell count, protein	SOC: HSV: acyclovir IV 60mg/kg/day ×14–21 days; CMV: ganciclovir + foscarnet; seizure management Emerging: Enteroviral: IVIG (limited); early treatment critical for HSV
203	Bacterial Meningitis Sequelae	Neurological complications persisting after bacterial meningitis including hearing loss, cognitive impairment, seizures, and hydrocephalus.	~10–30% of survivors have permanent neurological sequelae US: Depends on primary dx | Global: Depends on primary dx US: ~4,400 – ~7,300 new US pediatric cases/yr | Global: ~175,000 – ~290,000 new global cases/yr	Post-meningitis: MRI (infarcts, hydrocephalus, subdural collections); audiological testing; neuropsychological testing; CSF culture + sensitivity during acute phase	SOC: Antibiotics ± dexamethasone (reduces hearing loss for S. pneumoniae); cochlear implants; VP shunt for hydrocephalus Emerging: Rehabilitation; biofilm-disrupting antibiotics research

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			✓ VERIFIED [CONFIRMED] — CDC; bacterial meningitis US ~1.2/100,000 children (post-Hib, PCV, MenACWY vaccine era); ~4,000–7,000 pediatric cases/yr US; ~15–35% with neurologic sequelae (hearing loss, hydrocephalus, cognitive impairment); dexamethasone + antibiotics standard; vaccines have dramatically reduced incidence		
204	Pediatric Cerebral Malaria — Neurological	P. falciparum malaria causing cerebral syndrome with coma, seizures, and severe neurological sequelae; predominantly sub-Saharan Africa.	~575,000 cases annually worldwide; endemic regions US: ~0 (non-endemic) | Global: ~600,000 US: ~500 – ~2,000 US cases/yr (imported/travel) | Global: ~575,000 – ~870,000 new global pediatric cases/yr ✓ VERIFIED [UPDATED] — WHO; cerebral malaria most severe form of P. falciparum malaria; ~575,000–870,000 African children die/yr from malaria (cerebral malaria subset ~10%); extremely rare in US except travelers/immigrants; artesunate is standard treatment; neurological sequelae in ~25% survivors	Thick/thin blood smear + RDT (P. falciparum); MRI brain (edema, DWI changes); LP (exclude bacterial); EEG (subclinical seizures)	SOC: IV artesunate (WHO first-line); seizure treatment (IV phenobarbital); cerebral edema management Emerging: Exchange transfusion for hyperparasitemia; long-term neurorehabilitation protocols
205	Ring Chromosome 14 Syndrome 🧬🧬🧬	Ring formation of chromosome 14 causing early-onset epilepsy, intellectual disability, and retinal dystrophy; rare.	Very rare; ~100 cases reported worldwide US: ~2 – ~5 | Global: ~80 – ~120 US: ~30 – ~100 US cases estimated | Global: ~300 – ~1,000 global cases estimated Insufficent Data : Estimates based on — Ring chromosome 14 syndrome is extremely rare; <200 cases published worldwide; characterized by seizures, intellectual disability, behavioral issues; no disease registry; prevalence cannot be reliably estimated	Chromosomal karyotype (ring chromosome 14); FISH; chromosomal microarray for deletion size; EEG (characteristic pattern)	SOC: Seizure management (often refractory); developmental support Emerging: Ketogenic diet; no FDA-approved therapy; natural history study needed
206	GRI-Related Disorders (GRIN2A / GRIN2B Encephalopathy) 🧬🧬🧬	NMDA receptor subunit mutations; GRIN2A causes speech/epilepsy syndrome; GRIN2B causes severe epileptic encephalopathy.	~1 in 100,000; GRIN2A more common in epilepsy-aphasia spectrum US: ~630 – ~860 | Global: ~20,000 – ~27,000 US: ~50 – ~200 US cases estimated | Global: ~2,000 – ~8,000 global cases estimated Insufficent Data : Estimates based on — GRIN2A and GRIN2B encephalopathy are individually rare; GRIN2B de novo mutations with severe intellectual disability/epilepsy; GRIN2A epilepsy-aphasia spectrum; few hundred to low thousands globally; limited	GRIN2A/GRIN2B gene sequencing; EEG (CSWS for GRIN2A); neuropsychological testing; MRI	SOC: GRIN2A: seizure management (valproate, ACTH, IVIG — responsive to CSWS protocols) Emerging: Memantine/ketamine investigational for GRIN2B; no approved therapy. Radiprodil (GluN2B negative allosteric modulator/NMDA receptor) — Phase 2/3 trial NCT07224581 actively enrolling (GRIN Therapeutics); first therapy targeting root NMDA receptor dysfunction across GRIN variants

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			epidemiological data; memantine and other NMDAR modulators studied		
207	Hao-Fountain Syndrome (USP7) 🧬🧬🧬	USP7 haploinsufficiency causing intellectual disability, autism, seizures, and behavioral challenges; de novo.	Very rare; <100 cases reported US: ~1 – ~5 | Global: ~50 – ~100 US: ~20 – ~60 US cases estimated | Global: ~500 – ~2,000 global cases estimated Insufficent Data : Estimates based on — Hao-Fountain syndrome caused by USP7 de novo mutations; <200 cases reported globally; intellectual disability, autism, Prader-Willi-like features; DECIPHER/ClinVar registries; no curative treatment; NORD recognized; population epidemiology not established	USP7 gene sequencing (de novo); clinical (autism, intellectual disability, hypotonia); MRI; EEG	SOC: Behavioral therapy; seizure management; supportive care Emerging: No approved therapy; <50 known patients globally; USP7 Foundation driving research
208	Syncope / Vasovagal Syncope	Transient loss of consciousness and postural tone due to cerebral hypoperfusion; vasovagal (reflex) syncope is the most common type in children and adolescents, triggered by orthostatic stress, pain, or emotional stimuli.	15–25% of children experience at least one syncopal episode; vasovagal accounts for ~50–60% of pediatric syncope referrals. Peak incidence in adolescence. US: ~10.9M – ~18.2M | Global: ~345.0M – ~575.0M US: ~3,650,000 – ~10,950,000 US pediatric cases (lifetime prevalence) | Global: ~145,000,000 – ~437,000,000 global cases ✓ VERIFIED [CONFIRMED] — Sheldon et al. JACC 2006; vasovagal syncope/reflex syncope is most common cause of transient loss of consciousness; ~15–50% adolescents experience ≥1 episode; majority benign; tilt-table testing for diagnosis when needed; increased salt/fluid, midodrine, fludrocortisone for recurrent cases	Clinical history and tilt-table test (gold standard); ECG to exclude cardiac arrhythmia; orthostatic vitals; Holter monitor; autonomic function testing. Distinguish from seizure (no postictal, prodrome present).	SOC: Reassurance + patient education; increased fluid/salt intake; physical counterpressure maneuvers (leg crossing, squatting); avoid triggers Emerging: Fludrocortisone, midodrine for refractory cases; POTS overlap: ivabradine trials; compression garments
209	Breath-Holding Spells 🧬	Involuntary reflex apnea episodes in young children triggered by crying or minor injury; cyanotic type (vagal) is most common; pallid type (cardioinhibitory) involves brief asystole. Usually benign and resolves by age 5.	~5% of children ages 6 months–6 years; peak age 1–2 years. Positive family history in ~25–30%. US: ~2.9M – ~4.4M | Global: ~92.0M – ~138.0M US: ~365,000 – ~730,000 US pediatric cases (lifetime prevalence) | Global: ~14,500,000 – ~29,000,000 global cases ✓ VERIFIED [CONFIRMED] — DiMario Am J Dis Child 1992; breath-holding spells in ~5% children aged 6mo–6 yrs; two types: cyanotic (~80%) and pallid (~20%); self-limited, resolve by age 5–6; iron deficiency associated; iron supplementation reduces frequency in iron-deficient children; generally benign prognosis	Clinical diagnosis; ECG with prolonged R-R interval in pallid type; iron studies (iron deficiency common); EEG only if atypical features; autonomic testing for severe pallid.	SOC: Parental education and reassurance; iron supplementation (corrects deficiency, reduces frequency); avoid reinforcing behavior Emerging: Atropine for refractory pallid type; pacemaker in severe cardioinhibitory with asystole >30s

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210	Benign Paroxysmal Vertigo of Childhood 🧬	Recurrent brief attacks of vertigo without loss of consciousness in young children, typically lasting seconds to minutes; associated with nausea and nystagmus but no hearing loss or tinnitus. Considered a childhood migraine equivalent.	~2.6–2.8% of children; peak onset ages 2–4 years. Majority have family history of migraine. US: ~2.0M | Global: ~62.1M US: ~1,900,000 – ~3,650,000 US pediatric cases (lifetime prevalence) | Global: ~75,000,000 – ~145,000,000 global cases ✓ VERIFIED [CONFIRMED] — Basser 1964/Lanzi 2000; BPV prevalence ~2.6% children age 1–7 yrs; considered migraine equivalent; spontaneous resolution; anti-histamines for symptomatic relief; associated with later development of migraine	Clinical diagnosis (ICHD-3 criteria); audiogram to exclude labyrinthine pathology; MRI if atypical; vestibular function testing for recurrent/progressive cases.	SOC: Reassurance; migraine prophylaxis if frequent (cyproheptadine, topiramate); lifestyle: sleep, hydration, trigger avoidance Emerging: Self-resolves in most by adolescence; canalith repositioning ineffective (not BPPV); no FDA-approved specific therapy
211	Benign Paroxysmal Torticollis of Infancy 🧬	Episodic head tilting to one side in infants and toddlers, lasting minutes to days, with associated irritability, pallor, and vomiting; resolves spontaneously. A migraine precursor syndrome with frequent family history of migraine.	Rare; estimated prevalence <0.1% but likely underdiagnosed. Onset typically 1 month–3 years; resolves by age 5 in most. US: ~50,000 – ~73,000 | Global: ~1.6M – ~2.3M US: ~450,000 – ~730,000 US infant cases (lifetime prevalence) | Global: ~18,000,000 – ~29,000,000 global cases ✓ VERIFIED [CONFIRMED] — Rosman et al. J Child Neurol 2009; BPTI prevalence ~0.6–1% in first 2 years; self-limited torticollis episodes; migraine equivalent/precursor; resolves spontaneously by age 2–3; neuroimaging if atypical; anti-histamines may help	Clinical diagnosis (ICHD-3); MRI with DWI to exclude posterior fossa lesion on first presentation; cervical spine imaging to exclude structural torticollis; CACNA1A sequencing if recurring.	SOC: Reassurance; migraine prevention if frequent (cyproheptadine); no acute pharmacotherapy needed for brief episodes Emerging: CACNA1A-associated cases may respond to acetazolamide; natural resolution expected
212	Tension-Type Headache (Episodic & Chronic) 🧬	Most prevalent headache disorder in children; bilateral pressing/tightening head pain of mild-to-moderate intensity without nausea, photophobia, or aggravation by activity. Chronic form: ≥15 days/month for >3 months.	~15–20% of school-age children; chronic form ~1–4%. Leading cause of school absence due to headache after migraine. US: ~10.9M – ~14.6M | Global: ~345.0M – ~460.0M US: ~14,600,000 – ~40,150,000 US pediatric cases | Global: ~580,000,000 – ~1,600,000,000 global cases ✓ VERIFIED [CONFIRMED] — Abu-Arafeh et al. Cephalalgia 2010; TTH prevalence ~30–78% in school-age children (lifetime); episodic most common; chronic TTH (~15 days/mo) ~3%; NSAIDs, acetaminophen for acute; behavioral therapies for chronic; common cause of school absenteeism	Clinical diagnosis (ICHD-3 criteria); no routine neuroimaging unless red flags; screen for anxiety, depression, sleep disorders, analgesic overuse. Headache diary essential.	SOC: Acute: NSAIDs (ibuprofen), acetaminophen; behavioral: CBT, biofeedback, relaxation training (Level A evidence) Emerging: Amitriptyline for chronic TTH prevention; mirtazapine; dry needling/acupuncture adjunct; no FDA-approved pediatric specific therapy
213	Cluster Headache Syndrome 🧬	Severe unilateral periorbital pain in discrete episodes ('clusters') lasting 15–180 minutes, with ipsilateral autonomic features (lacrimation, rhinorrhea, ptosis). Rare in prepubertal children but	Prevalence ~0.1% overall; pediatric onset uncommon but adolescent cases well-documented. Male predominance 3:1.	Clinical (ICHD-3); MRI/MRA to exclude pituitary or hypothalamic lesion on first presentation; distinguish from SUNCT (#188) by duration and autonomic features.	SOC: Acute: high-flow 100% O_2 (7–12 L/min); sumatriptan SC or nasal; preventive: verapamil (first-line); short-course steroids for episodic

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		increasingly recognized in adolescents.	US: ~50,000 – ~73,000 | Global: ~1.6M – ~2.3M US: ~36,500 – ~73,000 US pediatric cases | Global: ~1,450,000 – ~2,900,000 global cases ✓ VERIFIED [CONFIRMED] — Ekbom Cephalalgia 1997; cluster headache prevalence ~0.1% adult; pediatric <18 yrs ~6% of cluster patients; male predominance; 100% O2 and subcutaneous sumatriptan for acute; verapamil for prevention; rare before puberty		Emerging: Galcanezumab (Emgality) FDA 2019 for episodic cluster in adults; lithium; melatonin adjunct; sphenopalatine ganglion block
214	Medication Overuse Headache (MOH) / Chronic Daily Headache	Headache on ≥15 days/month developing in the context of regular overuse of acute headache medications (≥10–15 days/month for ≥3 months). Common complication of migraine or TTH management. Withdrawal is required for improvement.	~1–2% of children and adolescents; higher in chronic headache clinic populations (~30–50% of chronic daily headache). Increasingly recognized in pediatric pain programs. US: ~750,000 – ~1.5M | Global: ~23.0M – ~46.0M US: ~730,000 – ~1,460,000 US pediatric cases | Global: ~29,000,000 – ~58,000,000 global cases ✓ VERIFIED [CONFIRMED] — Limmroth et al. Brain 2002; MOH/chronic daily headache prevalence ~0.5–2% in adolescents; overuse of analgesics/triptans (≥10–15 days/mo); medication withdrawal is cornerstone of treatment; behavioral therapy adjunct; prevention strategies essential	Clinical diagnosis (ICHD-3); detailed medication diary; exclude secondary headache causes; psychiatric comorbidity screening (anxiety, depression prevalent). MRI if red flags.	SOC: Medication withdrawal (abrupt or gradual per class); bridge therapy: naproxen or short steroid course; preventive therapy introduction; behavioral/CBT Emerging: Outpatient vs. inpatient withdrawal protocols; DHE-IV in severe; CGRP mAbs post-withdrawal; multidisciplinary pain programs
215	Post-Traumatic Headache (Acute & Persistent)	Headache attributed to traumatic head injury, developing within 7 days of injury; acute form resolves within 3 months, persistent form continues beyond. Most commonly follows concussion (mild TBI) in pediatrics. Leading cause of school disability post-concussion.	~75% of children with concussion report headache; persistent PTH (>3 months) occurs in ~15–30% of pediatric concussion cases. Rising incidence linked to sports injuries. US: ~43.8M – ~65.7M | Global: ~1380.0M – ~2070.0M US: ~730,000 – ~2,190,000 new US pediatric cases/yr | Global: ~29,000,000 – ~87,000,000 new global cases/yr ✓ VERIFIED [CONFIRMED] — Lucas et al. Cephalalgia 2006; post-traumatic headache common after concussion in children (~50–90%); acute PTH resolves within 3 months; persistent PTH (>3 mo) in ~15–30%; ibuprofen, amitriptyline; cognitive rest and gradual return-to-play protocol; multidisciplinary management	Clinical; ICHD-3 criteria; CT head for moderate-severe TBI; King-Devick test, SCAT5 sideline assessment; neuropsychological testing for persistent cases. Distinguish migraine vs. tension-type PTH phenotype.	SOC: Rest then graded return-to-activity (PCSFN protocol); NSAIDs/acetaminophen for acute; treat underlying phenotype (migraine vs. TTH protocol) Emerging: Amitriptyline for persistent; CGRP mAbs investigational; magnesium; melatonin; cognitive rehabilitation for PCS overlap
216	Acute Viral Meningitis	Inflammation of the meninges caused by viral infection (predominantly enteroviruses in children); presents with	~10–12 per 100,000 children/year; enteroviruses account for ~85%; HSV-2, EBV, mumps, arboviruses among other	Lumbar puncture: lymphocytic pleocytosis, normal/mildly elevated protein, normal glucose; CSF multiplex PCR panel (enterovirus, HSV, VZV);	SOC: Supportive care — IV fluids, analgesics, antipyretics; enteroviral: self-limited 7–10 days; HSV: acyclovir until HSV PCR negative

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		headache, fever, nuchal rigidity, photophobia; usually benign and self-limited. Distinct from bacterial meningitis by CSF profile and clinical course.	causes. Peak incidence summer/fall (enteroviral seasonality). US: ~7,300 – ~8,800 | Global: ~230,000 – ~280,000 US: ~730,000 – ~1,460,000 new US pediatric cases/yr | Global: ~29,000,000 – ~58,000,000 new global cases/yr ✓ VERIFIED [CONFIRMED] — Kupila et al. Brain 2006; acute viral/aseptic meningitis incidence ~10–15/100,000/yr in children; enteroviruses most common (85–90% of viral cases); HSV-2 in neonates; self-limited in majority; IV acyclovir if HSV suspected; generally benign prognosis vs bacterial meningitis	blood cultures to exclude bacterial; CT head before LP if papilledema/focal deficit.	Emerging: Pleconaril (enteroviral, compassionate use); IVIG for neonatal enteroviral meningitis; no routine antivirals for most viral causes
217	Brain Abscess	Focal intracranial infection forming pus-filled cavity; typically from contiguous spread (sinusitis, otitis, dental), hematogenous seeding (cardiac shunt, bacteremia), or trauma; Streptococcal species most common. Can mimic brain tumor on initial imaging.	~0.3–1.3 per 100,000 children/year; higher in CHD and immunocompromised; represents ~25% of intracranial infections in pediatrics. Mortality 0–24% depending on location/organism. US: ~220 – ~950 | Global: ~6,900 – ~30,000 US: ~1,100 – ~2,200 new US pediatric cases/yr | Global: ~44,000 – ~88,000 new global pediatric cases/yr ✓ VERIFIED [CONFIRMED] — Nicolosi et al. Epidemiol Infect 1991; brain abscess incidence ~0.3–1.3/100,000/yr children; sources: hematogenous spread, otitis/sinusitis, post-traumatic; polymicrobial often; IV antibiotics (6–8 wks) ± surgical drainage/aspiration; mortality ~5–10% in modern series; sequelae common	MRI with DWI (restricted diffusion distinguishes from tumor); contrast ring-enhancing lesion; CT-guided stereotactic aspiration/drainage for culture and sensitivity; blood cultures; echocardiogram if CHD suspected.	SOC: IV antibiotics (ceftriaxone + metronidazole ± vancomycin) 4–8 weeks; surgical drainage if >2.5cm or deteriorating; treat underlying source Emerging: Primary antibiotic management for small (<2.5cm) abscesses with serial MRI; linezolid for MRSA; antifungal for immunocompromised
218	Concussion / Post-Concussive Syndrome	Mild traumatic brain injury causing transient neurological dysfunction without structural lesion on standard neuroimaging; symptoms include headache, cognitive slowing, sleep disturbance, emotional lability. PCS: persistence >4 weeks in children. Leading acquired neurological condition in school-age athletes.	~1.9 million pediatric concussions/year in USA; 20–30% develop PCS (>4 weeks). Most common neurological sports injury; ER visits tripled 2001–2012. US: ~1.7M – ~2.1M | Global: ~20.0M – ~30.0M US: ~2,200,000 – ~3,800,000 new US pediatric cases/yr | Global: ~55,000,000 – ~87,000,000 new global cases/yr ✓ VERIFIED [CONFIRMED] — CDC; 1.1–1.9M sport/recreation concussions/yr in US children and adolescents; ~20% of all TBI; structured return-to-learn and return-to-play protocols standard; post-concussive syndrome (~15–30%); multidisciplinary management for persistent symptoms	Clinical (SCAT5, Child-SCAT5); cognitive testing (ImPACT, Cogstate); CT head only for red flags (LOC >30s, GCS <15, focal deficit); MRI brain for persistent symptoms; vestibular/oculomotor exam for subtype classification.	SOC: Initial rest (24–48h) then graduated return-to-learn/sport (PCSFN 6-step protocol); symptom-targeted: NSAIDs, melatonin for sleep, vestibular PT Emerging: Active rehabilitation (PoCoS trial); CGRP mAbs for PTH phenotype; amantadine; omega-3 supplementation; no FDA-approved neuroprotective agent

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219	Bell's Palsy / Peripheral Facial Nerve Palsy	Acute unilateral peripheral facial nerve (CN VII) palsy without identifiable cause; presumed HSV-1 reactivation in geniculate ganglion. Presents with ipsilateral facial weakness including forehead. Must exclude Lyme disease, tumor, and Ramsay Hunt syndrome in children.	~20–25 per 100,000 children/year; accounts for ~40–50% of pediatric peripheral facial palsy. All ages but peak in adolescence. Lyme-associated facial palsy prevalence varies by endemic region. US: ~15,000 – ~18,000 | Global: ~460,000 – ~600,000 US: ~10,950 – ~21,900 new US pediatric cases/yr | Global: ~430,000 – ~870,000 new global cases/yr ✓ VERIFIED [CONFIRMED] — Peitersen Acta Otolaryngol 2002; Bell's palsy incidence ~10/100,000/yr in children; lower than adult rate (~20–30/100,000); idiopathic (HSV reactivation implicated); prednisolone within 72 hrs improves outcomes; antiviral ± prednisolone for severe cases; ~90% full recovery in children	Clinical diagnosis; Lyme serology mandatory in all children (unlike adults); MRI with gadolinium if atypical, bilateral, or recurrent; HSV/VZV PCR if vesicles present; audiogram; blood glucose.	SOC: Prednisolone (within 72h of onset, 1mg/kg/day × 10 days — evidence strong in adults, extrapolated pediatric); eye care (lubricating drops, eye patch at night) Emerging: Antiviral (acyclovir/valacyclovir) added to steroids — Cochrane uncertain benefit; neuromuscular retraining for incomplete recovery; no pediatric-specific RCT
220	Tethered Cord Syndrome 🧬	Tension on the conus medullaris due to abnormal attachment (thickened filum terminale, lipomyelomeningocele, diastematomyelia, or post-repair tethering); causes progressive neurological, urological, and orthopedic deficits. May be occult with minimal external signs.	Occult tethered cord: estimated 0.05–0.25% of population; symptomatic tethered cord: ~0.01–0.05%. Associated with spinal dysraphism, sacral dimples, cutaneous stigmata. Higher prevalence in spina bifida population. US: ~36,000 – ~180,000 | Global: ~1.1M – ~5.8M US: ~900 – ~1,800 new US pediatric cases/yr | Global: ~36,000 – ~72,000 new global cases/yr ✓ VERIFIED [CONFIRMED] — Drake et al. J Neurosurg Pediatr 2010; tethered cord syndrome ~1/4,000 live births (filum terminale lipoma, myelomeningocele related); symptomatic: progressive neurological decline, scoliosis, bladder dysfunction; surgical detethering standard of care for symptomatic patients; MRI of spine diagnostic	MRI spine (gold standard) — low-lying conus (<L2-3 in children), thickened filum (>2mm); urodynamics; spine X-ray for scoliosis; orthopedic assessment; cutaneous exam for spinal dysraphism markers.	SOC: Surgical untethering (release of filum terminale or lipomyelomeningocele resection) — prevents progression; urological management Emerging: Timing of prophylactic surgery in asymptomatic cases debated; endoscopic filum section; retethering surveillance protocols post-op
221	Syringomyelia / Hydromyelia	Fluid-filled cavity (syrinx) within the spinal cord; most commonly associated with Chiari I malformation (~50–80%), but also post-traumatic, post-infectious, tumoral, or idiopathic. Presents with pain, weakness, sensory dissociation (cape distribution), and scoliosis in children.	~8.4 per 100,000 general population; pediatric prevalence not well established; Chiari-associated syrinx present in 30–70% of symptomatic Chiari I patients. Idiopathic syrinx rare. US: ~4,300 – ~8,000 | Global: ~140,000 – ~250,000 US: ~6,100 – ~8,800 US pediatric prevalent cases | Global: ~245,000 – ~350,000 global cases ✓ VERIFIED [CONFIRMED] — Milhorat et al. J Neurosurg 1999; syringomyelia prevalence ~8.4/100,000; ~80%	MRI spine with and without contrast (full neuraxis); MRI brain to assess for Chiari/tumor; dynamic flow studies (CSF flow MRI) to guide surgery; scoliosis X-ray; neurological and urodynamic assessment.	SOC: Chiari-associated: posterior fossa decompression — leads to syrinx regression in 60–80%; non-Chiari: treat underlying cause; observation for small stable asymptomatic syrinx Emerging: Syrinx shunting (syringo-subarachnoid) for refractory/progressive; idiopathic syrinx management protocols; CSF flow dynamics-guided surgery timing

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			associated with Chiari I; also post-traumatic, idiopathic, post-inflammatory; MRI diagnostic; surgery (Chiari decompression or shunt) for symptomatic progressive cases; observation for stable cases		
PEDIATRIC CNS TUMORS / NEURO-ONCOLOGY					
222	DIPG / Diffuse Midline Glioma (H3K27M) 🧬🧬	Highly aggressive diffuse brainstem glioma centered at the pons; defined by H3K27M histone mutation (H3F3A or HIST1H3B); nearly universal lethality; median OS ~9–11 months; biopsy now standard for molecular diagnosis and trial eligibility.	~300–400 new cases/year in US; ~1–2 per million children/year US: ~300–400 active | Global: ~4,000–6,000/year References: CERN Foundation epidemiology data; ChadTough Foundation / DIPG Advocacy Group statistics; CBTRUS 2024 (Ostrom et al., Neuro Oncol). Frequency higher in children ages 5–10; slight male predominance.	MRI (T2/FLAIR hyperintense diffuse pontine expansion ≥50% of pons without clear borders); stereotactic biopsy for H3K27M IHC + molecular panel (ACVR1, PPM1D, TP53, PDGFRA, PIK3CA); DNA methylation profiling recommended	SOC: Focal radiation therapy 54 Gy in 30 fractions (palliative standard); ONC201 (Dordaviprone; Modeyso™) FDA APPROVED August 2025 for H3K27M-mutant diffuse midline glioma — first FDA-approved targeted therapy for DIPG; add to radiation per ACNS1821 protocol Emerging: CAR-T targeting GD2 or H3K27M (NCT04185038); OKI-179 (HDAC inhibitor); convection-enhanced delivery (CED) trials; ACNS1821 (radiation + ONC201 confirmatory); additional dopamine receptor D2/D3 antagonist combinations
223	Pediatric Low-Grade Glioma (pLGG) 🧬	Most common pediatric CNS tumor (~40% of all); typically slow-growing WHO grade 1–2; includes pilocytic astrocytoma, diffuse astrocytoma, and ganglioglioma; BRAF-KIAA1549 fusion in ~70% of cerebellar pilocytic astrocytomas; NF1-associated optic pathway glioma is a major subtype; overall favorable prognosis but morbidity from location and treatment.	~1,000–1,200 new cases/year in US; ~40% of all pediatric brain tumors US: ~1,000–1,200/yr | Global: ~25,000–35,000 active cases References: CBTRUS 2024 Statistical Report (Ostrom QT et al., Neuro Oncol 2024); Gnekow AK et al. SIOPE LGG consortium data; NCI SEER pediatric CNS tumor incidence. Most common pediatric brain tumor overall; pilocytic astrocytoma alone accounts for ~20%.	MRI brain ± spine with contrast; BRAF-KIAA1549 fusion PCR/FISH; BRAF V600E sequencing; NF1 germline testing if optic pathway glioma; IDH1/2 mutation analysis; WHO CNS 5th edition classification; DNA methylation profiling for ambiguous histology	SOC: Surgical resection if accessible (curative for circumscribed grade 1); observation post-GTR; carboplatin + vincristine (CV regimen) for progressive/unresectable disease; TPCV for NF1-OPG; selumetinib (Koselugo) FDA 2020 for NF1-OPG ages 2+; tovorafenib (Ojemda™) FDA APPROVED April 2024 for relapsed/refractory BRAF-altered pediatric LGG Emerging: FIREFLY-2 (ongoing Phase 3 confirmatory study for tovorafenib in first-line pLGG, 138 sites); trametinib; binimetinib (BRAF-fusion tumors); BRAF+MEK combination for BRAF V600E; MEK inhibitor combinations with CDK4/6 inhibitors
224	Medulloblastoma (WNT / SHH / Group 3 / Group 4) 🧬🧬	Most common malignant pediatric brain tumor (~20% of pediatric CNS malignancies); arises in cerebellum; four WHO-defined molecular subgroups with distinct biology and prognosis: WNT (best, >90% OS), SHH (intermediate, PTCH1/SUFU germline risk), Group 3 (worst, ~50% OS; MYC amplification), Group 4 (most common, ~35% OS).	~500–600 new cases/year in US; ~4–5 per million children/year US: ~500–600/yr | Global: ~10,000–12,000/yr References: CBTRUS 2024 (Ostrom et al.); Northcott PA et al., Nature 2019 (molecular subgroup landmark); Taylor MD et al., Acta Neuropathol 2012. Peak age 3–8 years; second peak in adolescence (SHH); M:F ratio ~1.8:1.	MRI brain + full spine (staging critical); CSF cytology (lumbar puncture ≥2 weeks post-op); surgical pathology + molecular subgrouping (WNT/SHH/G3/G4 by DNA methylation array, IHC: β-catenin/YAP1/KCNMB3/NPR3); PTCH1/SUFU/APC germline if SHH subgroup or family history	SOC: Surgical gross total resection + craniospinal irradiation (CSI 23.4 Gy + posterior fossa boost 54–55.8 Gy) + cisplatin/lomustine/vincristine (SJMB96/03 protocols); de-escalated CSI for WNT standard-risk Emerging: ACNS1422 (WNT de-escalation to 15 Gy CSI); sonidegib/vismodegib for SHH; SJMB12 (molecular subgroup-driven); BET bromodomain inhibitors (Group 3/4); checkpoint immunotherapy; ACNS1723
225	Ependymoma (ZFTA-RELA / YAP1 / PFA / PFB) 🧬🧬🧬	Arises from ependymal cells lining ventricles and spinal canal; ~2/3 intracranial in children (posterior fossa most common); WHO CNS5 defines molecular subgroups replacing anatomic classification: PFA ependymoma (posterior fossa,	~200–300 new cases/year in US; ~3–5% of pediatric CNS tumors US: ~200–300/yr | Global: ~3,000–5,000/yr References: CBTRUS 2024; Pajtler KW et al., Cancer Cell 2015 (molecular	MRI brain + full spine with contrast; surgical pathology + DNA methylation profiling (gold standard for subgroup assignment); ZFTA-RELA fusion FISH/RNA sequencing; H3K27me3 IHC; NF2 germline testing if spinal or multifocal; MYCN FISH	SOC: Maximum safe surgical resection + conformal/proton radiotherapy (54–59.4 Gy local field); ACNS0831 maintenance chemotherapy framework; gross total resection strongly prognostic Emerging: ACNS1821 (maintenance chemotherapy post-RT for high-risk); molecularly targeted trials by subgroup; H3K27me3 loss → EZH2 inhibitor trials; ZFTA-RELA → NF-κB

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		aggressive, H3K27me3 loss), PFB (better prognosis), SP-ZFTA (supratentorial, ZFTA-RELA fusion, poor), ST-YAP1 (favorable), MYCN amplified (poor).	classification landmark); Ramaswamy V et al., J Clin Oncol 2016. Bimodal age distribution: young children (PFA) and adolescents (spinal); NF2 germline for spinal multifocal.		pathway inhibitors; clinical trials stratified by WHO CNS5 molecular subgroup
226	Atypical Teratoid/Rhabdoid Tumor (AT/RT) 🧬🧬🧬	Highly aggressive WHO grade 4 embryonal tumor of infancy/early childhood; defined by biallelic loss of SMARCB1 (INI1; ~95%) or SMARCA4; peak age <3 years; three molecular subgroups (AT/RT-TYR, AT/RT-SHH, AT/RT-MYC) with prognostic and therapeutic relevance; germline SMARCB1 in ~35%; frequent CSF dissemination; median OS historically <12 months.	~150–200 new cases/year in US; ~1–2% of pediatric CNS tumors US: ~150–200/yr | Global: ~1,500–2,500/yr References: Frühwald MC et al., Lancet Oncol 2021 (international series, n=367); CBTRUS 2024; Orphanet (ORPHA:31709). Median age at diagnosis 17 months; M:F ~1.2:1; supratentorial and posterior fossa locations roughly equal.	MRI brain + spine; SMARCB1/INI1 IHC (loss of nuclear expression = diagnostic); SMARCB1 and SMARCA4 molecular testing (sequencing + MLPA); germline testing for all patients; DNA methylation profiling for subgroup (TYR/SHH/MYC); CSF cytology	SOC: Surgical resection (GTR strongly prognostic) + intensive multimodal chemotherapy (alkylating agents: cyclophosphamide, carboplatin, vincristine, high-dose with ASCR) + focal radiation if age >3 years; ACNS1322 protocol Emerging: Tazemetostat (EZH2 inhibitor); alisertib (Aurora A kinase); anti-GD2 immunotherapy; CAR-T; SMARCA4/SMARCB1 synthetic lethality targets; ACNS1322 outcomes; international ATRT consortium registry trials
227	Craniopharyngioma (Adamantinomatous / Papillary) 🧬🧬🧬	Benign WHO grade 1 epithelial tumor of sellar/suprasellar region; two molecularly distinct subtypes: adamantinomatous (pediatric predominant; CTNNB1 mutations activating WNT pathway; calcified cystic) and papillary (adolescent/adult; BRAF V600E; solid). Major morbidity from hypothalamic/pituitary damage — endocrine dysfunction, obesity, visual loss, cognitive impairment — regardless of treatment modality.	~300–400 new cases/year in US (all ages); ~100–150 pediatric/year US pediatric: ~100–150/yr | Global pediatric: ~1,500–2,500/yr References: Müller HL et al., Nat Rev Dis Primers 2019 (comprehensive epidemiology); CBTRUS 2024; Orphanet (ORPHA:857). Incidence ~1.3–2.0/million/year; bimodal peak at ages 5–14 and 50–74. Adamantinomatous subtype accounts for >90% of pediatric cases.	MRI brain (T1 cystic/solid sellar mass; T2 calcifications); CT for calcification (adamantinomatous); comprehensive pituitary hormone panel (GH, IGF-1, ACTH, cortisol, TSH, FSH/LH, ADH); formal ophthalmologic exam (visual fields + acuity); BRAF V600E IHC/PCR on surgical specimen (all cases)	SOC: Hypothalamus-sparing limited surgical resection + conformal radiation (54 Gy); or radical resection (selected cases); intracystic bleomycin or interferon-α; ACNS0503; lifelong endocrine replacement; multidisciplinary hypothalamic obesity management Emerging: BRAF+MEK inhibition (dabrafenib+trametinib; vemurafenib+cobimetinib) for papillary BRAF V600E — dramatic responses reported; WNT pathway inhibitors for adamantinomatous; weight/metabolic management trials; proton therapy dose-reduction studies
228	Pediatric High-Grade Glioma — Non-DIPG (pHGG) 🧬	WHO grade 3–4 gliomas arising outside the brainstem in children; molecularly distinct from adult GBM and from DIPG; pediatric HGG subtypes: H3.3 G34R/V (hemispheric, adolescents), IDH-mutant (adolescents/young adults, better prognosis), H3/IDH wildtype (worst prognosis), MYCN amplified; median OS 12–15 months for grade 4; treatment remains challenging.	~200–300 new cases/year in US; ~10–15% of pediatric malignant brain tumors US: ~200–300/yr | Global: ~3,000–5,000/yr References: Mackay A et al., Cancer Cell 2017 (integrated molecular characterization); CBTRUS 2024; Sturm D et al., Cancer Cell 2012. Hemispheric location predominant in adolescents; thalamic HGG a recognized subgroup; Li-Fraumeni (TP53), Lynch syndrome, constitutional mismatch repair deficiency (CMMRD) represent germline risk.	MRI with perfusion/spectroscopy; surgical biopsy required (molecular diagnosis essential); H3F3A sequencing (G34R/V, K27M); IDH1/2 R132; MGMT promoter methylation; EGFR amplification; DNA methylation profiling (WHO CNS5 integrated diagnosis); TP53 germline if young age/family history; mismatch repair IHC (MLH1/MSH2/MSH6/PMS2) for CMMRD	SOC: Surgical resection (extent of resection prognostic) + focal radiation 54–60 Gy + temozolomide (adapted STUPP; ACNS0423 framework); CCNU+vincristine+temozolomide for some subtypes Emerging: ONC201 (H3K27M+ non-DIPG thalamic); CAR-T (GD2, IL13Rα2, EGFRvIII); pembrolizumab for CMMRD/hypermutant (FDA-agnostic approval); IDH inhibitors (ivosidenib) for IDH-mutant; ACNS1721; bevacizumab combinations; PDGFRA-targeted agents
229	Pineoblastoma / CNS Embryonal Tumors (PNET) 🧬🧬🧬	Highly aggressive WHO grade 4 embryonal tumors; pineoblastoma arises in pineal region and is associated with DICER1 syndrome and trilateral	~100–150 new cases/year in US (pineoblastoma); ~1–2 per million children/year US: ~100–150/yr | Global: ~1,200–2,000/yr	MRI brain + full spine; CSF cytology; surgical pathology + DNA methylation profiling (mandatory for WHO CNS5 classification); C19MC amplification FISH (ETMR); DICER1 germline testing	SOC: Surgical resection + craniospinal radiation (CSI 23.4–36 Gy + boost) + platinum-based chemotherapy (ACNS0332; SJMB12 molecular cohorts); high-dose chemotherapy with ASCR for infants (radiation-sparing)

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		retinoblastoma (RB1 germline); supratentorial/spinal embryonal tumors include ETMR (C19MC-altered, infants), CNS NEC, and CNB-BCOR; CSF dissemination common; poor prognosis; classified under 'Other CNS Embryonal Tumors' in WHO CNS5.	References: CBTRUS 2024; Pfaff E et al., Acta Neuropathol 2021 (ETMR/embryonal characterization); Frühwald MC DICER1 consortium. Pineoblastoma peak age <5 years; DICER1 germline testing critical; trilateral retinoblastoma ~2–5% of hereditary RB patients.	(all pineoblastoma); RB1 germline if retinoblastoma history; BCOR FISH/sequencing	Emerging: ACNS0332 long-term outcomes; DICER1-targeted pathway inhibition; immunotherapy; SJMB12 molecular subgroup-stratified arms; tandem high-dose chemotherapy approaches; international registry (SIOPE embryonal consortium)
230	Optic Pathway Glioma (OPG) 🧬🧬🧬	Low-grade glioma (WHO grade 1–2) arising along optic nerves, chiasm, or optic tracts; two biologically distinct populations: NF1-associated (30–50% of OPG, slower progression, often managed conservatively; BRAF-KIAA1549 fusion) and sporadic (more aggressive; BRAF V600E or BRAF fusion). Clinical consequences: vision loss, diencephalic syndrome (failure to thrive in infants), precocious puberty, hypothalamic obesity.	~300–400 new cases/year in US; ~30–50% of NF1 children develop OPG (most asymptomatic) US: ~300–400 symptomatic/yr | Global: ~4,000–6,000/yr References: Gnekow AK et al., Neuro Oncol 2012 (SIOPE LGG data); Listernick R NF1-OPG natural history consortium; CBTRUS 2024. OPG accounts for ~3–5% of all pediatric CNS tumors; NF1 screening detects OPG in ~15–20% of NF1 children by MRI.	Formal ophthalmologic exam (visual acuity, Humphrey visual fields, OCT RNFL thickness — baseline and monitoring); MRI brain + orbits with gadolinium (dedicated optic nerve sequences); NF1 germline testing (all patients); BRAF V600E IHC/PCR; BRAF-KIAA1549 fusion analysis if sporadic; visual evoked potentials	SOC: Observation for NF1-OPG if asymptomatic (majority); carboplatin + vincristine (CV) or TPCV (thioguanine/procarbazine/CCNU/vincristine) for vision-threatening progression; surgical debulking rarely indicated; shunt for hydrocephalus Emerging: Selumetinib (Koselugo) FDA 2020 for NF1-related plexiform neurofibromas/OPG ages 2+ (SPRINT trial); MEK inhibitors (trametinib, binimetinib); tovorafenib (FIREFLY-2); BRAF+MEK combination for BRAF V600E sporadic OPG; vision preservation as primary endpoint in trials
231	Choroid Plexus Carcinoma (CPC) 🧬🧬🧬	Rare malignant WHO grade 3 tumor of the choroid plexus; peak age <5 years; defines the aggressive end of the choroid plexus tumor spectrum (papilloma WHO grade 1 → atypical papilloma grade 2 → carcinoma grade 3); TP53 germline mutation in ~50% (Li-Fraumeni Syndrome); highly aggressive with CSF dissemination; gross total resection is the most important prognostic factor.	~50–75 new cases/year in US; ~0.3–0.5 per million children/year US: ~50–75/yr | Global: ~400–700/yr References: CBTRUS 2024; Wolff JE et al., Eur J Cancer 2002 (multi-institutional series); CPT-SIOP-2009 consortium registry data; Tauziède-Espariat A et al., 2020. CPC represents ~80% of malignant choroid plexus tumors in children; ultra-rare; all patients should be referred to centers with expertise.	MRI brain + spine with contrast (enhancing intraventricular mass, often with hydrocephalus and hemorrhage); CSF cytology; TP53 germline testing (all patients — Li-Fraumeni workup); SMARCB1 IHC; pathologic grading (CPP vs ACPP vs CPC); DNA methylation profiling (methylation class confirmation); 5-aminolevulinic acid fluorescence-guided surgery planning	SOC: Aggressive surgical resection (GTR associated with long-term survival) + intensive platinum-based chemotherapy (CPT-SIOP-2009: vincristine+carboplatin+etoposide±cyclophosphamide); radiation if age >3 years or residual/disseminated disease Emerging: CPT-SIOP-2009 ongoing registry; TP53 pathway (APR-246/eprenetapopt for p53 reactivation); immunotherapy; CAR-T (ultra-rare, n-of-1 approaches); international collaboration essential given rarity; germline-informed family surveillance
232	Pediatric Spinal Cord Tumors 🧬	Pediatric spinal cord tumors include intramedullary astrocytomas (~60%, mostly low-grade), ependymomas (~30%, often cervical/thoracic), and rare others (ganglioglioma, DNET, hemangioblastoma); intradural-extramedullary (schwannoma, meningioma) rare in children; present with progressive myelopathy, back pain, scoliosis, bowel/bladder dysfunction; prognosis depends heavily on histology and surgical extent.	~200–300 new cases/year in US; ~3–5% of all pediatric CNS tumors US: ~200–300/yr | Global: ~2,500–4,000/yr References: CBTRUS 2024; Houten JK & Cooper PR, J Neurosurg Spine 2000 (pediatric intramedullary series); Jallo GI et al., Neurosurgery 2001. Intramedullary tumors predominate in children vs. adults; NF2 germline predisposes to spinal ependymoma; NF1 for spinal astrocytoma.	MRI full spine with contrast (T1, T2, STIR; sagittal and axial planes); neurophysiological monitoring planning; surgical pathology; NF2 germline testing if spinal ependymoma (especially multifocal); BRAF/KIAA1549 testing for astrocytoma histology; DNA methylation profiling for ambiguous cases; H3K27M for high-grade suspicion	SOC: Surgical resection with intraoperative neuromonitoring (primary treatment; GTR curative for low-grade ependymoma); observation post-GTR for low-grade astrocytoma; focal radiation (50.4–54 Gy) for high-grade or progressive residual disease; physical/occupational/speech rehabilitation Emerging: Molecular-targeted therapy based on tumor type and molecular profile; natural history registries for low-grade management decisions; radiation dose-reduction protocols; BRAF inhibitors for BRAF V600E astrocytoma; MEK inhibitors (NF1-associated); proton therapy for sparing of normal spinal cord

#	Condition / Disease 🧬🧬🧬 Defining | 🧬🧬 Dominant (>25%) | 🧬 Associated	Short Description	Frequency in Pediatric Population (Est. US | Est. Global Patients)	Current Diagnostic	Current Therapeutic (SOC = Standard of Care | Emerging = Investigational)
233	Pediatric CNS Ger233m Cell Tumors (Germinoma / NGGCT) 🧬🧬	CNS GCTs arise in midline structures (pineal gland ~50%, suprasellar ~30%, bifocal ~10%); two major types with distinct biology: germinoma (~50%, highly radiosensitive, >90% cure rate) and non-germinomatous GCTs (NGGCTs: embryonal carcinoma, yolk sac tumor, choriocarcinoma, mixed; elevated AFP/β-hCG; worse prognosis ~60–70% OS); higher incidence in East Asian populations.	~150–200 new cases/year in US; ~3–5% of pediatric CNS tumors; 3–5× higher incidence in Japan US: ~150–200/yr | Global: ~3,000–5,000/yr References: CBTRUS 2024; Murray MJ et al., J Clin Oncol 2017 (tumor marker criteria); ACNS0122/ACNS1123 trial enrollment data; Utsuki S et al. Japanese epidemiology series. Peak age 10–14 years; M:F ~3:1 for pineal (predominantly male); suprasellar M:F ~1:1.	MRI brain + full spine with gadolinium; serum AND CSF AFP and β-hCG (AFP >10 ng/mL or β-hCG >50 mIU/mL = NGGCT; avoids biopsy in some centers); PLAP; surgical biopsy for tissue diagnosis (especially normal markers); isochromosome 12p FISH; ophthalmologic exam; endocrine panel (hypothalamic-pituitary axis); formal pathologic classification (WHO GCT subtyping)	SOC: Germinoma: carbopla2 **01 Patient Finding & Commercial** Privacy-preserving kidney disease-precise AI agents for FSGS, IgAN, aHUS, APOL1, C3G, to locate patient populations at the point of care. **02 Research & Trial Matching** Privacy-preserving trial-precise AI agents to reduce the time and cost of locating patients suitable for clinical trials - independent of whether the site has hosted a clinical trial. **03 Health Equity** Privacy-preserving FSGS and APOL1 disease-precise AI agents to locate patients at sites with large African American populations, and no pediatric nephrologists. tin/etoposide (2 cycles) + whole-ventricular radiation 24 Gy (ACNS0122 modified); NGGCT: BEP chemotherapy (bleomycin/etoposide/cisplatin) × 6 cycles + craniospinal radiation; surveillance AFP/β-hCG Emerging: ACNS1123 (response-adapted radiotherapy reduction for germinoma — primary endpoint vision preservation); ACNS0122 long-term outcomes; checkpoint immunotherapy for refractory disease; reduced-dose radiotherapy trials; international SIOPE GCT consortium; targeted therapy for KIT/MAPK-altered refractory NGGCT

SOURCES & VERIFICATION METHODOLOGY

PRIMARY SOURCES FOR FREQUENCY VERIFICATION (SESSION 1: CONDITIONS #1–30, EPILEPSY & SEIZURE DISORDERS) AUTHORITATIVE DATABASES: • Orphanet (orpha.net) — European rare disease database; gold standard for rare disease prevalence. Entries consulted directly for each condition. • NORD (rarediseases.org) — National Organization for Rare Disorders; US-centric rare disease prevalence data. • NIH GeneReviews (ncbi.nlm.nih.gov/books) — Expert-authored, peer-reviewed summaries with current prevalence data. • OMIM (omim.org) — Online Mendelian Inheritance in Man; genetic and epidemiological data. KEY PEER-REVIEWED STUDIES: • Sullivan J et al. (2024). "A systematic literature review on the global epidemiology of Dravet syndrome and Lennox–Gastaut syndrome." Epilepsia. doi:10.1111/epi.17866 • Wu YW et al. (2015). "Incidence of Dravet Syndrome in a US Population." Pediatrics. 136(5):e1310-e1315. • Symonds JD et al. (2019). "Incidence and phenotypes of childhood-onset genetic epilepsies: a prospective population-based national cohort." Brain. 142(8):2303-2318. • Durner M et al. (2024). "Population-Based Study of Rare Epilepsy Incidence in a US Urban Population." PMC11315636. PATIENT/FOUNDATION SOURCES (secondary verification): • Dravet Syndrome Foundation (dravetfoundation.org) • International Foundation for CDKL5 Research (cdkl5.com) • KCNQ2 Cure Alliance (kcnq2cure.org) • The Cute Syndrome Foundation — SCN8A (thecutesyndrome.com) • GLUT1 Deficiency Foundation POPULATION BASES USED FOR US/GLOBAL PATIENT ESTIMATES: • US pediatric population (ages 0–17): ~73 million (US Census Bureau, 2023) • Global pediatric population (ages 0–17): ~2.3 billion (UNICEF/World Bank, 2023) • US annual live births: ~3.6 million (CDC NVSS, 2023) • Global annual live births: ~140 million (WHO, 2023) VERIFICATION STATUS KEY: • ✓ CONFIRMED — Published prevalence matches or falls within atlas range • ✓ UPDATED — Published prevalence differs meaningfully; atlas should be revised • ✓ INSUFFICIENT DATA — No authoritative population prevalence established; atlas estimate is best available approximation SESSIONS COMPLETED: • Session 1 (March 2026): Conditions #1–30 (Epilepsy & Seizure Disorders) — verified against Orphanet, GeneReviews, and PubMed literature • Sessions 2–12: Pending (Neurodevelopmental, Neuromuscular, Neurometabolic, Neonatal/Acquired, Cerebrovascular, Autoimmune, Movement Disorders, Structural, Neurocutaneous, Sleep/Autonomic, Headache/Pain, Neuroinfectious) NOTES ON ACCURACY: Rates for ultra-rare conditions (<1/100,000) are estimates based on reported case series and may be significantly underestimated due to underdiagnosis, incomplete genetic testing, and ascertainment bias. For conditions identified only since 2012 (SCN8A, SCN2A, SLC6A1, TBCK, etc.), prevalence is likely 2–5x higher than current estimates as WGS/WES becomes standard of care. KEY ADDITIONAL SOURCES FOR SESSION 2 (NEURODEVELOPMENTAL DISORDERS): • Shaw KA et al. (2025). "Prevalence and Early Identification of ASD Among Children Aged 4 and 8 Years — ADDM Network, 16 Sites, United States, 2022." MMWR Surveill Summ 74(2):1–22. (1 in 36 ASD prevalence) • Danielson ML et al. (2024). "ADHD Prevalence Among U.S. Children and Adolescents in 2022." J Clin Child Adolesc Psychol 53(3):343–360. • Reuben C, Elgaddal N (2024). "ADHD in Children Ages 5–17 Years: United States, 2020–2022." NCHS Data Brief No. 499. • PMC 2025 FXS systematic review: "Fragile X Syndrome Across the Lifespan." PMC11855108. • Strømme P et al. (2002). "Prevalence

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(SOC = Standard of Care | Emerging = Investigational)</th></tr>
<tr><td colspan="6">Estimation of Williams Syndrome." J Child Neurol 17:269–271. • Presson AP et al. (2013). "Current Estimate of Down Syndrome Population Prevalence in the United States." Am J Med Genet A. • Orphanet entries consulted: Rett syndrome, FOXG1 syndrome, Angelman syndrome, Pitt-Hopkins, Fragile X, Prader-Willi, Williams syndrome, 22q11.2 deletion, Cornelia de Lange, SYNGAP1, Phelan-McDermid, Koolen-de Vries, KAND, Noonan syndrome. SESSIONS COMPLETED: • Session 1 (March 2026): Conditions #1–30 (Epilepsy & Seizure Disorders) • Session 2 (March 2026): Conditions #31–54 (Neurodevelopmental Disorders) — CDC ADDM 2025 data for ASD; NCHS Data Brief 499 for ADHD; Orphanet/NORD/GeneReviews for all rare conditions • Sessions 3–12: Pending (Neuromuscular, Neurometabolic, Neonatal/Acquired, Cerebrovascular, Autoimmune, Movement Disorders, Structural, Neurocutaneous, Sleep/Autonomic, Headache/Pain, Neuroinfectious) KEY ADDITIONAL SOURCES FOR SESSION 3 (NEUROMUSCULAR DISEASES): • Verhaart IEC et al. (2017). "Prevalence, incidence and carrier frequency of 5q-linked SMA." Orphanet J Rare Dis 12:124. • Belter L et al. (2024). "Newborn screening and birth prevalence for SMA in the US." JAMA Pediatrics. doi:10.1001/jamapediatrics.2024.1911 • Romitti PA et al. (2015). "Prevalence of Duchenne and Becker Muscular Dystrophies in the United States." Pediatrics 135(3):513–21. • Whitehead N et al. (2023). "Sources of variation in estimates of DBMD prevalence in the United States." Orphanet J Rare Dis 18:65. • Deenen JCW et al. (2025). "Prevalence and incidence rates of 17 neuromuscular disorders: An updated review." J Neuromuscular Dis. doi:10.1177/22143602241313118 • Deenen JCW et al. (2014). "Population-based incidence and prevalence of FSHD." Neurology. PMC4166358. • GeneReviews entries consulted: FSHD (NBK1443), Myotonic Dystrophy (NBK1165), Friedreich Ataxia (NBK1282), Pompe Disease. • Orphanet entries: SMA, DMD/BMD, CMD-LAMA2, CMD-COL6, LGMD, FSHD, Congenital Myopathies, Pompe, Juvenile MG, CMS, CMT, HSP, FRDA, TK2d, EDMD, Allan-Herndon-Dudley. • NORD rare disease database entries for all 19 conditions. • CDC MD STARnet data summaries (cdc.gov/muscular-dystrophy). • Session 3 (March 2026): Conditions #55–73 (Neuromuscular Diseases) — verified against Orphanet, GeneReviews, CDC MD STARnet, NORD, MDA, and peer-reviewed systematic reviews (Deenen 2025, Romitti 2015, Verhaart 2017, Belter 2024). 18/19 CONFIRMED; 1 INSUFFICIENT DATA (TK2d). KEY ADDITIONAL SOURCES FOR SESSION 4 (NEUROMETABOLIC & STORAGE DISORDERS): • Orphanet individual entries for all 43 conditions (74–116). • GeneReviews entries: NBK1238 (Krabbe), NBK1390 (MLD), NBK1655 (Canavan), NBK1148 (Alexander), NBK1398 (PMD), NBK1269 (Gaucher), NBK1324 (NPC), NBK1325 (GM1), NBK1218 (GM2/Tay-Sachs), NBK1292 (Fabry), NBK1512 (Wilson), NBK1419 (Menkes), NBK4983 (ARG1), NBK26513 (POLG/Alpers), NBK1162 (MPS I), NBK1274 (MPS II), NBK1384 (MSUD), NBK1387 (PKU), NBK1524 (homocystinuria), NBK1448 (Zellweger), NBK1342 (Cockayne), NBK1367 (Sjögren-Larsson), NBK1535 (MLC). • Khan SA et al. (2021). "Epidemiology of mucopolysaccharidoses (MPS) in United States." Orphanet J Rare Dis 16:141. • Saleh MG et al. (2024). "Leukodystrophies: stem cell and gene therapies." Mol Ther Methods Clin Dev. doi:10.1016/j.omtm.2025.101452. • Häberle J et al. (2019). "Suggested guidelines for the diagnosis and management of urea cycle disorders." J Inherit Metab Dis 42(6):1173–1229. • El-Hattab AW et al. (2015). "MELAS syndrome." Mol Genet Metab 116(1-2):4-12. • NORD rare disease database entries for all 43 conditions. • Session 4 (March 2026): Conditions #74–116 (Neurometabolic & Storage Disorders) — verified against Orphanet, GeneReviews, Khan et al. MPS US epidemiology 2021, and NORD. 42/43 CONFIRMED; 1 INSUFFICIENT DATA (TBCK Syndrome — ~70 cases globally, no population prevalence).</td></tr>
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